CONGRUENCE DATA AND COMPACT INDUCTION IN SMALL CHARACTERISTIC

This note gives congruence data that produce supercuspidal representations by compact induction for every connected reductive group over every nonarchimedean local field. Theorem 11.1 imposes no restriction on the residue characteristic or the ramification of a splitting field. On a central extension with simply connected derived group, elliptic toral functionals give deep congruence characters whose intertwining is contained in an open subgroup compact modulo center. A finite induction supplies the inducing representations. A smooth central twist then lets the entire inducing pair descend to the original group.

The proof identifies group intertwining with transport of the full coadjoint coset, using exact equivariant coordinates on unipotent radicals and a fixed depth margin for the remaining analytic coordinates. Properness of the action on rationally elliptic covectors supplies the required compact subgroup. The earlier sections also compute root subgroups with different depths that restore a perfect finite commutator pairing. For Sp_4 over every local field of characteristic two, we compute the inducing subgroup and all representations containing an explicit family of characters, including characters whose leading reductions are not elliptic.

The Heisenberg and Clifford steps follow the architecture of Yu's construction and its refinements [1, 2, 3]. The occurrence statements for this construction exhaust the representations containing the specified data. They do not supply extraction from an arbitrary supercuspidal representation. The depth margin can always be arranged for the constructed data; it cannot be imposed on a character already known to occur without a further proof.

Existence of supercuspidal representations for all these groups is already known by Kret's finite-group construction [8]. The subject here is the specified congruence construction and its intertwining calculation. Its outputs do not exhaust all supercuspidals: Proposition 12.2 and Corollary 12.3 give an explicit depth-zero counterexample to universal occurrence in this family, even after central extension and twisting.

Theorem 30.3 compares any two supplied compact inducing pairs by finitely many matrix equations with an explicit Cartan bound. Thus equivalence of successful inducing data is already controlled. For extraction, Proposition 28.3 gives a finite refinement that preserves occurrence under an explicit precision hypothesis. The remaining coordinates retain the radical of the commutator pairing. Their compatibility with earlier stages is not established. Propositions 13.2 and 13.4 also show why increasing the level at a fixed vertex cannot serve as a universal extraction procedure.

Proposition 31.1 gives the exact formal-degree criterion for an occurring representation of an open subgroup compact modulo center to induce the prescribed supercuspidal. It also distinguishes an irreducible compact induction from a compact induction consisting of several copies of one supercuspidal. The latter has a matrix algebra of endomorphisms; choosing a primitive idempotent in that algebra does not by itself produce a new inducing subgroup.

Theorem 15.2 makes closure under the coefficient blocks a finite decision for every supplied irreducible local model. A radius depending on the local subgroup and fiber dimension bounds every possible finite block. The procedure either finds inducing data or certifies failure of that model. Theorem 32.2 gives a positive extraction criterion: an occurring type at a full special vertex induces the given supercuspidal if its restriction to the intersection with a minimal parabolic is irreducible. Neither result asserts that every supercuspidal has a local model satisfying its successful alternative.

Theorem 16.1 also makes a change of embedding within a fixed finite multiplicity space a finite system of polynomial equations. Proposition 16.4 shows that this system can have no solution even when the representation is already compactly induced. Its proof gives an explicit change of local type that succeeds in that example. Thus changing the embedding alone does not provide universal extraction.

Theorem 18.1 computes the global complement of the outgoing coefficient blocks. Their total rank satisfies $R = n$ or $R \geq 2n$ when the coefficient projection is nontrivial. Thus the previously considered strict-descent interval $n < R < 2n$ is empty. The case $R = n$ gives compact induction directly. Proposition 18.2 also extracts an inducing pair at either endpoint of the diagonal bounds for every R. At $R = 2n$, Theorem 19.1 gives an explicit successful rotation when $c = 1/2$ and the complement has the original local type. These

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results apply in every residue characteristic and impose no splitting hypothesis. Theorems 20.1 and 20.2 remove that type hypothesis whenever the active coefficient blocks have two neighbor orbits at $R = 2n$. They construct equivariant unitary operators directly from the blocks. Lemma 21.1 identifies reducing subspaces forced by equality in the rank inequality for the projection equations. It yields actual compact induction at $R = 2n$ if every range projection has full central support in the local range algebra, if every individual block has rank at most two, or if the fiber dimension is prime (Theorems 21.2, 22.4, and Corollary 22.6). In these cases the closure of an original fiber is supported on at most $4n$ coordinates. Theorem 21.3 also handles a single active orbit of degree three. Theorem 23.1 treats every commutative local range algebra, with arbitrary multiplicities, by an explicit decomposition into triangles or four-cycles. Theorems 24.2 and 24.7 give further criteria for noncommutative algebras acting with multiplicity one. They recover central incidence spaces from the images of other range projections, allowing finite sums and intersections. This permits arbitrarily large individual block ranks. Corollary 25.5 also handles the spanning condition when the local factors have prime multiplicity or multiplicity four. Corollary 26.2 completes extraction from every boundary model with fiber dimension at most eleven. These assertions concern supplied coefficient models; they do not produce such models for every supercuspidal representation.

Proposition 27.1 gives further changes of local data from individual coefficient orbits, without a bound on the total rank R. Theorem 27.3 bounds the number of steps that strictly decrease the formal-degree ratio and then the fiber dimension. The bound uses only the initial fiber dimension and compact subgroup volumes. Corollary 27.4 extracts every terminal model with $R \le 2n$. Models with larger rank can remain, so this finite descent does not yet give universal exhaustion.

Theorem 33.3 and Corollary 33.4 give an exhaustive construction under the single condition $p \nmid |W|$, where W is the absolute Weyl group. They require no tame splitting hypothesis on G over F. This consequence of the known exhaustion and equivalence theorems uses Yu data over the fields defining the absolute simple factors, a compatible central character, and finite Clifford theory. It includes wildly ramified restriction of scalars and arbitrary central tori.

1. The integral obstruction to changing a representative

Let $\mathcal{O}$ be a discrete valuation ring with uniformizer ϖ, fraction field F, and normalized valuation val. Let L be a finite free $\mathcal{O}$-module, and let $R \subseteq L$ be a submodule of rank s. Write $R^{\mathrm{sat}} = (FR) \cap L$. Choose Smith bases so that L has basis $e_1, \dots, e_l$ and R has basis $r_i = \varpi^{a_i} e_i$, where $a_i \ge 0$ and $1 \le i \le s$.

Lemma 1.1. *For $\lambda \in L^\vee$ and an integer $q \ge 1$, the coset $\lambda + \varpi^q L^\vee$ contains a functional vanishing on R if and only if*

$$\lambda(r_i) \in \varpi^{q+a_i}\mathcal{O} \quad (1 \le i \le s).$$

Equivalently, $\lambda(R^{\mathrm{sat}}) \subseteq \varpi^q\mathcal{O}$. If $\delta = \max_i a_i$, the stronger condition $\lambda(R) \subseteq \varpi^{q+\delta}\mathcal{O}$ suffices.

Proof. Vanishing on r_i is equivalent to vanishing on e_i, since $\mathcal{O}$ is a domain. A correction in $\varpi^q L^\vee$ can change $\lambda(e_i)$ to zero exactly when $\lambda(e_i) \in \varpi^q\mathcal{O}$. The coordinates of the correction on the remaining basis vectors can be zero. Since $\lambda(r_i) = \varpi^{a_i}\lambda(e_i)$, this proves every assertion. □

Thus divisibility on the original coroot lattice is insufficient when that lattice is not saturated. A bounded denominator gives a sufficient margin, but it does not make that margin hold in a prescribed character coset. In equal characteristic p, the differential of a coroot divisible by p can vanish. Dividing the abstract cocharacter by p then changes the actual differential and the group commutator; it is not an integral correction of a representative.

2. A finite refinement using actual root subgroups

Let F be a nonarchimedean local field with ring of integers $\mathcal{O}$, residue field k of cardinality q, and uniformizer ϖ. Fix an additive character $\psi : F \to \mathbf{C}^\times$ whose annihilator of $\mathcal{O}$ is $\mathcal{O}$. Let $\mathcal{G}$ be a split connected reductive group scheme over $\mathcal{O}$, with split maximal torus $\mathcal{T}$ and Chevalley root homomorphisms x_α. Put $G = \mathcal{G}(F)$ and $L = \mathrm{Lie}(\mathcal{G})(\mathcal{O})$. Use the decomposition

$$L = \mathrm{Lie}(\mathcal{T})(\mathcal{O}) \oplus \bigoplus_{\alpha \in \Phi} \mathcal{O}E_\alpha, \qquad E_\alpha = dx_\alpha(1).$$

Choose signs so that $[E_\alpha, E_{-\alpha}] = H_\alpha$, where $H_\alpha = d\alpha^\vee(1)$. The signs in the alternating pairings below can be changed by changing root coordinates.

For an integer $j \geq 1$, let $G_j = \ker(\mathcal{G}(\mathcal{O}) \to \mathcal{G}(\mathcal{O}/\varpi^j\mathcal{O}))$. Choose $\beta \in L^\vee$ vanishing on all root spaces. Define

$$c_\alpha = \beta(H_\alpha), \qquad \Gamma = \{\alpha \in \Phi : c_\alpha \neq 0\}, \qquad d_\alpha = \mathrm{val}(c_\alpha) \quad (\alpha \in \Gamma).$$

Then $d_{-\alpha} = d_\alpha$. Set $\delta = \max_{\alpha\in\Gamma} d_\alpha$, with $\delta = 0$ if Γ is empty. Choose an integer $m \geq 2\delta + 1$, and put

$$n = 2m + 1, \qquad H = G_{m+1}, \qquad a_\alpha = m - d_\alpha.$$

Since $2(m + 1) \geq n$, the square-zero congruence quotient gives a canonical map

$$\ell : H/G_n \xrightarrow{\sim} \varpi^{m+1}L/\varpi^n L.$$

Define the character θ of H by $\theta(h) = \psi(\varpi^{-n}\beta(\ell(h)))$. This is independent of the lift of $\ell(h)$ to $\varpi^{m+1}L$. Finally, let P be the subgroup generated by H and $x_\alpha(\varpi^{a_\alpha}\mathcal{O})$ for $\alpha \in \Gamma$.

Proposition 2.1. *The group P is compact and open, H is normal in P, and θ is P-invariant. Root coordinates identify its abelian quotient with*

$$V = P/H \simeq \bigoplus_{\alpha\in\Gamma} \mathcal{O}/\varpi^{d_\alpha+1}\mathcal{O}.$$

The commutator pairing B_θ on V is trivial between nonopposite root summands. On a pair of opposite summands it is

$$B_\theta\big(x_\alpha(\varpi^{a_\alpha}u), x_{-\alpha}(\varpi^{a_\alpha}v)\big) = \psi\left(\varpi^{-(d_\alpha+1)}(\varpi^{-d_\alpha}c_\alpha)uv\right). \tag{1}$$

In particular, this pairing is perfect. There is a unique irreducible smooth representation ω_θ of P whose restriction to H is θ-isotypic, and

$$\dim \omega_\theta = \prod_{\alpha\in\Gamma\cap\Phi^+} q^{d_\alpha+1}.$$

Let C be the inverse image in P of the sum of the positive-root summands of V. The character θ extends to a character η of C, and $\omega_\theta \simeq \mathrm{Ind}_C^P \eta$ for every such extension. For every smooth representation π of G, evaluation gives an isomorphism of P-representations

$$\omega_\theta \otimes \mathrm{Hom}_P(\omega_\theta, \pi) \xrightarrow{\sim} \pi^\theta. \tag{2}$$

Consequently θ occurs in π if and only if ω_θ occurs in π. Moreover,

$$I_G(\omega_\theta) \subseteq I_G(\theta). \tag{3}$$

Proof. Put $a = m - \delta \geq 1$. Since $2a \geq m + 1$, the quotient G_a/G_{m+1} is an additive congruence quotient. The images of the indicated root subgroups are its independent root submodules, with the indicated lengths. This proves the description of P/H. It also proves compactness and openness: P is a union of finitely many cosets of H in $\mathcal{G}(\mathcal{O})$. The group H is normal in $\mathcal{G}(\mathcal{O})$.

For a toral functional, the integral adjoint action of a root element satisfies

$$\beta \circ \mathrm{Ad}(x_\alpha(t)) - \beta = \pm t c_\alpha E^\vee_{-\alpha}.$$

Indeed, the only root space that can acquire a toral component under this action is the $-\alpha$ root space, and all subsequent terms lie in root spaces. Conjugation acts on the square-zero congruence quotient by this integral adjoint action. For $t \in \varpi^{a_\alpha}\mathcal{O}$, the change in its pairing with an element of $\varpi^{m+1}L$ has valuation at least $a_\alpha + d_\alpha + m + 1 = n$. Thus each root generator preserves θ. Invariance under H follows because θ is a character.

The integral Chevalley commutator formula for nonopposite roots is a product of root elements with parameter valuations at least $a_\alpha + a_\gamma \geq 2m - 2\delta \geq m + 1$. Each factor belongs to H, and θ is trivial on every root subgroup in H. Their pairing is therefore trivial.

For opposite roots use the root homomorphism from SL_2. Write $u' = \varpi^{a_\alpha}u$, $v' = \varpi^{a_\alpha}v$, and $s = u'v'$. In SL_2 the commutator is

$$\begin{pmatrix} 1 & u' \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ v' & 1 \end{pmatrix}\begin{pmatrix} 1 & -u' \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -v' & 1 \end{pmatrix} = \begin{pmatrix} 1+s+s^2 & -(u')^2v' \\ u'(v')^2 & 1-s \end{pmatrix}.$$

Its lower-diagonal-upper factorization has torus $\mathrm{diag}(A, A^{-1})$, with $A = 1 + s + s^2$, and root parameters $u'(v')^2/A$ and $-(u')^2v'/A$. Those root parameters have valuation at least $3a_\alpha \geq m + 1$, so their images lie in H and have trivial θ-value. The remaining factor is $\alpha^\vee(A)$. In its congruence linearization,

$$\ell(\alpha^\vee(A)) = u'v'H_\alpha \pmod{\varpi^{4a_\alpha}L}.$$

Here both the term s^2 and the nonlinear terms of the integral cocharacter have valuation at least $4a_\alpha$. The bound on m implies $4a_\alpha \geq 4m - 4\delta \geq n$. Applying θ therefore gives $\psi(\varpi^{-n} c_\alpha u' v')$, which is (1). Because $\varpi^{-d_\alpha} c_\alpha$ is a unit and the annihilator of $\mathcal{O}$ under ψ is $\mathcal{O}$, each opposite-root pairing is perfect.

Let $N = \ker \theta$. Invariance makes N normal in P, and P/N is finite. The subgroup H/N is central, with faithful character induced by θ. Perfection of the commutator pairing makes H/N the whole center of P/N. The central-character summand of $\mathbf{C}[P/N]$ has dimension $|P/H|$ and center $\mathbf{C}$: the coefficient of a noncentral coset must vanish in any central element, since some commutator is detected by θ. It is therefore a single matrix algebra. This proves uniqueness and the dimension formula.

The positive-root summands form a maximal isotropic subgroup of V. Consequently C/N is abelian, so its central subgroup H/N has an extension of θ to C/N. Inflate an extension to C and call it η. The induced representation $\operatorname{Ind}_C^P \eta$ is θ-isotypic on H and has dimension $[P : C] = \sqrt{|P/H|}$. Uniqueness and the dimension formula just proved show that it is ω_θ.

The space π^θ is P-stable and is a module for that matrix algebra. Evaluation proves (2), including for an infinite-dimensional smooth π. Finally, a nonzero intertwiner for ω_θ restricts to $H \cap gHg^{-1}$, where the two representations act by the scalar characters θ and ${}^g\theta$. These characters must agree, proving (3). □

3. A characteristic-two example in type C_2

Take $F = k((t))$ of characteristic two, $\mathcal{G} = \mathrm{Sp}_4$ over $k[[t]]$, and the diagonal torus with coordinates $\operatorname{diag}(z_1, z_2, z_1^{-1}, z_2^{-1})$. Let h_i be the differential of its ith coordinate cocharacter. Choose $\beta(h_1) = 1$ and $\beta(h_2) = 1 + t$. The long roots $2e_i$ have coroots h_i, whereas the short roots $e_1 - e_2$ and $e_1 + e_2$ have coroots $h_1 - h_2$ and $h_1 + h_2$. Thus

$$d_\alpha = 0 \text{ for long roots}, \qquad d_\alpha = 1 \text{ for short roots}.$$

For every $m \geq 3$, put $H = G_{m+1}$ and $n = 2m + 1$. The group P is generated by H, the short root subgroups at depth $m - 1$, and the long root subgroups at depth m. Proposition 2.1 gives

$$P/H \simeq (\mathcal{O}/t^2\mathcal{O})^4 \oplus k^4, \qquad \dim \omega_\theta = q^6.$$

All commutator assertions use the actual group. In particular, nonopposite short-root commutators that have a long-root term with coefficient two vanish in characteristic two. Mixed short-long commutators are already in H. For opposite short roots, the possible nonlinear torus error has depth at least $4(m - 1) \geq 2m + 1$; for long roots it has depth at least $4m$.

For $k = \mathbf{F}_2$, the group $P/\ker\theta$ is an extraspecial group of order 2^{13}. Its center is $H/\ker\theta$ of order two, and its quotient by that center is the twelve-dimensional $\mathbf{F}_2$-space V. Define a quadratic form $Q : V \to \mathbf{F}_2$ by $(-1)^{Q(v)} = \theta(\widetilde{v}^2)$ for a lift $\widetilde{v} \in P$. Each root element has order two. The positive-root half of V is isotropic for the commutator pairing and is generated by these square-one lifts. It is therefore a totally singular subspace of dimension six for Q. Thus Q has Arf invariant zero. The preimage of this half in $P/\ker\theta$ is elementary abelian; choose the extending character η with values in $\{1, -1\}$. Its induced representation gives a real model of the irreducible Heisenberg representation of dimension 64. This assertion does not include an extension of that representation to a normalizer.

The roots on which the leading reduction of β vanishes are precisely the short roots. They form a root system of type D_2, which is not the root system of a proper Levi subgroup of Sp_4. Nevertheless the refinement above has a perfect commutator pairing and preserves occurrence. This is a statement about this finite step, not an exclusion of possible data with several stages. No connectedness assertion about the full coadjoint stabilizer is used. In characteristic two the Weyl sign changes act trivially on the toral Lie algebra and can give additional stabilizer components.

The same valuation calculation should not be transferred unchanged to mixed characteristic. Over $\mathbf{Q}_2$ with $\varpi = 2$, the values 1 and $1 + \varpi = 3$ give short-root values -2 and 4, of valuations one and two. Here the displayed general bound is $m \geq 5$ and the Heisenberg dimension is q^7.

4. Congruence characters and coadjoint transporters

Retain the split connected reductive group scheme $\mathcal{G}$ over $\mathcal{O}$, and put $K = \mathcal{G}(\mathcal{O})$, $G = \mathcal{G}(F)$ and $L = \mathrm{Lie}(\mathcal{G})(\mathcal{O})$. Write $L^\vee = \mathrm{Hom}_{\mathcal{O}}(L, \mathcal{O})$, and retain the congruence subgroups G_j. In this section β may be any element of $L^\vee$. Choose integers $1 \leq a < n \leq 2a$, and set $m = n - a$. The ideal $\varpi^a\mathcal{O}/\varpi^n\mathcal{O}$ has square zero, so

smoothness gives the canonical additive identification

$$\ell_{a,n} : G_a/G_n \xrightarrow{\sim} \varpi^a L/\varpi^n L. \tag{4}$$

Define

$$\theta_\beta(h) = \psi\big(\varpi^{-n}\beta(\ell_{a,n}(h))\big), \qquad \Omega_\beta = \beta + \varpi^m L^\vee. \tag{5}$$

The value of the character is independent of the lift in (4). Our convention for the coadjoint action is $(g\beta)(X) = \beta(\mathrm{Ad}(g^{-1})X)$. For a character θ of a compact open subgroup H, write $\theta^g(h) = \theta(g^{-1}hg)$ on gHg^{-1}, and let $I_G(\theta)$ be the set of g for which the restrictions of θ and θ^g to $H \cap gHg^{-1}$ agree.

Proposition 4.1. *For every $g \in G$, one has*

$$g \in I_G(\theta_\beta) \quad \Longleftrightarrow \quad \Omega_\beta \cap g\Omega_\beta \neq \varnothing. \tag{6}$$

This holds for every split root datum and in every residue characteristic.

Proof. Choose opposite Borel subgroups containing the split maximal torus $\mathcal{T}$. Fix orders on the positive and negative roots giving the usual product coordinates on their unipotent radicals $\mathcal{U}^+$ and $\mathcal{U}^-$. Choose a $\mathbf{Z}$-basis $\lambda_1, \dots, \lambda_r$ of $X_*(\mathcal{T})$, and put $H_i = d\lambda_i(1)$. The H_i form an $\mathcal{O}$-basis of $\mathrm{Lie}(\mathcal{T})(\mathcal{O})$; together with the root vectors $E_\alpha = dx_\alpha(1)$ they form an $\mathcal{O}$-basis of L. Here the torus basis uses the full cocharacter lattice, so it does not require any coroot to be primitive.

The multiplication map $\mathcal{U}^- \times \mathcal{T} \times \mathcal{U}^+ \to \mathcal{G}$ is an open immersion whose image contains the identity. Every $h \in G_a$ therefore has a unique expression

$$h = \left(\prod_{\alpha\in\Phi^-} x_\alpha(z_\alpha)\right)\left(\prod_{i=1}^{r} \lambda_i(1+y_i)\right)\left(\prod_{\alpha\in\Phi^+} x_\alpha(z_\alpha)\right), \tag{7}$$

where all z_α, y_i belong to $\varpi^a\mathcal{O}$. Indeed, reduction of h is the identity, so h belongs to this open cell; uniqueness of its coordinates then gives the stated congruences. Conversely, those congruences imply $h \in G_a$. Consequently

$$c : G_a \xrightarrow{\sim} \varpi^a L, \qquad c(h) = \sum_{\alpha\in\Phi} z_\alpha E_\alpha + \sum_{i=1}^{r} y_i H_i \tag{8}$$

is a bijection of sets. Modulo $\varpi^n L$, it agrees with (4): over $\mathcal{O}/\varpi^n\mathcal{O}$, each parameter belongs to the square-zero ideal $\varpi^a\mathcal{O}/\varpi^n\mathcal{O}$, and multiplication of the factors adds their tangent vectors. This also proves that c modulo $\varpi^n L$ is K-equivariant under conjugation.

For $t \in \mathcal{T}(F)$, conjugation by t^{-1} multiplies z_α by $\alpha(t)^{-1}$ and leaves every y_i unchanged. It preserves the order of the factors in (7). Thus, whenever h and $t^{-1}ht$ belong to G_a, the exact equality

$$c(t^{-1}ht) = \mathrm{Ad}(t^{-1})c(h) \tag{9}$$

holds before passing to a quotient. It follows that

$$c(G_a \cap tG_at^{-1}) = \varpi^a L \cap \mathrm{Ad}(t)(\varpi^a L). \tag{10}$$

In fact, the torus coordinates on both sides have depth a, while the α-coordinate has depth $\max\{a, a + \mathrm{val}(\alpha(t))\}$. Every tuple with these bounds gives an element of the group intersection, because applying t^{-1} puts all its parameters back in $\varpi^a\mathcal{O}$. This proves both inclusions in (10).

Now take $\beta_1, \beta_2 \in L^\vee$. By (9) and (10), equality of θ_{β_1} and $\theta^t_{\beta_2}$ on $G_a \cap tG_at^{-1}$ is equivalent to

$$\psi\big(\varpi^{-n}(\beta_1 - t\beta_2)(X)\big) = 1 \quad \text{for every } X \in \varpi^a L \cap \mathrm{Ad}(t)(\varpi^a L). \tag{11}$$

For two $\mathcal{O}$-lattices A, B in a finite-dimensional F-vector space, let $A^\perp$ and $B^\perp$ denote their annihilators under the evaluation pairing followed by ψ. Then

$$(A \cap B)^\perp = A^\perp + B^\perp. \tag{12}$$

To check this, choose elementary-divisor bases that make both lattices diagonal. In each coordinate, intersection takes the maximum of the two exponents, while addition of the annihilator lattices takes the minimum of their negatives. Applying (12) to (11) gives

$$\beta_1 - t\beta_2 \in \varpi^m L^\vee + t(\varpi^m L^\vee), \tag{13}$$

$$(\beta_1 + \varpi^m L^\vee) \cap t(\beta_2 + \varpi^m L^\vee) \neq \varnothing. \tag{14}$$

These two conditions are equivalent. The argument uses only the evaluation pairing between L and $L^\vee$.

For general g, use the hyperspecial Cartan decomposition to write $g = k_1 t k_2$ with $k_1, k_2 \in K$ and $t \in \mathcal{T}(F)$. Since conjugation by K acts on (4) through the adjoint action, the original character equality becomes the equality just considered with $\beta_1 = k_1^{-1}\beta$ and $\beta_2 = k_2\beta$. The lattice $L^\vee$ is K-stable. Applying k_1 to (14) gives exactly $\Omega_\beta \cap g\Omega_\beta \neq \varnothing$, proving both implications. □

The exact equality (9) is what permits an arbitrary Cartan element in this argument. Conjugating a map known only modulo $\varpi^n L$ by a nonintegral t could magnify its error. The ordered coordinates avoid that problem because their transformation under t is exact.

Proposition 4.2. *Suppose that*

$$\{g \in G : \Omega_\beta \cap g\Omega_\beta \neq \varnothing\} \subseteq K. \tag{15}$$

Decompose the finite-dimensional representation of K as

$$W = \operatorname{Ind}_{G_a}^K \theta_\beta \simeq \bigoplus_{i=1}^{s} \sigma_i^{\oplus d_i}, \tag{16}$$

where the σ_i are distinct irreducible smooth representations. Then

$$\pi_i = \text{c-Ind}_K^G \sigma_i \tag{17}$$

are irreducible supercuspidal representations and are pairwise inequivalent. They are exactly the irreducible smooth representations of G containing θ_β.

Proof. Proposition 4.1 and (15) give $I_G(\theta_\beta) \subseteq K$. The support description of the compact-induction endomorphism algebra therefore gives

$$\operatorname{End}_G\big(\text{c-Ind}_{G_a}^G \theta_\beta\big) = \operatorname{End}_K(W) \simeq \bigoplus_{i=1}^{s} M_{d_i}(\mathbf{C}). \tag{18}$$

The first identification is the natural one from induction in stages: every endomorphism is supported inside K. Applying compact induction to (16) shows that the endomorphisms already supported inside K account for the entire right side of (18). Mackey decomposition consequently supplies no intertwiner between the σ_i outside K and no cross-intertwiner between distinct σ_i anywhere in G. In particular, $I_G(\sigma_i) = K$ for every i.

We give the irreducibility argument explicitly. By Frobenius reciprocity and the absence of other Mackey intertwiners, the σ_i-isotypic space in $\pi_i|_K$ is exactly its canonical inducing fiber. Let V be a nonzero G-subrepresentation of π_i. Translate a nonzero inducing function in V so that its value at 1 is nonzero, and project it onto the σ_i-isotypic space by compact averaging. Evaluation at 1 is K-equivariant, so this projection retains the nonzero value. Thus V meets the inducing fiber nontrivially. Irreducibility of σ_i makes V contain the whole fiber, and its G-translates generate π_i. Hence $V = \pi_i$. The absence of cross-intertwiners also proves pairwise inequivalence.

The same argument applies to $\sigma_i^\vee$. The natural pairing of the compact inductions from σ_i and $\sigma_i^\vee$ has compactly supported matrix coefficients, since both inducing functions have compact support modulo the compact group K. Irreducibility and admissibility identify the second compact induction with the smooth dual of the first. The compact-mod-center matrix coefficient criterion therefore makes each π_i supercuspidal.

Finally, if an irreducible smooth representation π contains θ_β, compact Frobenius reciprocity gives a nonzero map $\text{c-Ind}_{G_a}^G \theta_\beta \to \pi$. By induction in stages and (16), its source is the finite direct sum $\bigoplus_i \pi_i^{\oplus d_i}$. One summand maps nontrivially to π and is therefore isomorphic to it. Conversely, each σ_i contains θ_β, so each π_i contains it as well. □

The inducing representations in (16) are obtained from an explicit finite quotient. Set

$$N = \bigcap_{k \in K/G_a} k \ker(\theta_\beta) k^{-1}, \tag{19}$$

where one representative is chosen for each left coset. Since $\ker(\theta_\beta)$ is normal in G_a, this is its full normal core in K. It is open and normal, and W factors through K/N. Thus ordinary finite-group representation theory determines all the σ_i after the transporter bound has been established. The proposition exhausts the representations containing the specified congruence character; occurrence of such a character in an arbitrary supercuspidal representation remains a separate problem.

5. An integral toral slice

Retain the split integral model of the preceding section, and suppose $\mathcal{G}$ is semisimple. In this and the next section, G denotes its algebraic generic fiber. Put $K = \mathcal{G}(\mathcal{O})$, write K_j for its principal congruence subgroups, and write $L^* = \mathrm{Hom}_{\mathcal{O}}(L, \mathcal{O})$. Let $\mathcal{S} \subset \mathcal{G}$ be a maximal torus splitting over a finite unramified extension E/F. In this section the torus $S = \mathcal{S}_F$ need not be split over F. Assume

$$N_G(S)(F) \subseteq K. \tag{20}$$

Descent of the toral and root summands over $\mathcal{O}_E$ gives a direct sum $L = \mathfrak{s}_{\mathcal{O}} \oplus L_{\mathrm{rt}}$ of free $\mathcal{O}$-modules. Embed $\mathfrak{s}^*_{\mathcal{O}}$ in L^* by extending functionals by zero on L_{rt}. Choose $\beta \in \mathfrak{s}^*_{\mathcal{O}}$ such that $c_\alpha = \beta(H_\alpha) \neq 0$ for every absolute root, and put $\delta = \max_\alpha \mathrm{val}_E(c_\alpha)$, with $\mathrm{val}_E(\varpi) = 1$.

Proposition 5.1. *For $m \geq 2\delta + 1$, the coset $\Omega = \beta + \varpi^m L^*$ satisfies*

$$\Omega \subseteq \mathrm{Ad}^*(K_{m-\delta})(\beta + \varpi^m \mathfrak{s}^*_{\mathcal{O}}). \tag{21}$$

Its full coadjoint transporter is contained in K.

Proof. Smoothness gives integral formal coordinates $g(u)$ through the identity whose differential identifies the parameter space with L_{rt}. One can obtain them by choosing formal parameters dual to a basis of $\mathfrak{s}_{\mathcal{O}} \oplus L_{\mathrm{rt}}$ and setting the toral parameters to zero. The resulting integral power series converge when the parameters lie in $\varpi\mathcal{O}$. Write

$$F(u, \eta) = \mathrm{Ad}^*(g(u))(\beta + \eta) - \beta = A(u, \eta) + R(u, \eta), \qquad A = (D_\beta, 1),$$

where $\eta \in \mathfrak{s}^*_{\mathcal{O}}$ and R has integral coefficients and no terms of degree below two. The differential D_β maps L_{rt} to L^*_{rt}. After extending scalars to $\mathcal{O}_E$, it pairs opposite root spaces with coefficients $\pm c_\alpha$. Therefore

$$\varpi^\delta D_\beta^{-1} \in \mathrm{Hom}_{\mathcal{O}}(L^*_{\mathrm{rt}}, L_{\mathrm{rt}}).$$

This integrality assertion descends by faithful flatness. It does not require division by $[E : F]$.

For $y \in \varpi^m L^*$, consider the iteration $x \mapsto A^{-1}(y - R(x))$ on the complete set

$$x = (u, \eta), \qquad u \in \varpi^{m-\delta} L_{\mathrm{rt}}, \qquad \eta \in \varpi^m \mathfrak{s}^*_{\mathcal{O}}.$$

Put $r = m - \delta$. The quadratic bound gives $R(x) \in \varpi^{2r} L^*$, where $2r \geq m + 1$. The iteration consequently preserves this set, including the stronger precision on η. If two inputs differ by ϖ^j times the indicated lattices, their remainders differ by an element of $\varpi^{2r+j} L^*$. Applying A^{-1} improves their relative precision by at least $m - 2\delta \geq 1$. Thus the iteration converges to a solution of $F(u, \eta) = y$, proving (21).

Every $\lambda \in \beta + \varpi^m \mathfrak{s}^*_{\mathcal{O}}$ still has nonzero values on every H_α. Its coadjoint stabilizer contains S, and the root-space calculation gives Lie algebra exactly $\mathfrak{s}$. After algebraic closure, equality of tangent dimension and dimension proves smoothness; the identity component of this stabilizer is S. The full stabilizer need not be connected. An element transporting two such toral covectors therefore normalizes S. Using (21) for two points of Ω gives

$$\{g \in G(F) : \mathrm{Ad}^*(g)\Omega \cap \Omega \neq \varnothing\} \subseteq K_{m-\delta} N_G(S)(F) K_{m-\delta} \subseteq K.$$

□

For $a \geq m$ and $n = a + m$, the character of K_a/K_n represented by $\varpi^{-n}\beta$ therefore has all its intertwining in K, by the exact coset theorem of the preceding section. Decomposing its finite induction to K gives irreducible supercuspidal compact inductions and exhausts precisely the irreducible representations containing that character. This construction has no restriction on the residue characteristic. Its hypotheses concern the specified integral torus, covector, and precision.

6. Explicit compact induction for Sp_4 in characteristic two

Let $F = k((t))$, where k has cardinality q and characteristic two. Choose $c \in k$ with $\mathrm{Tr}_{k/\mathbf{F}_2}(c) = 1$, and set

$$E = F(\zeta), \qquad \zeta^2 + \zeta + c = 0, \qquad M = \begin{pmatrix} 0 & c \\ 1 & 1 \end{pmatrix}, \qquad J_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The extension E/F is unramified quadratic. Write τ for its nontrivial automorphism. Use the basis $(1, \zeta)$ on each copy of E, and put

$$G = \mathrm{Sp}(J_2 \oplus J_2), \qquad K = G(k[[t]]), \qquad B = M \oplus (1 + t)M, \qquad \beta(X) = \mathrm{tr}(BX).$$

This defines an integral functional on $L = \mathrm{Lie}(G)(k[[t]])$; no nondegeneracy of the trace form on L is used. Let S be the product of the two norm-one tori acting by multiplication on $E \oplus E$. It is an integral maximal torus. Since B commutes with S, the functional β annihilates its root spaces. The two coordinate coroot differentials h_i are the identity matrix on their respective two-dimensional blocks, so

$$\beta(h_1) = 1, \qquad \beta(h_2) = 1 + t.$$

Thus the long-root values are units, every short-root value is t, and $\delta = 1$. The integral root decomposition can also be checked directly: the eigenvector matrix of M over $\mathcal{O}_E$ has columns $(\zeta + 1, 1)^T$ and $(\zeta, 1)^T$, and determinant 1.

The symplectic form in these coordinates is

$$h(x, y) = \mathrm{Tr}_{E/F}(x_1\tau(y_1) + x_2\tau(y_2)).$$

An element of $N_G(S)(F)$ permutes the two E-summands, and on each summand has the form $x \mapsto ax$ or $x \mapsto a\tau(x)$. Preservation of h forces $N_{E/F}(a) = 1$. Such an a is an $\mathcal{O}_E$-unit. Consequently every normalizer element preserves $\mathcal{O}_E^2$, and (20) holds.

Fix $\psi(x) = (-1)^{\mathrm{Tr}_{k/\mathbf{F}_2}([t^{-1}]x)}$. For $m \geq 3$, put $H = K_{m+1}$ and $n = 2m + 1$, and define

$$\theta_m(h) = \psi\left(t^{-n}\,\mathrm{tr}(B(h-1))\right) \qquad (h \in H). \tag{22}$$

The product error has valuation at least $2m + 2 > n$, so this is a character, trivial on K_n. The symplectic equation also shows that $h - 1$ modulo t^n is its canonical Lie algebra linearization. Proposition 5.1 and the exact coset theorem therefore give $I_{G(F)}(\theta_m) \subseteq K$.

Proposition 6.1. *For each fixed $m \geq 3$, let $\mathcal{R}_m$ be the set of irreducible representations ρ of the finite group K/K_{2m+1} such that $\mathrm{Hom}_{K_{m+1}}(\theta_m, \rho) \neq 0$. Inflation followed by compact induction gives a bijection*

$$\mathcal{R}_m \longrightarrow \{\pi \in \mathrm{Irr}(G(F)) : \mathrm{Hom}_{K_{m+1}}(\theta_m, \pi) \neq 0\}, \qquad \rho \longmapsto \mathfrak{c}\text{-}\mathrm{Ind}_K^{G(F)}\,\rho.$$

The set is nonempty, and every representation on the right is supercuspidal.

Proof. The set $\mathcal{R}_m$ consists exactly of the irreducible constituents of $\mathrm{Ind}_H^K\,\theta_m$, so it is nonempty. Compact containment of the intertwining and the reconstruction result of the preceding section prove all the assertions. □

The leading reduction of this datum does not pass the single-layer ellipticity test. For $v = (x, y) \in E \oplus E$, its quadratic covector is

$$q_\beta(v) = \beta(vv^{\mathsf{T}}(J_2 \oplus J_2)) = N_{E/F}(x) + (1 + t)N_{E/F}(y).$$

Modulo t it vanishes on the diagonal Lagrangian $\{(x, x)\}$. The higher precision in (22) is what separates the two toral values and yields compact intertwining.

6.1. **The inducing subgroup and the Heisenberg factor.** Write $L = L_{\mathrm{tor}} \oplus L_{\mathrm{long}} \oplus L_{\mathrm{short}}$ for the integral decomposition obtained from S; the summands have ranks $2, 4, 4$. The square-zero identification for K_{m-1}/K_{m+1} is available because $m \geq 3$. Define P as the inverse image in K_{m-1} of

$$\left(t^m L_{\mathrm{long}} + t^{m-1} L_{\mathrm{short}} + t^{m+1} L\right)/t^{m+1} L.$$

This definition uses an actual congruence quotient over F. After unramified base change it agrees with the root subgroup construction above. Invariance of θ_m and perfection of the commutator form descend: both are integral identities, and the associated map to the $F/\mathcal{O}$-valued module dual is an isomorphism after faithfully flat base change. Applying ψ then gives the perfect character pairing over F. In particular

$$|P/H| = q^{12}, \qquad \dim \omega_m = q^6, \tag{23}$$

where ω_m is the unique irreducible representation of P above θ_m. One must descend the form before applying ψ: restricting $\psi \circ \mathrm{Tr}_{E/F}$ to F would instead give the trivial character in this characteristic.

Put $C = \mathrm{Stab}_G(\beta)(F)$ and $S_j = S(F) \cap K_j$. The full stabilizer is smooth with identity component S, and

$$C = S(F) \rtimes (\mathbf{Z}/2\mathbf{Z})^2.$$

The two involutions are the independent automorphisms τ on the two summands. Weyl sign changes fix the toral covector in characteristic two; the remaining Weyl elements interchange 1 and $1 + t$ and do not fix it. Let $J_m = K_{\theta_m}$.

Proposition 6.2. *For every $m \geq 3$,*

$$I_{G(F)}(\theta_m) = J_m = CP, \qquad C \cap P = S_{m+1}.$$

In particular $J_m/P \simeq C/S_{m+1}$ is a finite group of order $4(q+1)^2q^{2m}$.

Proof. Since H is normal in K and all intertwining is contained in K, $I_{G(F)}(\theta_m) = K_{\theta_m}$. Both C and P preserve the character, and C normalizes P, so $CP \subseteq J_m$.

If $k \in J_m$, then $k\beta \in \beta + t^m L^*$. The slice gives $k\beta = u(\beta + \eta)$ with $u \in K_{m-1}$ and $\eta \in t^m\mathfrak{s}_{\mathcal{O}}^*$. The element $u^{-1}k$ transports two toral covectors whose stabilizer identity components are S, so it normalizes S. Its Weyl action either fixes β or interchanges its two coordinates. The latter changes a coordinate by t, and is impossible modulo t^m. Therefore $\eta = 0$ and $u^{-1}k \in C$.

For $u \in K_{m-1}$, its coadjoint action on β modulo t^m is its linear action D_β; the nonlinear error has valuation at least $2m-2 \geq m+1$. Hence $u \in J_m$ precisely when its long-root component has depth at least m. The toral component is unrestricted and the short-root component already has the required depth $m-1$. Smoothness of $\mathcal{S}$ allows the toral component modulo t^{m+1} to be lifted by an element of S_{m-1}. It follows that $J_m \cap K_{m-1} = S_{m-1}P$, proving $J_m = CP$.

The nonidentity components of C remain distinct from S after reduction. Thus an element of $C \cap P$ lies in S_{m-1}. Its congruence coordinate is toral, whereas the defining module of P/H has zero toral component; it must belong to S_{m+1}. The reverse inclusion is immediate. Each norm-one torus has $|S^{(i)}(F)/S^{(i)}_{m+1}| = (q+1)q^m$, proving the order formula. □

This gives smaller finite data in the form used in Heisenberg and Clifford constructions. Set $A_m = J_m/P$. Choose intertwiners for the action of J_m on the unique P-representation ω_m. They give a projective extension $\widetilde{\omega}_m$ to J_m with multiplier κ_m inflated from A_m. All choices can be made in the finite group J_m/K_{2m+1}: choose a coset section, solve the finite intertwiner equations, and compare products of intertwiners. This prescribes κ_m by finite matrix calculations.

For each irreducible projective representation ν of A_m with multiplier κ_m^{-1}, define

$$\pi_{m,\nu} = \mathfrak{c}\text{-Ind}_{J_m}^{G(F)}(\widetilde{\omega}_m \otimes \nu). \tag{24}$$

The multipliers cancel, so the tensor product is an ordinary representation of J_m. Finite Clifford theory, or the action on $\mathrm{Hom}_P(\omega_m, \rho)$, identifies these tensor products with all irreducible J_m-representations above θ_m. The exact intertwining identity proves that (24) is irreducible and supercuspidal. For fixed m, different ν give inequivalent representations, and every irreducible representation containing θ_m occurs. No extension of the Heisenberg representation as an ordinary representation of J_m is assumed.

7. Deep congruence intertwining for arbitrary reductive groups

Let $\mathbf{G}$ be a connected reductive group over the nonarchimedean local field F, and write $G = \mathbf{G}(F)$. Choose a special maximal parahoric $K = \mathcal{P}(\mathcal{O})$, with smooth affine parahoric group scheme $\mathcal{P}$. Put

$$L = \mathrm{Lie}(\mathcal{P})(\mathcal{O}), \qquad L^\vee = \mathrm{Hom}_{\mathcal{O}}(L, \mathcal{O}), \qquad K_j = \ker\big(\mathcal{P}(\mathcal{O}) \to \mathcal{P}(\mathcal{O}/\varpi^j\mathcal{O})\big).$$

Retain the additive character ψ whose annihilator of $\mathcal{O}$ is $\mathcal{O}$. For integers $1 \leq a < n \leq 2a$ and $\beta \in L^\vee$, the canonical square-zero identification defines

$$\ell_{a,n} : K_a/K_n \xrightarrow{\sim} \varpi^a L/\varpi^n L, \qquad \theta_\beta(h) = \psi\big(\varpi^{-n}\beta(\ell_{a,n}(h))\big). \tag{25}$$

Set $\Omega_\beta = \beta + \varpi^{n-a}L^\vee$, using the coadjoint convention $(g\beta)(X) = \beta(\mathrm{Ad}(g^{-1})X)$.

Proposition 7.1. *There are integers $a_0 \geq 1$ and $c \geq 0$, depending on $\mathbf{G}$ and the chosen parahoric, such that for every $\beta \in L^\vee$ and $a \geq a_0$ with $a < n \leq 2a - c$, one has*

$$I_G(\theta_\beta) = \{g \in G : \Omega_\beta \cap g\Omega_\beta \neq \varnothing\}. \tag{26}$$

No restriction on the characteristic or splitting field of $\mathbf{G}$ is required.

Proof. Choose a maximal F-split torus S whose apartment contains the chosen special point, and put $M = Z_{\mathbf{G}}(S)$. Choose opposite minimal F-parabolics with common Levi subgroup M and unipotent radicals U^- and U^+. Their multiplication map

$$U^- \times M \times U^+ \longrightarrow \mathbf{G} \tag{27}$$

is an open immersion, and its tangent map gives the direct sum $\mathrm{Lie}(\mathbf{G}) = \mathrm{Lie}(U^-) \oplus \mathrm{Lie}(M) \oplus \mathrm{Lie}(U^+)$. These statements follow from the dynamic open cell over F; see [5, Appendix A.2]. A cocharacter of S that is strictly positive on the roots of U^+ contracts U^+ to the identity. Its negative contracts U^-.

We first construct coordinates on either radical U. Write $A = F[U]$ and let I be its augmentation ideal. The contracting action extends to $\mathbf{A}^1 \times U$, with value the identity at parameter zero. It therefore gives a nonnegative grading of A with $A_0 = F$ and $I = \bigoplus_{r>0} A_r$. Since S is split, choose homogeneous S-weight vectors $f_1, \dots, f_d \in I$ lifting a basis of I/I^2, where $d = \dim U$. They generate A: for a homogeneous element, subtract its linear expression modulo I^2; the remainder is a sum of products of positive-degree terms of smaller degree, to which induction applies. The resulting surjection $F[X_1, \dots, X_d] \to A$ has prime kernel, since U is smooth and geometrically connected. Both rings have dimension d, so this kernel is zero. Thus the chosen functions, normalized by their differentials, give an S-equivariant isomorphism of varieties

$$f_U : U \xrightarrow{\sim} \mathrm{Lie}(U), \qquad f_U(1) = 0, \qquad df_U|_1 = 1. \tag{28}$$

This construction uses the full unipotent radical and applies unchanged when the relative root system is nonreduced.

Choose an analytic chart $f_M : W_M \to B_M$ from a neighborhood of 1 in $M(F)$ to a neighborhood of 0 in $\mathrm{Lie}(M)(F)$, with derivative the identity. The group S acts trivially on M. Combining this chart with (27) and (28) gives an analytic bijection

$$f : W = U^-(F) W_M U^+(F) \xrightarrow{\sim} \mathrm{Lie}(U^-)(F) \oplus B_M \oplus \mathrm{Lie}(U^+)(F), \qquad df|_1 = 1. \tag{29}$$

Both its domain and its image are $S(F)$-stable, and

$$f(shs^{-1}) = \mathrm{Ad}(s) f(h) \quad (s \in S(F),\ h \in W) \tag{30}$$

holds exactly. The global coordinates on U^- and U^+ ensure this domain statement even when s is unbounded.

The special-parahoric Cartan decomposition gives $G = KM(F)K$ by [4, Theorem 1.0.3]. The group M/S is anisotropic reductive, so $(M/S)(F)$ is compact by [5, Theorem I.1.3]. Hilbert theorem 90 for the split torus S identifies $M(F)/S(F)$ with $(M/S)(F)$; this is a topological identification because the quotient morphism is smooth and open on local points. The image of $M(F) \cap K$ is open in this compact quotient. Hence there is a finite set $D \subset M(F)$, containing 1, such that

$$M(F) = \bigcup_{d \in D} S(F) d (M(F) \cap K), \qquad G = \bigcup_{d \in D} KS(F)dK. \tag{31}$$

Smoothness of $\mathcal{P}$ gives integral formal coordinates q at the identity with derivative the identity on L. They identify K_a with $\varpi^a L$ for all sufficiently large a. To obtain them, choose functions in the augmentation ideal lifting an $\mathcal{O}$-basis of its conormal module. The resulting morphism is étale along the identity section, and Hensel lifting gives the asserted bijections. Modulo $\varpi^n L$, for $n \le 2a$, these coordinates agree with $\ell_{a,n}$ in (25).

For each $d \in D$, define near zero

$$F_d(X) = \mathrm{Ad}(d)^{-1} f\big(d q^{-1}(X) d^{-1}\big) = X + R_d(X). \tag{32}$$

Every R_d is analytic with no terms of degree less than two. There are integers a_0 and $c \ge 0$ such that these expressions are defined on $\varpi^{a_0} L$, and, for $a \ge a_0$,

$$R_d(\varpi^a L) \subseteq \varpi^{2a-c} L, \tag{33}$$

$$R_d(X) - R_d(Y) \in \varpi^{a-c+b} L \quad \text{if } X, Y \in \varpi^a L,\ X - Y \in \varpi^b L,\ b \ge a. \tag{34}$$

Here the constants are uniform because D is finite. More explicitly, choose a common convergence ball $\varpi^r L$ and a lower valuation bound $-B$ on the coefficients after rescaling to that ball. A monomial of degree $j \ge 2$ on $\varpi^a L$ has valuation at least $j(a-r) - B \ge 2a - 2r - B$ for $a \ge r$. Telescoping the difference of that monomial at X, Y gives valuation at least $b + a - 2r - B$. Thus $c = 2r + B$, enlarged to be nonnegative if necessary, gives (33)–(34).

Enlarge a_0 so that $a_0 > c$. For every $Y \in \varpi^a L$, the map $X \mapsto Y - R_d(X)$ is a strict contraction of $\varpi^a L$ to itself. Its unique fixed point proves that F_d bijects this ball with itself. Writing $L_d = \mathrm{Ad}(d) L$, we obtain the exact equality

$$f(d K_a d^{-1}) = \varpi^a L_d. \tag{35}$$

Moreover, if $n \le 2a - c$, then for $h \in d K_a d^{-1}$ and $\beta \in L^\vee$,

$$\theta^d_\beta(h) = \psi\big(\varpi^{-n}(d\beta)(f(h))\big). \tag{36}$$

Indeed, write $h = dq^{-1}(X)d^{-1}$ in (32). The difference of the two exponents is $\varpi^{-n}\beta(R_d(X))$, which is integral by (33). The case $d = 1$ gives the same expression for θ_β itself on K_a.

For $s \in S(F)$ and $d \in D$, equations (30) and (35) imply

$$f(K_a \cap sdK_a d^{-1}s^{-1}) = \varpi^a L \cap \mathrm{Ad}(sd)(\varpi^a L). \tag{37}$$

For the converse inclusion, take X in the right side and put $h = f^{-1}(X) \in K_a$. The element $s^{-1}hs$ remains in W by $S(F)$-stability, and its coordinate is $\mathrm{Ad}(s^{-1})X \in \varpi^a L_d$. Equation (35) and injectivity of f on W therefore place it in $dK_a d^{-1}$. This proves equality of the full sets.

For $\beta_1, \beta_2 \in L^\vee$, the two characters θ_{β_1} and $\theta^{sd}_{\beta_2}$ agree on the group intersection in (37) if and only if

$$\psi\big(\varpi^{-n}(\beta_1 - sd\beta_2)(X)\big) = 1 \quad \text{for every } X \in \varpi^a L \cap \mathrm{Ad}(sd)(\varpi^a L).$$

For the evaluation pairing followed by ψ, elementary divisors give $(A \cap B)^\perp = A^\perp + B^\perp$ for two lattices. The displayed condition is therefore equivalent to

$$\beta_1 - sd\beta_2 \in \varpi^{n-a}L^\vee + sd(\varpi^{n-a}L^\vee),$$

and hence to intersection of the corresponding coadjoint cosets. Finally write $g = k_1 sdk_2$ using (31). Conjugation by K acts exactly on the square-zero quotient and preserves L. The original character equality reduces to the case just proved with $\beta_1 = k_1^{-1}\beta$ and $\beta_2 = k_2\beta$. Applying k_1 to the resulting coset intersection proves (26). □

For any prescribed precision $m \geq 1$, the proposition applies with $n = a + m$ once $a \geq \max\{a_0, m + c\}$. Thus its fixed loss of precision can be accommodated when constructing a deep character from a chosen coadjoint coset. It does not improve the precision of a character already known to occur in a given representation.

8. Rational ellipticity and compact induction

Retain the connected reductive F-group $\mathbf{G}$, put $G = \mathbf{G}(F)$ and $\mathfrak{g} = \mathrm{Lie}(\mathbf{G})(F)$, and write $Z = Z(\mathbf{G})(F)$ and $Q = G/Z$. Define

$$\mathcal{E} = \{\beta \in \mathfrak{g}^* : \beta|_{\mathrm{Lie}(U_P)(F)} \neq 0 \text{ for every proper } F\text{-parabolic } P \subset \mathbf{G}\}, \tag{38}$$

where U_P denotes the unipotent radical. If $\mathbf{G}$ has no proper F-parabolic, this condition is empty and $\mathcal{E} = \mathfrak{g}^*$.

Proposition 8.1. *The set $\mathcal{E}$ is open in the local-field topology and is G-invariant. For compact subsets $A, B \subset \mathcal{E}$, the transporter*

$$T(A, B) = \{g \in G : gB \cap A \neq \varnothing\} \tag{39}$$

has compact image in Q. In particular, the stabilizer of every $\beta \in \mathcal{E}$ is compact modulo Z.

Proof. There are finitely many absolute parabolic types. For each type with F-points, its projective parabolic variety has compact F-points. The spaces $\mathrm{Lie}(U_P)$ form a subbundle of the constant Lie algebra bundle on this variety: over a splitting field it is the bundle associated to the P-stable subspace $\mathrm{Lie}(U_P)$, and this construction descends. Consequently the annihilator incidence set

$$\{(\beta, P) : \beta|_{\mathrm{Lie}(U_P)} = 0\}$$

is closed. Its F-points project to a closed subset of $\mathfrak{g}^*$, since the parabolic factor is compact. Equivalently, for convergent covectors one can pass to a convergent subsequence of their parabolics and preserve the annihilation condition. The finite union over proper types is closed, proving openness of $\mathcal{E}$. Conjugation takes U_P to $U_{gPg^{-1}}$, proving invariance. Nonzero scalar multiplication also preserves $\mathcal{E}$.

Choose a special maximal parahoric K, a maximal F-split torus S whose apartment contains its special point, and $M = Z_{\mathbf{G}}(S)$. As in (31), Cartan decomposition and compactness of $M(F)/S(F)$ give a finite set $D \subset M(F)$ with

$$G = \bigcup_{d \in D} KS(F)dK. \tag{40}$$

The structural inputs are [4, Theorem 1.0.3] and [5, Theorem I.1.3]. Choose positive relative roots Φ^+ and their simple roots Δ. The relative root system may be nonreduced. Every positive root is a nonnegative integral combination of Δ, and

$$\mathfrak{g} = \mathrm{Lie}(M)(F) \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha. \tag{41}$$

For $s \in S(F)$, let $\nu(s) \in X_*(S)$ be characterized by $\langle \alpha, \nu(s) \rangle = \mathrm{val}(\alpha(s))$ for every character α. Modulo $S(\mathcal{O})$, one has $s = \nu(s)(\varpi)$.

Let $A_{\mathbf{G}}$ be the maximal F-split central torus. The common kernel of the relative root forms on $X_*(S)$ is $X_*(A_{\mathbf{G}})$. Indeed, a cocharacter in this kernel is trivial under every absolute root after base change, so it is central; the converse is immediate. This is a saturated sublattice, and the map

$$X_*(S)/X_*(A_{\mathbf{G}}) \longrightarrow \mathbf{Z}^{\Phi}, \qquad \lambda \longmapsto (\langle \alpha, \lambda \rangle)_{\alpha \in \Phi} \tag{42}$$

is injective. Thus bounded root valuations allow only finitely many classes of $\nu(s)$ modulo $X_*(A_{\mathbf{G}})$. Such elements s range in a compact set modulo $A_{\mathbf{G}}(F)$, and hence modulo Z.

The finite relative Weyl group has representatives in $N_{\mathbf{G}}(S)(F)$. Using these representatives in (40), we may require

$$\mathrm{val}(\alpha(s)) \geq 0 \quad (\alpha \in \Delta). \tag{43}$$

Absorbing the finitely many representatives into the outer factors keeps those factors in fixed compact sets, and conjugating D by them still gives a finite subset of $M(F)$. The representatives need not belong to K.

Suppose that $T(A,B)/Z$ is not relatively compact. Choose g_j whose classes escape every compact subset of Q, and choose $\beta_j \in A$, $\beta'_j \in B$ with $\beta_j = g_j \beta'_j$. Write $g_j = k_{1,j} s_j d_j k_{2,j}$ in the form just described. Pass to a subsequence so that $d_j = d$ is constant, both outer factors converge, and both covector sequences converge. Put

$$\gamma_j = k_{1,j}^{-1} \beta_j, \qquad \delta_j = d k_{2,j} \beta'_j.$$

These sequences lie in compact subsets of $\mathcal{E}$, and $\gamma_j = s_j \delta_j$. Their limits therefore belong to $\mathcal{E}$.

The classes of s_j still escape compact sets modulo Z. By (42), some simple-root valuations are unbounded. After a further subsequence, each nonnegative integer sequence $\mathrm{val}(\alpha(s_j))$ is either constant or tends to positive infinity. Let $I \subsetneq \Delta$ consist of the simple roots with bounded valuations, and let P_I be the corresponding standard F-parabolic. Then

$$\mathrm{Lie}(U_{P_I})(F) = \bigoplus_{\substack{\alpha \in \Phi^+ \\ \alpha \notin \mathrm{span}(I)}} \mathfrak{g}_\alpha. \tag{44}$$

Every root in this sum has a positive coefficient outside I, so $\mathrm{val}(\alpha(s_j)) \to +\infty$. For $X \in \mathfrak{g}_\alpha$, our coadjoint convention gives

$$\gamma_j(X) = \alpha(s_j)^{-1} \delta_j(X), \qquad \delta_j(X) = \alpha(s_j) \gamma_j(X). \tag{45}$$

The values $\gamma_j(X)$ remain bounded, whereas $\alpha(s_j) \to 0$. Thus the limit of δ_j annihilates every root space in (44). This contradicts its membership in $\mathcal{E}$, since P_I is proper. Multiple relative roots cause no change: their valuations are the corresponding multiples, and they occur as their own weight spaces in (41). The argument uses actual S-characters, so it also applies when some root or coroot differentials vanish in characteristic p.

This proves relative compactness. The transporter $T(A,B)$ is closed: for convergent transporters, compactness of B supplies a convergent subsequence of source covectors, and closedness of A preserves the transporting relation. Since $T(A,B)$ is Z-saturated, its image in Q is closed and hence compact. The quotient Q is locally profinite and second countable, so the sequence argument suffices. Its local profinite structure follows, for example, from the compact open images $V/(V \cap Z)$ of compact open subgroups $V \subset G$. □

Proposition 8.2. *Let $\beta \in \mathcal{E}$, and let $L^\vee$ be any lattice in $\mathfrak{g}^*$. There is an open subgroup $C \subset G$, compact modulo Z, such that for all sufficiently large m,*

$$\{ g \in G : (\beta + \varpi^m L^\vee) \cap g(\beta + \varpi^m L^\vee) \neq \varnothing \} \subseteq C. \tag{46}$$

Proof. Let H be the stabilizer of β. Proposition 8.1 shows that H/Z is compact. Choose a compact open subgroup $V \subset Q$. Its normalizer is open, so H/Z has only finitely many conjugates of V. Their intersection W is compact open and normalized by H/Z. Consequently $\overline{C} = (H/Z)W$ is a compact open subgroup of Q. Its inverse image C in G is open, compact modulo Z, and contains H.

Choose a compact neighborhood Ω_0 of β in $\mathcal{E}$. Its transporter has compact image in Q. Let $\Omega_j \subset \Omega_0$ be compact neighborhoods shrinking to β. If each had a transporter $g_j \notin C$, a subsequence of $g_j Z$ would converge to some $q \in Q$. The transporting source and target covectors both converge to β, so q fixes β. The complement of $\overline{C}$ is closed, hence $q \notin \overline{C}$, contradicting $H/Z \subset \overline{C}$. Taking Ω_j to be sufficiently small lattice cosets proves (46). □

Proposition 8.3. *Suppose $\beta \in \mathcal{E}$. After multiplying it by a nonzero scalar, regard it as an element of the special-parahoric lattice $L^\vee$ from Proposition 7.1. Choose m as in Proposition 8.2, and take $a \geq \max\{a_0, m+c\}$ and $n = a + m$. Let $\theta = \theta_\beta$ be the character of K_a in* (25)*. There is a smooth unitary character χ of Z extending $\theta|_{Z\cap K_a}$. For any such χ, put*

$$J = ZK_a, \qquad \lambda(zh) = \chi(z)\theta(h) \quad (z \in Z,\ h \in K_a). \tag{47}$$

With C as in Proposition 8.2, decompose

$$\operatorname{Ind}_J^C \lambda \simeq \bigoplus_{i=1}^{s} \sigma_i^{\oplus d_i} \tag{48}$$

into distinct irreducible representations. Then c-$\operatorname{Ind}_C^G \sigma_i$ *are irreducible supercuspidal and pairwise inequivalent. They are exactly the irreducible smooth representations with central character χ containing θ.*

Proof. Nonzero scalar multiplication preserves $\mathcal{E}$, so the integral normalization is allowed. Propositions 7.1 and 8.2 give $I_G(\theta) \subseteq C$. In particular, $K_a \subseteq C$. The circle group is divisible, so the unitary character $\theta|_{Z\cap K_a}$ extends algebraically to Z. Its kernel still contains the open kernel of the original character, making the extension smooth. Agreement on $Z \cap K_a$ makes (47) well defined. The subgroup J is open and has finite index in C, since C/Z is compact. Thus the induction in (48) is finite-dimensional and unitary, and it is completely reducible. Each σ_i has central character χ.

Restriction gives $I_G(\lambda) \subseteq I_G(\theta) \subseteq C$. Consequently all endomorphisms of c-$\operatorname{Ind}_J^G \lambda$ are supported inside C. Induction in stages gives

$$\operatorname{End}_G\big(\text{c-}\operatorname{Ind}_J^G \lambda\big) = \operatorname{End}_C\big(\operatorname{Ind}_J^C \lambda\big) \simeq \bigoplus_{i=1}^{s} M_{d_i}(\mathbf{C}). \tag{49}$$

The endomorphisms already supported inside C account for the whole right side. Mackey decomposition therefore supplies no intertwining between the σ_i outside C and no cross-intertwining between distinct σ_i anywhere in G.

Put $\pi_i =$ c-$\operatorname{Ind}_C^G \sigma_i$. Its σ_i-isotypic space under C is exactly its canonical inducing fiber. The isotypic projector is obtained by averaging over C/Z: the central scalars cancel because both representations have central character χ. Given a nonzero G-subrepresentation, translate a nonzero inducing function so that its value at 1 is nonzero, and apply this projector. Evaluation at 1 is C-equivariant, so the projected value remains nonzero. Irreducibility of σ_i then puts the whole inducing fiber in the subrepresentation, and its translates generate π_i. This proves irreducibility. The same Mackey computation proves pairwise inequivalence.

Applying the argument to $\sigma_i^\vee$ and using the natural pairing of the compact inductions shows that the matrix coefficients are compact modulo Z. Irreducibility and admissibility identify the paired representation with the smooth dual, and the usual matrix coefficient criterion proves supercuspidality. Finally, for a representation with central character χ, occurrence of θ is equivalent to occurrence of λ. Frobenius reciprocity, induction in stages, and (48) therefore give the stated exact occurrence result. □

These propositions apply to every supplied element of $\mathcal{E}$ without a smoothness assumption on its stabilizer. Nonemptiness of $\mathcal{E}$ and occurrence of one of the resulting congruence characters in an arbitrary supercuspidal representation are separate questions. The construction above does not assert universal exhaustion.

9. Elliptic input with simply connected derived group

Let $\mathbf{G}$ be a connected reductive F-group and T a maximal F-torus. Over a finite separable splitting field E, write

$$\mathfrak{g}_E = \mathfrak{t}_E \oplus \bigoplus_{\alpha\in\Phi(\mathbf{G}_E,T_E)} \mathfrak{g}_{\alpha,E}.$$

The projection onto $\mathfrak{t}_E$ descends to F. Thus a functional in $\mathfrak{t}^*(F)$ extends uniquely to a T-invariant functional on $\mathfrak{g}$ by vanishing on the sum of the root spaces. We call this its toral extension. For an absolute root α, put $H_\alpha = d\alpha^\vee(1) \in \mathfrak{t}_E$.

Proposition 9.1. *Suppose that T is elliptic, meaning that $T/Z(\mathbf{G})$ is anisotropic, and that the toral functional β satisfies $\beta(H_\alpha) \neq 0$ for every absolute root. Then β belongs to the set $\mathcal{E}$ in* (38)*.*

Proof. Work first over an algebraic closure $\overline{F}$. In the projective variety of parabolic subgroups of any fixed type, consider the closed subscheme

$$Y_\beta = \{P' : \beta|_{\mathrm{Lie}(U_{P'})} = 0\}.$$

The scheme structure is the zero scheme of the restriction of β to the tautological subbundle of unipotent Lie algebras. Since β is T-invariant, Y_β is T-stable.

At a T-fixed parabolic P', the torus T is contained in P'. Choose a Borel subgroup of P' containing T. Let Ψ denote the positive roots occurring in $\mathrm{Lie}(U_{P'})$. The tangent space to the parabolic variety at P' is

$$\mathfrak{g}_{\overline{F}}/\mathfrak{p}' = \bigoplus_{\alpha\in\Psi} \mathfrak{g}_{-\alpha,\overline{F}}.$$

The linearized equations for Y_β are

$$\beta([X,Y]) = 0 \quad \text{for every } Y \in \mathrm{Lie}(U_{P'}).$$

Choose root vectors with $[E_{-\alpha}, E_\alpha] = \pm H_\alpha$. For $X = \sum_{\alpha\in\Psi} x_\alpha E_{-\alpha}$, evaluation at $Y = E_\gamma$ gives $\pm x_\gamma \beta(H_\gamma) = 0$. Every other bracket has nonzero root weight, or is zero, and is annihilated by the toral functional. The hypotheses therefore give $X = 0$, so the tangent space to Y_β at every T-fixed point is zero.

The connected torus T preserves each irreducible component of Y_β. Each component is projective and has a T-fixed point by the fixed point theorem for connected solvable groups. A component of positive dimension would give a positive-dimensional tangent space at this point. Hence Y_β has finitely many geometric points. The connected group T fixes each of them, so every parabolic in Y_β contains T.

If a proper F-parabolic P annihilated β on its unipotent Lie algebra, the preceding conclusion would give $T \subset P$. A Levi subgroup of P containing T has a central F-split torus strictly larger modulo $Z(\mathbf{G})$ than the split center of $\mathbf{G}$. This contradicts ellipticity of T. There is therefore no such P, proving $\beta \in \mathcal{E}$. □

Proposition 9.2. *If $\mathbf{G}_{\mathrm{der}}$ is simply connected, then $\mathbf{G}$ admits an elliptic maximal F-torus and a toral functional satisfying the hypotheses of Proposition 9.1. This holds in every characteristic.*

Proof. Every semisimple group over a nonarchimedean local field has an anisotropic maximal F-torus. This statement includes local fields of positive characteristic; see [6, Exercise 5.18 and Remark 5.20], which refers to [7, Theorem 10.5.1]. Choose such a torus $T_0 \subset \mathbf{G}_{\mathrm{der}}$ and put $T = Z_{\mathbf{G}}(T_0)$. Over a splitting field, the centralizer calculation with roots shows that T is a maximal torus, that $T \cap \mathbf{G}_{\mathrm{der}} = T_0$, and that T/T_0 is a torus. Moreover, $T/Z(\mathbf{G})$ is isogenous to a quotient of T_0 by a finite central subgroup. It is anisotropic, so T is elliptic.

The exact sequence of tori gives a saturated inclusion $X_*(T_0) \subset X_*(T)$. Simple connectedness of the derived group identifies $X_*(T_0)$ with its coroot lattice. Each coroot is Weyl-conjugate to a simple coroot of the same length, and a simple coroot is a member of a lattice basis. Consequently every coroot is primitive in $X_*(T_0)$ and in $X_*(T)$. In positive characteristic p, the differential of a cocharacter of a split torus vanishes exactly when that cocharacter belongs to $pX_*(T)$. Thus $H_\alpha \neq 0$ for every absolute root in every characteristic.

Choose a finite separable splitting field E. For each root, the condition $\beta(H_\alpha) = 0$ defines a proper F-linear subspace of $\mathfrak{t}^*(F)$: the F-linear dual spans $\mathfrak{t}_E^*$ after scalar extension and therefore detects the nonzero vector H_α. The field F is infinite, so finitely many such proper subspaces do not cover $\mathfrak{t}^*(F)$. Choose a functional outside their union and take its toral extension. Proposition 9.1 applies. If the derived group is trivial, there are no roots or proper parabolics and the same assertion holds with $T = \mathbf{G}$ and $\beta = 0$. □

The use of the simply connected derived group here is limited to nonvanishing of the actual coroot differentials. The next section removes this restriction on the group for the construction of representations by descending an inducing pair after a central twist.

10. Central twists and descent of inducing data

Let F be a nonarchimedean local field of any characteristic. An induced F-torus is a finite product of tori of the form $\mathrm{Res}_{E/F}\,\mathbf{G}_m$, where E/F is finite and separable. We first record the central extension used in the construction.

Proposition 10.1. *Every connected reductive F-group G admits an exact sequence*

$$1 \longrightarrow S \longrightarrow \widetilde{G} \xrightarrow{q} G \longrightarrow 1 \tag{50}$$

in which S is an induced torus and $\widetilde{G}_{\mathrm{der}}$ is simply connected. The map on F-points is an open surjection with kernel $S(F)$. Inflation identifies smooth representations of $G(F)$ with smooth representations of $\widetilde{G}(F)$ trivial on $S(F)$, and preserves irreducibility and supercuspidality.

Proof. Let $G_{\mathrm{sc}} \to G_{\mathrm{der}}$ be the simply connected cover and put $G' = G_{\mathrm{sc}} \times \mathrm{Rad}(G)$, where $\mathrm{Rad}(G)$ is the central radical torus. The multiplication map $G' \to G$ is a central isogeny with finite kernel D_0 of multiplicative type. Choose a finite Galois extension splitting D_0. The finite character module of D_0 is a quotient of a finite free permutation module: take a finite set of generators and all their Galois translates. Dually, this gives a closed embedding $i : D_0 \hookrightarrow S$ into an induced torus. Define

$$\widetilde{G} = (G' \times S)/\{(d, i(d)^{-1}) : d \in D_0\}.$$

This finite central quotient is connected reductive. Its map to G has kernel the embedded copy of S, giving (50). The map $G_{\mathrm{sc}} \to \widetilde{G}$ has trivial scheme-theoretic kernel, since i is a closed embedding. Its image is the derived group, which is therefore simply connected. The construction allows D_0 to be nonsmooth.

Hilbert theorem 90 gives $H^1(F, S) = 1$, so q is surjective on F-points. The kernel S is smooth, and q is a smooth morphism. The local implicit function theorem therefore makes its map on F-points open. Thus $G(F)$ has the quotient topology from $\widetilde{G}(F)/S(F)$, proving the assertions about smooth representations and irreducibility. Parabolic subgroups correspond under the central quotient, and their unipotent radicals map isomorphically. The corresponding Jacquet modules are the same vector-space quotients. Their vanishing proves the assertion about supercuspidality. □

The character extension needed below also applies when an isogeny of tori is inseparable. We give the topological details.

Lemma 10.2. *Let A be a locally compact totally disconnected abelian group, and let $B \subseteq A$ have the subspace topology. Every smooth character $\eta : B \to \mathbf{C}^\times$ extends to a smooth character of A. A unitary character has a unitary extension.*

Proof. Choose a compact open subgroup $U \subseteq A$ such that $U \cap B \subseteq \ker \eta$. On the open subgroup BU, the formula $\eta_0(bu) = \eta(b)$ is well-defined and defines a character trivial on U. Divisibility of $\mathbf{C}^\times$ extends this character algebraically to A. Its kernel still contains U, so the extension is smooth. For a unitary character use divisibility of the unit circle instead. □

Lemma 10.3. *Let $f : S \to A_0$ be an isogeny of F-tori, with kernel D. Its map on F-points is proper, its image B is closed in $A_0(F)$, and the induced map $S(F)/D(F) \to B$ is a homeomorphism. Consequently, every smooth character of $S(F)$ trivial on $D(F)$ descends to a smooth character of B. These conclusions do not require f to be separable.*

Proof. Choose affine embeddings of S and A_0. Since f is finite, every coordinate function on S is integral over the coordinate ring of A_0. It therefore satisfies a monic equation whose coefficients are regular functions on A_0. On a compact subset of $A_0(F)$ these coefficients are bounded. The nonarchimedean bound for a root of a monic polynomial bounds each source coordinate. Taking an affine embedding that includes the required inverse coordinates, the inverse image of the compact set is a closed bounded subset of a finite-dimensional F-vector space, hence is compact.

Thus f is proper on F-points and has closed image. It induces a continuous proper bijection $S(F)/D(F) \to B$, which is a homeomorphism. The quotient map by $D(F)$ is open. A smooth character trivial on $D(F)$ therefore descends smoothly through this homeomorphism. □

Fix (50), and put

$$D = S \cap \widetilde{G}_{\mathrm{der}}, \qquad A = \widetilde{G}/\widetilde{G}_{\mathrm{der}}, \qquad Z = Z(\widetilde{G})(F). \tag{51}$$

The intersection defining D is scheme-theoretic. Since it lies in the center of the semisimple group $\widetilde{G}_{\mathrm{der}}$, it is finite, but it need not be smooth. The image of $S \to A$ is a subtorus A_0, and $S \to A_0$ is an isogeny with kernel D.

Lemma 10.4. *Let $H \subseteq \widetilde{G}(F)$ be compact and open, with $H \cap D(F) = \{1\}$, and let θ be a smooth character of H. There is a smooth central character $\omega : Z \to \mathbf{C}^\times$ such that*

$$\omega|_{Z \cap H} = \theta|_{Z \cap H}, \qquad \omega|_{D(F)} = 1.$$

The character ω can be chosen unitary. The condition on H holds for all sufficiently small compact open subgroups.

Proof. On $D(F)(Z \cap H)$ define $\omega_0(dh) = \theta(h)$. The intersection hypothesis makes this well-defined. It is a smooth character, trivial on $D(F)$. Since H is compact, θ has finite image, so ω_0 is unitary. Apply Lemma 10.2 in the abelian group Z. Finally, $D(F)$ is finite, so a sufficiently small neighborhood of the identity meets it only at 1. □

Proposition 10.5. *Suppose an irreducible supercuspidal representation $\widetilde{\pi}$ of $\widetilde{G}(F)$ has central character ω trivial on $D(F)$ and is given by an actual compact-inducing pair*

$$\widetilde{\pi} \simeq \text{c-Ind}_{\widetilde{J}}^{\widetilde{G}(F)} \widetilde{\lambda}, \tag{52}$$

where $\widetilde{J}$ is open and compact modulo Z, and $\widetilde{\lambda}$ is irreducible. There is a smooth character χ of $\widetilde{G}(F)$ factoring through $A(F)$ such that $\widetilde{\pi} \otimes \chi$ is trivial on $S(F)$. The twisted inducing pair descends to an irreducible representation λ of an open subgroup $J \subseteq G(F)$ compact modulo $Z(G)(F)$, and

$$(\text{c-Ind}_{J}^{G(F)} \lambda) \circ q \simeq \widetilde{\pi} \otimes \chi. \tag{53}$$

In particular, the compact induction downstairs is irreducible and supercuspidal.

Proof. By Lemma 10.3, $\omega|_{S(F)}$ descends to a smooth character of the closed subgroup $B = \text{im}(S(F) \to A(F))$. Here $A_0(F)$ is closed in $A(F)$, so the topology is the one required in Lemma 10.2. Extend the inverse character from B to a smooth character χ_A of $A(F)$, and let χ be its pullback to $\widetilde{G}(F)$. Then $(\omega\chi)|_{S(F)} = 1$. Twisting (52) gives

$$\widetilde{\pi} \otimes \chi \simeq \text{c-Ind}_{\widetilde{J}}^{\widetilde{G}(F)} \widetilde{\lambda}', \qquad \widetilde{\lambda}' = \widetilde{\lambda} \otimes \chi|_{\widetilde{J}}.$$

We verify that the actual inducing pair descends. Use the compact-induction model $f(jg) = \widetilde{\lambda}'(j)f(g)$ with right translation. For a nonzero vector v, let f_v be supported on $\widetilde{J}$ with $f_v(1) = v$. If $s \in S(F)$ were outside $\widetilde{J}$, its right translation would satisfy $f_v(s) = 0 \neq f_v(1)$, contradicting the trivial action of $S(F)$ on $\widetilde{\pi} \otimes \chi$. Thus $S(F) \subseteq \widetilde{J}$. Evaluating the trivial action at 1 now gives $\widetilde{\lambda}'(s)v = v$ for every $s \in S(F)$ and every v.

Put $J = q(\widetilde{J})$ and let λ be the descended representation. The openness of q makes J open and λ smooth. Irreducibility of $\widetilde{\lambda}'$ implies irreducibility of λ. The natural map from the compact space $\widetilde{J}/(\widetilde{J} \cap Z)$ onto $J/(J \cap Z(G)(F))$ proves compactness modulo center.

Since $S(F) \subseteq \widetilde{J}$, one has $q^{-1}(J) = \widetilde{J}$, and q identifies the discrete coset spaces $\widetilde{J}\backslash\widetilde{G}(F)$ and $J\backslash G(F)$. Pullback $f \mapsto f \circ q$ therefore identifies their compact-induction spaces. Conversely, every upstairs inducing function is constant on the fibers of q, because $S(F)$ is central and acts trivially through $\widetilde{\lambda}'$. This proves (53). Twisting preserves irreducibility and supercuspidality, and Proposition 10.1 gives these properties downstairs. □

In a construction from a deep character θ on H, choose H small enough to avoid $D(F)$ and choose the central character by Lemma 10.4. Every compact-inducing pair constructed with that central character then satisfies Proposition 10.5. The twist is applied to the representation and its inducing pair; it need not preserve depth or a chosen coadjoint coset. Exhaustion would additionally require occurrence of suitable input data in every supercuspidal representation.

For an explicit calculation of the change in conductor, take $F = \mathbf{F}_2((t))$ and the central extension $\text{GL}_2 \to \text{PGL}_2$. Normalize ψ by taking the coefficient of t^{-1}, and put

$$M = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad H_a = 1 + t^a M_2(\mathcal{O}), \qquad \theta(1 + X) = \psi(t^{-n} \operatorname{tr}(MX)), \quad a < n \leq 2a.$$

Write $U_j = 1 + t^j\mathcal{O} \subset F^\times$. Since $\operatorname{tr}(M) = 1$, a central character extending the scalar restriction satisfies

$$\omega(1 + x) = \psi(t^{-n}x) \quad (x \in t^a\mathcal{O}).$$

It is trivial on U_n and nontrivial on $1 + t^{n-1}$, so its conductor exponent is n. The map from the scalar center to the abelianization is $z \mapsto z^2$. On the closed subgroup of squares define $\eta(z^2) = \omega(z)^{-1}$. In characteristic two the square map is injective, and

$$(F^\times)^2 \cap U_{2n-1} = (U_n)^2.$$

Hence η extends to a character χ of $F^\times$ trivial on U_{2n-1}. It satisfies

$$\chi(1+t^{2n-2}) = \omega(1+t^{n-1})^{-1} = -1,$$

so its conductor exponent is exactly $2n-1$. Twisting by $\chi \circ \det$ cancels the original scalar character through $\det((1+x)I) = 1+x^2$. Since $2n-1 > 2a$, this cancellation uses the quadratic determinant term outside the original linearization interval. It is consistent with the vanishing differential of the determinant on scalar directions in characteristic two.

11. Compact-induction data for every connected reductive group

Let G now denote any connected reductive group over a nonarchimedean local field F. The following data use the group laws and congruence quotients of smooth parahoric models. They impose no restriction on the residue characteristic or on the ramification of a splitting field.

Choose a central extension $1 \to S \to \widetilde{G} \xrightarrow{q} G \to 1$ as in Proposition 10.1, and put

$$Z = Z(\widetilde{G})(F), \qquad D = S \cap \widetilde{G}_{\mathrm{der}}, \qquad A = \widetilde{G}/\widetilde{G}_{\mathrm{der}}.$$

Write $\mathrm{ab} : \widetilde{G}(F) \to A(F)$ for the natural map. Choose an elliptic maximal torus $T \subset \widetilde{G}$ and a toral functional β satisfying $\beta(H_\alpha) \neq 0$ for all absolute roots. Such a choice exists by Proposition 9.2. Choose a special maximal parahoric model $\mathcal{P}$ for $\widetilde{G}$, with Lie lattice L, and multiply β by a nonzero scalar to put it in $L^\vee$.

Choose an open subgroup $C \subset \widetilde{G}(F)$ containing Z and compact modulo Z, and an integer $m \geq 1$, such that

$$\{g : (\beta + \varpi^m L^\vee) \cap g(\beta + \varpi^m L^\vee) \neq \varnothing\} \subseteq C. \tag{54}$$

Propositions 9.1 and 8.2 guarantee these choices. With the constants of Proposition 7.1, choose $a \geq \max\{a_0, m+c\}$ sufficiently large that $K_a \cap D(F) = \{1\}$. Put $H = K_a$, $n = a+m$, and

$$\theta(h) = \psi\big(\varpi^{-n}\beta(\ell_{a,n}(h))\big). \tag{55}$$

Choose a smooth unitary character ω of Z with $\omega|_{Z\cap H} = \theta|_{Z\cap H}$ and $\omega|_{D(F)} = 1$. Lemma 10.4 guarantees this choice.

The character $\lambda_0(zh) = \omega(z)\theta(h)$ is well-defined on ZH. Choose an irreducible constituent σ of the finite-dimensional representation

$$V_{\mathcal{D}} = \mathrm{Ind}_{ZH}^{C} \lambda_0. \tag{56}$$

Finally, choose a smooth unitary character χ_A of $A(F)$ whose restriction to $\mathrm{ab}(S(F))$ is the inverse of the character induced by $\omega|_{S(F)}$. Lemmas 10.2 and 10.3 give such a character, including when the torus isogeny is inseparable. These choices form a datum $\mathcal{D}$.

Theorem 11.1. *For every connected reductive G over every nonarchimedean local field F, data $\mathcal{D}$ as above exist. Each datum defines an open subgroup $J_{\mathcal{D}} = q(C)$ compact modulo $Z(G)(F)$ and an irreducible smooth representation $\lambda_{\mathcal{D}}$ by*

$$\lambda_{\mathcal{D}}(q(c)) = \sigma(c)\chi_A(\mathrm{ab}(c)) \quad (c \in C). \tag{57}$$

The formula is well-defined, and

$$\pi(\mathcal{D}) = \text{c-Ind}_{J_{\mathcal{D}}}^{G(F)} \lambda_{\mathcal{D}} \tag{58}$$

is irreducible and supercuspidal. For all choices other than σ fixed, distinct irreducible constituents in (56) *give inequivalent outputs.*

Proof. The preceding propositions guarantee every choice except the existence of a constituent in (56). Equations (54) and (26) give $I_{\widetilde{G}(F)}(\theta) \subseteq C$, hence $H \subseteq C$. The subgroup ZH has finite index in C. Its finite induction is nonzero and unitary, so it has an irreducible constituent σ. Proposition 8.3 shows that

$$\widetilde{\pi}_\sigma = \text{c-Ind}_{C}^{\widetilde{G}(F)} \sigma$$

is irreducible supercuspidal with central character ω.

Put $\chi = \chi_A \circ \mathrm{ab}$. For $s \in S(F)$, centrality gives

$$\sigma(s)\chi(s) = \omega(s)\chi_A(\mathrm{ab}(s))\,\mathrm{id} = \mathrm{id}\,.$$

Here $S(F) \subseteq Z \subseteq C$. Thus $\sigma \otimes \chi|_C$ factors through $q(C)$, giving exactly (57). The map q is open and surjective on F-points, so $J_{\mathcal{D}}$ is open. The quotient $J_{\mathcal{D}}/(J_{\mathcal{D}} \cap Z(G)(F))$ is a continuous image of the compact group C/Z. Irreducibility of σ gives irreducibility of $\lambda_{\mathcal{D}}$.

Twisting and pullback of inducing functions give the explicit chain

$$\begin{aligned} \pi(\mathcal{D}) \circ q &\simeq \text{c-Ind}_C^{\widetilde{G}(F)}(\sigma \otimes \chi|_C) \\ &\simeq \left(\text{c-Ind}_C^{\widetilde{G}(F)} \sigma\right) \otimes \chi = \widetilde{\pi}_\sigma \otimes \chi. \end{aligned} \tag{59}$$

The first identification uses $q^{-1}(q(C)) = C$, since the kernel $S(F)$ is contained in C; thus the two discrete coset spaces and the finite support conditions agree. Proposition 10.1 now proves irreducibility and supercuspidality downstairs. For fixed auxiliary choices, Proposition 8.3 distinguishes the $\widetilde{\pi}_\sigma$. Twisting by the same character and descending through a surjection preserve inequivalence. □

There is also an exact occurrence statement for the fixed data. For a smooth representation π of $G(F)$ define

$$\Pi = (\pi \circ q) \otimes \chi^{-1}.$$

Among irreducible π for which Π has central character ω, the outputs obtained by varying σ in (56) are exactly those satisfying

$$\text{Hom}_H(\theta, \Pi) \neq 0. \tag{60}$$

Indeed, Proposition 8.3 identifies all such Π with the displayed compact inductions upstairs, and (59) identifies their descents. In particular, every irreducible representation satisfying these fixed occurrence conditions is supercuspidal.

The theorem gives a nonempty construction for every group. It does not exhaust all supercuspidal representations: the next section gives a representation that cannot satisfy (60) for any datum. The finite algebra at this step is the endomorphism algebra of the explicitly given finite-dimensional induction (56); its simple constituents are used to build the actual inducing representation (57). Theorem 30.3 gives a finite equivalence test across all choices of $\mathcal{D}$ by comparing their final inducing pairs on $G(F)$. A universal extraction algorithm is not asserted.

12. A depth-zero obstruction to extraction from one linear character

The construction from a coadjoint coset with compact transporter does not exhaust supercuspidal representations, even after passage to a central extension and a smooth abelian twist. The following example is in characteristic two. The representation in the example already has a compact inducing pair; the obstruction concerns the proposed extraction from a single deep linear character.

Put $F = \mathbf{F}_2((t))$, $\mathcal{O} = \mathbf{F}_2[[t]]$, and $G = \text{SL}_2(F)$. Write $K = \text{SL}_2(\mathcal{O})$ and $K_a = \ker(K \to \text{SL}_2(\mathcal{O}/t^a\mathcal{O}))$ for $a \geq 1$. The reduction of K is $\text{SL}_2(\mathbf{F}_2) \simeq S_3$. Let σ be its sign character, inflated to K, and set

$$\pi = \text{c-Ind}_K^G \sigma. \tag{61}$$

Lemma 12.1. *The representation π in* (61) *is irreducible, admissible, and supercuspidal. It has depth zero.*

Proof. The unipotent radical of a proper parabolic in $\text{SL}_2(\mathbf{F}_2)$ is a group of order two. Its nonidentity element is a transposition under the identification with S_3, and hence acts by -1 in σ. Thus σ has no invariants under any such unipotent radical and is cuspidal as a representation of the finite group.

We give the compact induction argument directly. The Cartan decomposition is

$$G = \coprod_{r \geq 0} K d_r K, \qquad d_r = \text{diag}(t^r, t^{-r}).$$

For $r > 0$, the subgroup $U_-(\mathcal{O})$ is contained in $K \cap d_r K d_r^{-1}$. The character σ is nontrivial on this subgroup, whereas its d_r-conjugate is trivial there, since

$$d_r^{-1} x_-(u) d_r = x_-(t^{2r}u) \in K_1 \qquad (u \in \mathcal{O}).$$

It follows that the intertwining set of σ is exactly K. The Mackey decomposition then shows that the σ-isotypic subspace of $\pi|_K$ is precisely the canonical inducing fiber. Any nonzero subrepresentation contains a function whose value at the identity is nonzero after translation. Applying the σ-isotypic projector preserves that value

and gives a nonzero vector in the inducing fiber. The fiber is irreducible under K, and its G-translates span π. This proves irreducibility.

For admissibility, fix $a \geq 1$. A K_a-invariant function supported in Kd_rK must vanish when $2r \geq a$. Indeed, $K \cap d_r K_a d_r^{-1}$ then contains $U_-(\mathcal{O})$, on which σ is nontrivial. Only the finitely many indices with $2r < a$ remain, and each Kd_rK is a union of finitely many right K_a-cosets. Hence π^{K_a} is finite-dimensional. Since the K_a form a neighborhood basis at the identity, this proves admissibility. Matrix coefficients of this compact induction are compactly supported. The coefficient criterion for an irreducible admissible representation therefore gives supercuspidality. Finally, the inducing fiber is fixed by K_1, so the depth is zero. □

Fix a nontrivial additive character $\psi : F \to \mathbf{C}^\times$ whose annihilator of $\mathcal{O}$ is $\mathcal{O}$. Let K' be any vertex parahoric of G, let L' be the Lie lattice of its smooth parahoric model, and let K'_a be its congruence subgroups. Write $(L')^\vee = \mathrm{Hom}_{\mathcal{O}}(L', \mathcal{O})$. For integers $a < n \leq 2a$ and $\beta \in (L')^\vee$, the canonical square-zero congruence quotient defines

$$\theta_\beta(h) = \psi\big(t^{-n}\beta(\ell_{a,n}(h))\big), \qquad \ell_{a,n} : K'_a/K'_n \simeq t^a L'/t^n L'. \tag{62}$$

Put $m = n - a$ and $\Omega_\beta = \beta + t^m (L')^\vee$. For a subset Ω of $\mathfrak{g}^*$, write

$$\mathrm{Trans}_G(\Omega) = \{g \in G : g\Omega \cap \Omega \neq \varnothing\},$$

where $(g\gamma)(X) = \gamma(\mathrm{Ad}(g^{-1})X)$.

Proposition 12.2. *If θ_β in (62) occurs in $\pi|_{K'_a}$, then Ω_β contains a covector whose stabilizer contains a conjugate of $U_-(F)$. In particular, $\mathrm{Trans}_G(\Omega_\beta)$ is not compact modulo the center. This holds for every vertex parahoric, including both G-orbits of vertices.*

Proof. Choose $\eta \in \mathrm{PGL}_2(F)$ such that $K' = \eta K \eta^{-1}$. Such an η exists for every vertex, and it identifies $K'_a = \eta K_a \eta^{-1}$ and $L' = \mathrm{Ad}(\eta)\mathfrak{sl}_2(\mathcal{O})$. The Mackey decomposition of (61) restricted to K'_a shows that occurrence of θ_β gives some $g \in G$ and a nonzero intertwining map on $K'_a \cap gKg^{-1}$. Since σ is trivial on K_1, we necessarily have

$$\theta_\beta|_{K'_a \cap gK_1g^{-1}} = 1. \tag{63}$$

Apply the Cartan decomposition in $\mathrm{PGL}_2(F)$ to write

$$\eta^{-1}g = k_1 d k_2, \qquad d = \mathrm{diag}(t^r, 1), \qquad r \geq 0, \qquad k_1, k_2 \in \mathrm{PGL}_2(\mathcal{O}).$$

Integral conjugation preserves K_a, K_1, and $L = \mathfrak{sl}_2(\mathcal{O})$. Thus conjugation by ηk_1 identifies the intersection in (63) with $K_a \cap dK_1 d^{-1}$. Define $\widehat{\beta} \in L^\vee$ by $\widehat{\beta}(X) = \beta(\mathrm{Ad}(\eta k_1)X)$. The same change of coordinates identifies the character and its coset with $\theta_{\widehat{\beta}}$ and $\widehat{\beta} + t^m L^\vee$.

In characteristic two the matrices

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

form an $\mathcal{O}$-basis of L. For every $z \in t^a\mathcal{O}$, $x_-(z)$ lies in $K_a \cap dK_1 d^{-1}$, because $d^{-1}x_-(z)d = x_-(t^r z) \in K_1$. Its linearization is zf, so (63) and the conductor of ψ imply

$$\widehat{\beta}(f) \in t^m\mathcal{O}. \tag{64}$$

For every $x \in t^a\mathcal{O}$, the element $s(x) = \mathrm{diag}(1+x, (1+x)^{-1})$ also belongs to this intersection: it commutes with d and belongs to $K_a \subset K_1$. Since $n \leq 2a$, its canonical linearization modulo t^n is xh. Indeed, $(1+x)^{-1} - 1 \equiv x \pmod{t^{2a}}$ in characteristic two. Consequently

$$\widehat{\beta}(h) \in t^m\mathcal{O}. \tag{65}$$

Define $\beta_0 \in L^\vee$ by $\beta_0(e) = \widehat{\beta}(e)$ and $\beta_0(f) = \beta_0(h) = 0$. Equations (64) and (65) show that $\beta_0 \in \widehat{\beta} + t^m L^\vee$. For every $u \in F$, direct multiplication gives

$$\mathrm{Ad}(x_-(u))e = e + uh + u^2 f, \qquad \mathrm{Ad}(x_-(u))h = h, \qquad \mathrm{Ad}(x_-(u))f = f. \tag{66}$$

It follows that $U_-(F)$ fixes β_0. Therefore $U_-(F)$ is contained in the transporter of $\widehat{\beta} + t^m L^\vee$. This subgroup is closed and noncompact, while the center of $\mathrm{SL}_2(F)$ is trivial. Conjugating back by ηk_1 proves the assertion. In particular, the original coset cannot be contained in the rationally elliptic locus, since the transported covector β_0 annihilates $\mathrm{Lie}(U_-)$. □

Corollary 12.3. *The representation π cannot be obtained from the construction using a single deep coadjoint coset with compact transporter by first passing to a central extension*

$$1 \longrightarrow S \longrightarrow \widetilde{G} \xrightarrow{q} \mathrm{SL}_2 \longrightarrow 1$$

by an induced torus with $\widetilde{G}^{\mathrm{der}}$ simply connected, then twisting by an arbitrary smooth character and descending.

Proof. The restriction of q to $\widetilde{G}^{\mathrm{der}}$ is a central isogeny onto SL_2. Since the target is simply connected, this restriction is an isomorphism. Its scheme-theoretic intersection with S is therefore trivial, and multiplication gives an isomorphism

$$\widetilde{G} \simeq \mathrm{SL}_2 \times S$$

under which q is the first projection. In particular, the abelianization is S, so its characters are trivial on the SL_2 factor. The same holds for every smooth character of $\widetilde{G}(F)$: the group $\mathrm{SL}_2(F)$ is perfect. To see this directly, choose $c \in F^\times$ with $c^2 \neq 1$. Then

$$[\mathrm{diag}(c, c^{-1}), x_+(v)] = x_+((c^2-1)v),$$

so the upper root group lies in the commutator subgroup. The corresponding formula with $c^{-2}-1$ gives the lower root group, and these two root groups generate $\mathrm{SL}_2(F)$.

If a representation upstairs twists and descends to π, it is consequently of the form $\pi \otimes \nu$ for a smooth character ν of $S(F)$. Parahoric models, their congruence subgroups, and their Lie lattices for this product are products. Occurrence of a deep linear character in $\pi \otimes \nu$ therefore implies occurrence of its SL_2 factor in π. By Proposition 12.2, the corresponding factor of the coadjoint coset contains a covector fixed by a conjugate of $U_-(F)$. Combining it with any point in the torus factor gives a point of the full coset fixed by this same unipotent subgroup. The full transporter is therefore noncompact modulo $Z(\widetilde{G})(F)$, contradicting the required compact-transporter condition. □

Thus an exhaustion argument must use data beyond these single deep linear sectors. In this example the needed datum is already the cuspidal representation σ of the finite reductive quotient, and (61) provides the compact inducing pair explicitly. The example disproves universal occurrence for the one-stage construction, not the compact-induction conjecture.

The required depth-zero data have an established exhaustion theorem for every connected reductive group over every nonarchimedean local field. Choose a maximal parahoric P, a cuspidal irreducible representation $\overline{\sigma}$ of its finite reductive quotient P/P^+, and an irreducible representation τ of $N_{G(F)}(P)$ whose restriction to P contains the inflation of $\overline{\sigma}$. Moy and Prasad prove that

$$\text{c-Ind}_{N_{G(F)}(P)}^{G(F)} \tau$$

is irreducible supercuspidal of depth zero and that every depth-zero supercuspidal is obtained in this way [9, Propositions 6.6 and 6.8]. These results impose no restriction on the residue characteristic or splitting field. The representation τ is allowed to contain several conjugates of $\overline{\sigma}$ on restriction; an extension of a single $\overline{\sigma}$ is not required.

Corollary 12.4. *Let π_1 be a positive-depth irreducible supercuspidal representation of $\mathrm{SL}_2(F)$, with $F = \mathbf{F}_2((t))$, and let π_0 be* (61)*. Then $\pi_0 \boxtimes \pi_1$ is a positive-depth supercuspidal representation of $\mathrm{SL}_2(F) \times \mathrm{SL}_2(F)$ which cannot occur in the family of Theorem 11.1.*

Proof. The external tensor product is irreducible and admissible. Parabolic subgroups and their Jacquet functors factor over the two groups. Every proper parabolic has a proper factor, whose Jacquet module vanishes by supercuspidality. Thus the tensor product is supercuspidal. Buildings and Moy–Prasad filtrations also factor, so its depth is $\max\{0, \mathrm{depth}(\pi_1)\} > 0$.

Every vertex parahoric, Lie lattice, and congruence subgroup in the product is a product. Any occurring deep linear character therefore restricts to an occurring deep linear character of π_0. By Proposition 12.2, the first factor of its coadjoint coset contains a covector fixed by a conjugate of $U_-(F)$. Pair it with any covector in the second factor. The full coset has a point fixed by the same root group in the first factor, so its transporter is noncompact. As in Corollary 12.3, every relevant central extension splits off its torus kernel because the target is semisimple and simply connected. Smooth twists are trivial on both $\mathrm{SL}_2(F)$ factors. Hence extension and twisting do not change the obstruction. □

The positive-depth hypothesis on π_1 is nonempty in this example. The three irreducible complex representations of $\mathrm{SL}_2(\mathbf{F}_2) \simeq S_3$ are the trivial, sign, and two-dimensional standard representations. Restriction to a unipotent subgroup of order two has invariant dimensions 1, 0, and 1, respectively. Thus sign is its only cuspidal representation. The depth-zero theorem just quoted and the two vertex orbits imply that every depth-zero supercuspidal of $\mathrm{SL}_2(F)$ is one of the two vertex versions of (61). The obstruction applies to both after conjugation in $\mathrm{PGL}_2(F)$. On the other hand, Propositions 9.2 and 8.3 give a nonempty family of supercuspidals containing characters with compact transporter for this group itself. None has depth zero, so any of them may be used as π_1.

The product example explains why adding only representations of depth zero to the single-character family is insufficient. Finite cuspidal data must also be allowed as part of a representation of positive depth. This argument does not address whether a proposed enlargement already exhausts positive-depth representations of an absolutely simple group.

13. Deeper types of a maximal compact subgroup

The following example concerns irreducible representations of a maximal compact subgroup occurring in an actual supercuspidal representation. Let $F = \mathbf{F}_2((t))$, let $\mathcal{O} = \mathbf{F}_2[[t]]$, and put

$$G = \mathrm{PGL}_2(F), \qquad K = \mathrm{PGL}_2(\mathcal{O}), \qquad K_r = \ker\bigl(K \to \mathrm{PGL}_2(\mathcal{O}/t^r\mathcal{O})\bigr) \quad (r \geq 1).$$

The group K is the stabilizer of a vertex of the Bruhat–Tits tree and is maximal compact. Inflate the sign character of $K/K_1 \simeq \mathrm{PGL}_2(\mathbf{F}_2) \simeq S_3$ to a character ε of K, and define

$$\pi = \text{c-Ind}_K^G \varepsilon, \qquad a_n = \begin{pmatrix} t^n & 0 \\ 0 & 1 \end{pmatrix} \in G.$$

Let w_0 be a nonzero vector in the canonical inducing fiber. For $n \geq 0$, put $w_n = \pi(a_n)w_0$ and $W_n = \mathrm{span}\{\pi(k)w_n : k \in K\}$. Let B_0^- and B_0^+ denote the lower and upper triangular subgroups of K, respectively.

Lemma 13.1. *Let τ be a finite-dimensional smooth representation of K, trivial on K_r, with $\tau^{B_0^-} \neq 0$. Then a_N intertwines τ for every $N \geq r$. In particular, its intertwining set is unbounded in G.*

Proof. The intersection $L_N = K \cap a_N K a_N^{-1}$ consists of classes having representatives

$$h = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathcal{O}), \qquad b \in t^N\mathcal{O}.$$

For $N \geq 1$, both a and d are units. If $N \geq r$, the image of h modulo K_r lies in the image of B_0^-. On the other hand,

$$a_N^{-1} h a_N = \begin{pmatrix} a & t^{-N}b \\ t^N c & d \end{pmatrix}$$

has image modulo K_r in the image of B_0^+. These statements follow by replacing the indicated off-diagonal entry by zero; the resulting triangular matrix is invertible and congruent to the original one.

Choose a K-invariant positive Hermitian form on τ. There are nonzero vectors v_- and v_+ fixed by B_0^- and B_0^+, since these groups are conjugate in K. The functional $\ell_-(v) = \langle v, v_- \rangle$ is invariant under L_N, while v_+ is invariant under the conjugate representation τ^{a_N} on L_N. Thus $v \mapsto \ell_-(v)v_+$ is a nonzero intertwiner. The elements a_N are unbounded, proving the last assertion. □

Proposition 13.2. *The representation π is irreducible and supercuspidal. Its restriction to K has the algebraic direct sum decomposition*

$$\pi|_K = \bigoplus_{n \geq 0} W_n, \qquad W_n \simeq \mathrm{Ind}_{L_n}^K(\varepsilon^{a_n}|_{L_n}), \qquad L_n = K \cap a_n K a_n^{-1}. \tag{67}$$

Here $W_0 = \varepsilon$. For every $n \geq 1$, the representation W_n is trivial on K_{n+1} and satisfies $W_n^{K_n} = 0$. Every irreducible constituent τ of W_n has nonzero B_0^--invariants and is intertwined by every a_N with $N \geq n+1$.

Consequently every irreducible K-type in π of level greater than one has unbounded intertwining. The only irreducible K-type in π with intertwining contained in K is ε.

Proof. The Cartan double cosets Ka_nK, $n \geq 0$, partition G. The corresponding inducing fibers give the direct sum (67). The stabilizer in K of the fiber at a_n is L_n, acting by ε^{a_n}. Each L_n has finite index in K, so W_n is finite-dimensional and is generated by w_n.

For $n \geq 1$, the group K_{n+1} is contained in L_n, and $a_n^{-1}K_{n+1}a_n \subseteq K_1$. Thus K_{n+1} fixes w_n. Normality in K shows that it fixes all of W_n. The group K_n is also contained in L_n, but its character on w_n is nontrivial:

$$a_n^{-1}\begin{pmatrix}1 & t^n\\ 0 & 1\end{pmatrix}a_n = \begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix},$$

whose reduction is a transposition in S_3. Averaging over K_n therefore kills w_n. Since this averaging operator commutes with K, it kills W_n, proving $W_n^{K_n} = 0$. In particular every constituent of W_n has exact congruence level $n+1$.

For $b \in B_0^-$, the matrix $a_n^{-1}ba_n$ reduces to the identity in $\mathrm{PGL}_2(\mathbf{F}_2)$: its lower off-diagonal entry is divisible by t^n, and its two diagonal units reduce to 1. Hence w_n is fixed by B_0^-. If τ is any irreducible constituent of W_n, its isotypic projection of w_n is nonzero, since w_n generates W_n. This projection commutes with B_0^-, so $\tau^{B_0^-} \neq 0$. Lemma 13.1 now supplies all the asserted unbounded intertwiners.

We also verify irreducibility and supercuspidality of the representation used in the example. The preceding level calculation shows that its ε-isotypic space under K is exactly W_0. Given a nonzero subrepresentation, translate a nonzero inducing function until its value at 1 is nonzero. Projection to the ε-isotypic space preserves this value, because evaluation at 1 is K-equivariant. The subrepresentation therefore contains W_0, whose translates generate π. This proves irreducibility. The same calculation gives

$$\pi^{K_r} = \bigoplus_{0 \leq n < r} W_n,$$

so π is admissible.

Let U be the upper unipotent subgroup of G. Iwasawa decomposition $G = KB$, where B is the upper triangular Borel, shows that every gUg^{-1} is K-conjugate to U. The character ε is nontrivial on $K \cap U$, and hence on $K \cap gUg^{-1}$ for every g. The Mackey summands of $\pi|_U$ are compact inductions from these compact open intersections, with nontrivial characters. Their U-coinvariants vanish, so $\pi_U = 0$. All proper parabolics of G are conjugate to B, which proves supercuspidality.

Finally, $I_G(\varepsilon) = K$ can also be seen directly. For $N > 0$, the lower root element with parameter 1 belongs to $K \cap a_N K a_N^{-1}$. Its ε-value is -1, whereas its value under ε^{a_N} is 1, since conjugation by a_N^{-1} makes its lower parameter t^N. Cartan decomposition excludes every double coset outside K. Together with the unbounded intertwining of all W_n, $n \geq 1$, this proves the last assertion. □

Proposition 13.3. *Let τ be an irreducible constituent of some W_n, $n \geq 1$, and let $V_\tau = \text{c-Ind}_K^G \tau$. Its actual occurrence in π gives a surjection $V_\tau \to \pi$. If R is its kernel and U^- is the lower unipotent subgroup, then*

$$0 \longrightarrow R_{U^-} \longrightarrow (V_\tau)_{U^-} \longrightarrow \pi_{U^-} \longrightarrow 0, \qquad R_{U^-} \simeq (V_\tau)_{U^-} \neq 0. \tag{68}$$

Proof. Compact Frobenius reciprocity gives a nonzero map $V_\tau \to \pi$, and irreducibility makes it surjective. The U^--orbit of the canonical inducing fiber supplies the Mackey summand

$$\text{c-Ind}_{K\cap U^-}^{U^-}(\tau|_{K\cap U^-})$$

in $V_\tau|_{U^-}$. Its coinvariants are $\tau_{K\cap U^-}$, which are nonzero because $\tau^{B_0^-} \neq 0$. Coinvariants commute with the direct sum of Mackey summands, so $(V_\tau)_{U^-} \neq 0$.

Over $\mathbf{C}$, coinvariants by U^- are exact. Indeed, U^- is the increasing union of compact open subgroups; coinvariants by each such subgroup are exact by averaging, and filtered direct limits are exact. Apply this to $0 \to R \to V_\tau \to \pi \to 0$ and use $\pi_{U^-} = 0$. This gives (68). □

Thus Jacquet vanishing of the given supercuspidal representation does not force Jacquet vanishing of compact induction from an occurring irreducible K-type. In these examples the kernel of the natural surjection contains its entire nonzero Jacquet module.

13.1. **A fixed vertex can miss every inducing type.** The depth-1/2 affine-generic construction for residue field two gives a stronger obstruction to fixing a vertex subgroup. We repeat its elementary intertwining calculation to specify the example. Let $I \subset K$ be the upper Iwahori. Its elements have representatives

$$h = \begin{pmatrix} a & b \\ tc & d \end{pmatrix}, \qquad a, d \in \mathcal{O}^\times, \quad b, c \in \mathcal{O},$$

and define $\theta(h) = (-1)^{\bar{b}+\bar{c}}$. Multiplication adds the two indicated residues. Multiplying a representative by a scalar unit does not change them, since its residue is 1. Thus θ is a character of I. Put

$$w = \begin{pmatrix} 0 & 1 \\ t & 0 \end{pmatrix} \in G, \qquad J = I \rtimes \langle w \rangle.$$

Conjugation by w interchanges b and c, and $w^2 = 1$ in G. Consequently θ extends to a character λ of J by either choice $\lambda(w) = 1$ or $\lambda(w) = -1$.

Proposition 13.4. *For either extension λ, the representation $\pi_{\mathrm{aff}} = \text{c-Ind}_J^G \lambda$ is irreducible and supercuspidal. No irreducible K-type occurring in π_{aff} has intertwining contained in K, at any congruence level. The subgroup J is the full setwise stabilizer of an edge and is a maximal compact subgroup distinct from a vertex stabilizer.*

Proof. Write $s = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and let $u_+(x)$ and $u_-(x)$ be the upper and lower unipotent matrices. Iwahori double cosets are represented by a_k and $a_k s$, for $k \in \mathbf{Z}$. The following root elements belong to $I \cap nIn^{-1}$:

n	condition	u	$n^{-1}un$
a_k	$k > 0$	$u_-(t)$	$u_-(t^{k+1})$
a_k	$k < 0$	$u_+(1)$	$u_+(t^{-k})$
$a_k s$	$k \geq 0$	$u_-(t)$	$u_+(t^{k+1})$
$a_k s$	$k \leq -2$	$u_+(1)$	$u_-(t^{-k})$

In each row $\theta(u) = -1$ and $\theta(n^{-1}un) = 1$. The only remaining representatives are 1 and $a_{-1}s = w$ in G. They preserve θ, so $I_G(\theta) = J$. Restricting intertwiners gives $I_G(\lambda) = J$. The canonical inducing fiber is therefore the unique λ-isotypic copy under J, and projection to this fiber proves irreducibility as above.

For completeness, Iwasawa decomposition and the two Iwahori orbits on G/B give $G = IB \cup IsB = JB$, since $wB = sB$. The character λ is nontrivial on $J \cap U$, as witnessed by $u_+(1)$. The same Mackey coinvariant argument used for π therefore gives $(\pi_{\mathrm{aff}})_U = 0$, proving supercuspidality.

Normalize Haar measure by $\mathrm{vol}(K) = 1$. Since $[K : I] = 3$ and $[J : I] = 2$, one has $\mathrm{vol}(J) = 2/3$. The formal-degree formula for irreducible compact induction gives

$$d(\pi_{\mathrm{aff}}) = \frac{\dim \lambda}{\mathrm{vol}(J)} = \frac{3}{2}.$$

If an occurring irreducible K-type τ satisfied $I_G(\tau) \subseteq K$, its compact induction would be irreducible. Frobenius reciprocity would identify it with π_{aff}, and the same formal-degree formula would give $\dim \tau = 3/2$, a contradiction.

The Iwahori fixes its edge pointwise. At either endpoint its reduction interchanges the other two outgoing edges, so its fixed set is exactly the given edge. The element w exchanges the endpoints. Their common fixed set is its midpoint, so any compact subgroup containing J lies in the full stabilizer of that midpoint, which is J. Thus J is maximal compact. The example requires changing the facet, not merely increasing the level at a fixed vertex. □

14. A horospherical compression need not remain a projection

Retain the representation $\pi = \text{c-Ind}_K^G \varepsilon$ of Proposition 13.2, with $G = \mathrm{PGL}_2(\mathbf{F}_2((t)))$ and $K = \mathrm{PGL}_2(\mathbf{F}_2[[t]])$. Let $\mathcal{T}$ be its Bruhat–Tits tree, which has degree three, and let X be the vertex set. The canonical inducing lines give an orthogonal decomposition of the Hilbert completion of π as $\widehat{\pi} = \widehat{\bigoplus}_{z \in X} \ell_z$. The group G transports these lines. For each vertex x, put

$$W_x = \bigoplus_{z \sim x} \ell_z. \tag{69}$$

At the base vertex o, this is the annular representation W_1 introduced in Proposition 13.2.

Lemma 14.1. *The representation W_o of the full vertex stabilizer K is irreducible and has dimension three.*

Proof. The group K_1 fixes the three neighboring vertices, so it preserves their three inducing lines. Identify K_1/K_2 with $M_2(\mathbf{F}_2)/\mathbf{F}_2 I$ by writing a representative as $1+tA$ modulo t^2. On the line at $a_1 o$, conjugation by $a_1 = \mathrm{diag}(t,1)$ gives

$$a_1^{-1}(1+tA)a_1 \equiv \begin{pmatrix} 1 & A_{12} \\ 0 & 1 \end{pmatrix} \pmod{t}.$$

The sign character on this line is therefore $(-1)^{A_{12}}$. Conjugation by $K/K_1 \simeq S_3$ gives the other two characters. The three linear functionals on $M_2(\mathbf{F}_2)/\mathbf{F}_2 I$ are

$$A_{12}, \qquad A_{21}, \qquad A_{11}+A_{12}+A_{21}+A_{22}.$$

They are well defined modulo scalar matrices and are pairwise distinct. Hence a K_1-stable subspace is a sum of some of these three character lines. The group K permutes the lines transitively, so a K-stable subspace is either zero or all of W_o. □

Let $\mathcal{M} = \widehat{\bigoplus}_{x\in X} W_x$, with its natural unitary G-action. Its algebraic direct sum is $\text{c-Ind}_K^G W_o$. Define

$$S : \widehat{\pi} \longrightarrow \mathcal{M}, \qquad (Sv)_x = \frac{1}{\sqrt{3}} \operatorname{pr}_{W_x} v, \qquad P = SS^*. \tag{70}$$

Each line ℓ_z occurs in exactly three of the spaces W_x. Thus $\sum_x \|\operatorname{pr}_{W_x} v\|^2 = 3\|v\|^2$, so S is a G-equivariant isometry. The operator P is an orthogonal projection whose smooth algebraic image is isomorphic to the actual supercuspidal representation π.

Proposition 14.2. *The projection P has propagation two and diagonal blocks $P_{x,x} = \frac{1}{3}\mathrm{id}_{W_x}$. Fix an end ξ of the tree, and let b be a Busemann function decreasing by one along each edge toward ξ. For each x, define the common kernel of all maximally descending blocks by*

$$N_\xi(x) = \bigcap_{b(y)=b(x)-2} \ker(P_{y,x}).$$

Let Q be the diagonal orthogonal projection onto $\widehat{\bigoplus}_x N_\xi(x)$. Then every $N_\xi(x)$ has dimension two, but

$$QPQ = \frac{1}{3}Q, \qquad QPQ - (QPQ)^2 = \frac{2}{9}Q \neq 0. \tag{71}$$

In particular, removing all maximally descending blocks does not leave an orthogonal projection, even on a genuine Bruhat–Tits tree with irreducible full-vertex fibers and supercuspidal image.

Proof. The definition of S gives

$$P_{x,y} = \frac{1}{3} \operatorname{pr}_{W_x} |_{W_y}.$$

If $x = y$, this is one third of the identity. If $x \neq y$, it is nonzero exactly when x and y have distance two. Their common neighbor z is then unique, and $P_{x,y}$ is one third of the identification of their copies of ℓ_z, with zero action on the other two lines.

Write $f(x)$ for the unique neighbor of x toward ξ. Among the nonzero blocks out of x, the maximal decrease of b is two. It occurs only at $y = f(f(x))$, and this block acts on the line $\ell_{f(x)}$. Consequently

$$N_\xi(x) = \bigoplus_{\substack{z\sim x \\ z\neq f(x)}} \ell_z. \tag{72}$$

Consider a nonzero off-diagonal block $P_{x,y}$, and let z be the common neighbor of x, y. The copy of ℓ_z at x survives Q exactly when $z \neq f(x)$, equivalently $x = f(z)$. Survival at y would likewise require $y = f(z)$. These requirements cannot both hold when $x \neq y$. Thus every off-diagonal block of QPQ is zero, while its diagonal block is $\frac{1}{3}Q_x$. This proves both identities in (71). □

For arbitrary orthogonal projections P, Q, the obstruction is measured by the exact positive identity

$$QPQ - (QPQ)^2 = QP(1-Q)PQ. \tag{73}$$

It vanishes precisely when Q reduces P. Vanishing of the extreme Busemann blocks only removes selected matrix entries; it does not imply this reducing-subspace condition. In the example, Q is equivariant under the stabilizer of ξ, but not under G.

The example has an explicit finite central block decomposition, so it does not obstruct compact induction itself. For each z, take the three copies of ℓ_z in the fibers at its three neighbors. These form a three-dimensional space $\mathcal{H}_z$; the spaces $\mathcal{H}_z$ are mutually orthogonal and are permuted by G. On $\mathcal{H}_z$, the operator P is one third of the all-ones matrix and has rank one. The projection onto $\mathcal{H}_z$ belongs to the algebra generated by the coordinate projections and the blocks of P: for distinct neighbors x, y of z, the product $9P_{x,y}P_{y,x}$ is the projection onto the copy of ℓ_z at x, and summing these coordinate projections gives the asserted block. It commutes with all the generators. The horospherical projection Q retains just the copy at $f(z)$, cutting across this existing block. Recovering the P-invariant block requires restoring the other two coordinates. Thus this compression does not supply a new projection with smaller support.

15. A finite decision for closure under the projection blocks

Let G be the group of rational points of a connected reductive group over a nonarchimedean local field, let Z be its center, and let $\mathcal{B}$ be its reduced Bruhat–Tits building. Let $J \supset Z$ be open and compact modulo Z. Its action on $\mathcal{B}$ fixes a point, hence preserves the facet containing that point in its relative interior. Choose the barycenter o of this facet, so J fixes o. Set $X = G/J$ and $x_0 = J$. The equivariant map

$$p : X \longrightarrow \mathcal{B}, \qquad p(gJ) = go$$

has finite fibers: the full point stabilizer G_o is compact modulo Z, and $[G_o : J] < \infty$. Its image consists of facet barycenters and meets bounded subsets of $\mathcal{B}$ in finite sets.

Let V_x be a smooth unitary G-equivariant bundle of dimension d on X, with V_{x_0} irreducible under J. Put $\mathcal{M} = \bigoplus_{x \in X} V_x$. Suppose that P is a nonzero G-equivariant orthogonal projection on its Hilbert completion and has finite propagation with respect to the distance between $p(x)$ and $p(y)$. Write E_x for the coordinate projection and $P_{x,y} = E_x P E_y$. Let $\mathcal{A}$ be the algebra generated by the E_x and $P_{x,y}$ using finite sums and products. All its elements have finite rank and finite coordinate support.

For $W \subset \mathcal{M}$, its closure under the blocks means the algebraic space $\mathcal{A}W$, not a Hilbert space closure. A finitely supported subspace means a subspace contained in a finite sum of coordinate fibers. When $W \subset V_{x_0}$, its closure is computed by the increasing sequence of finite-dimensional diagonal spaces

$$C_0 = W, \qquad C_{r+1} = C_r + \bigoplus_{x \in X} \sum_{y \in X} P_{x,y}(C_r \cap V_y). \tag{74}$$

Each step uses finitely many blocks, and $\bigcup_r C_r = \mathcal{A}W$.

Lemma 15.1. *For every positive integer d, there is a radius $R > 0$ such that every facet meeting the metric sphere of radius R about o has more than d distinct J-translates. Any point with at most d distinct J-translates then lies strictly inside that sphere.*

Proof. First, the set of facets with at most d distinct J-translates is finite. Otherwise their barycenters have an unbounded sequence. The building is a proper locally finite Euclidean building, so a subsequence converges in its visual compactification to a point ξ at infinity. The orbit $J \cdot \xi$ has at most d points: if $j_1\xi, \dots, j_{d+1}\xi$ were distinct, disjoint neighborhoods of these points would separate the corresponding translates of every sufficiently late barycenter. The stabilizer of ξ in J is closed and has finite index, so it is open in J. It is contained in the proper rational parabolic subgroup stabilizing ξ. This is impossible, since an open subgroup of G is Zariski dense in the connected reductive group.

The finitely many facets just obtained are bounded. Choose R larger than the distance from o of every point in their closures. Then every facet meeting the sphere has more than d translates. If a point y on or outside the sphere had at most d translates, the point z where $[o, y]$ first meets the sphere would also have at most d translates: J fixes o, and the radial projection is J-equivariant. The unique facet containing z in its relative interior would consequently have at most d translates, a contradiction. If the reduced building is a point, any positive R works. □

Choose such an R and put

$$S_* = p^{-1}\big(\overline{B}(o, 2R)\big), \qquad D = d|S_*|. \tag{75}$$

The finite set S_* depends on G, J, d, o and the chosen geometric certificate, but not on P.

Theorem 15.2. *For every nonzero $W \subset V_{x_0}$, the sequence* (74) *has one of the following two outcomes. It stabilizes at a finite space supported on S_*, or it leaves $\bigoplus_{x\in S_*} V_x$. The outcome is detected after at most $D+1$ steps. Leaving this finite sum proves that $\mathcal{A}W$ is infinite-dimensional.*

Moreover, either every finitely supported subspace of $\mathcal{M}$ has finite closure under $\mathcal{A}$, or no nonzero finitely supported vector has finite closure. The first case is equivalent to $Z(\mathcal{A}) \neq 0$.

Proof. Suppose first that some nonzero finitely supported vector has finite closure. Applying a coordinate projection gives a nonzero vector in one fiber with finite closure. Translate that fiber to x_0. Irreducibility of V_{x_0} gives finitely many J-translates of this vector spanning V_{x_0}. Their closures are finite-dimensional, so

$$C = \mathcal{A}V_{x_0}$$

is finite-dimensional. Translating by G shows that every fiber has finite closure, and taking finite sums proves the asserted dichotomy.

The space C reduces $\mathcal{A}$, since $\mathcal{A}$ is a star algebra. The two-sided star ideal

$$I = \mathcal{A}E_{x_0}\mathcal{A}$$

acts on C and annihilates $C^\perp$. It is a finite-dimensional star algebra with image C, so its identity is the orthogonal projection onto C. That identity is central in $\mathcal{A}$, because I is a two-sided ideal. Its minimal central projections $Q_1, \dots, Q_s$ are consequently central in $\mathcal{A}$ as well. Every Q_j meets the base fiber: if $Q_j E_{x_0} = 0$, then $Q_j I = 0$, contrary to $Q_j \in I$.

Conjugation by J preserves I and permutes the Q_j. The sum over any J-orbit of these projections restricts to a nonzero J-invariant projection on V_{x_0}. Irreducibility makes that restriction the identity. Two different orbits would give orthogonal identities on V_{x_0}, which is impossible. Thus J is transitive on the Q_j, and $s \leq d$.

Fix $Q = Q_j$ and let $L = \mathrm{Stab}_G(Q)$. Then

$$[J : J \cap L] = s \leq d. \tag{76}$$

Its coordinate support is exactly $L \cdot x_0$. To see this, let x belong to the support and choose g with $gx_0 = x$. Translating the preceding argument shows that G_x is transitive on the minimal central blocks meeting V_x. Indeed, a minimal central block meeting V_x belongs to the ideal $\mathcal{A}E_x\mathcal{A}$, since its intersection with that ideal is nonzero. Both Q and gQg^{-1} meet that fiber. Hence some $k \in G_x$ satisfies $kgQg^{-1}k^{-1} = Q$, so $kg \in L$ and $kg \cdot x_0 = x$.

The subgroup L is closed: the finite-rank projection Q has a norm-continuous conjugation orbit under the strongly continuous unitary action. Equation (76) now makes $J \cap L$, and hence L, open. The center acts by the same scalar on every fiber, so $Z \subset L$. The group L preserves the finite coordinate support of Q. A stabilizer of a finite subset of G/J is contained in a finite union of cosets of conjugates of J, so L/Z is compact. The fixed point theorem supplies a point fixed by L in the reduced building. Its containing facet is preserved setwise, and its barycenter b is fixed by L. Equation (76) gives $|J \cdot b| \leq d$. Lemma 15.1 yields $d(o, b) < R$. For $l \in L$,

$$d(o, lo) \leq d(o, b) + d(b, lo) = 2d(o, b) < 2R.$$

Thus the support of every Q_j is contained in S_*, and the same holds for $C = \mathcal{A}V_{x_0}$.

Conversely, a nonzero central element of $\mathcal{A}$ has finite support. Applying finite-dimensional spectral calculus to its product with its adjoint gives a nonzero finite-rank central projection. Its image is a nonzero finite-dimensional reducing space. The first part of the proof then gives finite closure for every finitely supported vector. This proves the equivalence with $Z(\mathcal{A}) \neq 0$.

Finally, if the sequence starting at $W \subset V_{x_0}$ ever leaves the space supported on S_*, its closure cannot be finite by the bound just proved. If it remains there, each strict inclusion increases dimension, which is bounded by D. Equality at one step gives invariance under every block, so the sequence then stabilizes. This proves the claimed finite decision. □

This result gives an algorithm when the finite building calculations and the matrix entries of the projection blocks are supplied exactly, together with either a known propagation bound or complete finite incident-block lists. Exact pointwise access to matrix entries alone does not supply those lists. To find R, increase an integer radius and compute the finitely many facets meeting its sphere and their J-orbits. Stop when all orbit sizes exceed d. Lemma 15.1 proves termination. One may certify each strict inequality by exhibiting $d+1$ distinct translates. The subsequent steps are finite matrix calculations using (74); they require only the blocks meeting the currently computed finite supports. Exact arithmetic and equality tests are essential to this assertion.

If the algebraic image $P\mathcal{M}$ is an irreducible supercuspidal representation π, a positive outcome gives actual inducing data. Compute $C = \mathcal{A}V_{x_0}$, its finite algebra I, and a minimal central projection Q of I. Schur's lemma gives $P_{x,x} = c\,\mathrm{id}$ with $c > 0$, so

$$\operatorname{tr}(PQ) = c\operatorname{tr}(Q) > 0.$$

The finite space $\rho = PQ\mathcal{M}$ is therefore nonzero. The distinct G-translates of Q are orthogonal minimal central projections; their images under P are orthogonal and span π. With $L = \operatorname{Stab}_G(Q)$ this gives

$$\pi \simeq \text{c-Ind}_L^G \rho.$$

The representation ρ is irreducible, since a proper L-subrepresentation would induce a proper G-subrepresentation of this orthogonal direct sum.

The subgroup and its representation also have finite descriptions once the local group actions are supplied. Let T be the setwise stabilizer of the finite coordinate support of Q. It is open and compact modulo Z. Choose a compact open normal subgroup U of T fixing that finite coordinate space pointwise. Then $T/(ZU)$ is finite, and $L/(ZU)$ is the stabilizer of the matrix Q in its projective action on that space. The matrices of ρ are the restrictions of these actions to $PQ\mathcal{M}$, together with the original central character. This does not require a finite image for the central character.

For the projection in Proposition 14.2, this procedure repairs the failed compression explicitly. Starting with the entire three-dimensional fiber W_o, one step adjoins its six nonzero images in the fibers at distance two. The resulting space has dimension nine and is already invariant under every block. It is the direct sum of the three spaces $\mathcal{H}_z$ for $z \sim o$ described after (73). On each such space the coordinate projections and the blocks generate $M_3(\mathbf{C})$, so

$$\mathcal{A}E_o\mathcal{A} \simeq M_3(\mathbf{C}) \oplus M_3(\mathbf{C}) \oplus M_3(\mathbf{C}).$$

The three minimal central projections are the projections onto $\mathcal{H}_z$. Each has a one-dimensional image under P, and its stabilizer is the full vertex stabilizer G_z. The extracted representation on this line is the conjugate of ε. Thus the finite closure recovers the actual compact inducing pair in this example, while the horospherical compression alone did not remain a projection.

A negative outcome proves that this particular coordinate model has no nonzero finite reducing block. It does not prove that π has no compact inducing pair in another model. In particular, applying the procedure to a horospherical kernel replaces an unbounded closure search by a finite decision, but it does not establish that the decision is always positive.

16. The finite equations for changing an embedding

Let π be an irreducible unitary supercuspidal representation of G, with formal degree d_π. Fix an open subgroup $J \supset Z$, compact modulo Z, and an irreducible unitary representation τ of J of dimension d. Suppose that

$$i_1, \dots, i_m : \tau \longrightarrow \pi, \qquad i_s^* i_t = \delta_{s,t}\,\mathrm{id}_\tau,$$

are J-equivariant embeddings. One may take an orthonormal basis of the whole multiplicity space, which is finite-dimensional by admissibility, or any specified nonzero subspace of it. Write $i_u = \sum_s u_s i_s$ for $u \in \mathbf{C}^m$ with $\sum_s |u_s|^2 = 1$. Its image depends only on $[u] \in \mathbf{P}^{m-1}(\mathbf{C})$.

Put $c = d_\pi \operatorname{vol}(J/Z)/d$. The coefficient projection attached to i_u has blocks

$$P(u)_{gJ,hJ} = c \sum_{s,t} \overline{u_s} u_t\, i_s^* \pi(g^{-1}h) i_t. \tag{77}$$

Let $\mathcal{A}(u)$ be its algebra of coordinate projections and projection blocks. Call $[u]$ successful if $Z(\mathcal{A}(u)) \neq 0$.

Theorem 16.1. *The successful points in $\mathbf{P}^{m-1}(\mathbf{C})$ form a real algebraic closed subset defined by finitely many explicit equations. With S_* and $D = d|S_*|$ from Theorem 15.2, these equations can be chosen as entries of products of at most $D+1$ projection blocks whose paths start at J and end outside S_*. Each equation is bihomogeneous in $u, \overline{u}$, of bidegree at most $(D+1, D+1)$.*

Suppose that the finite group and support calculations are effective, and the real and imaginary parts of the needed mixed matrix coefficients lie in an effectively presented ordered real field with effective computations in its real closure, including quantifier elimination and real root isolation. Then nonemptiness of this locus is decidable. A positive answer produces an exact embedding and compact inducing data for π.

Proof. All the mixed coefficients $i_s^*\pi(g)i_t$ have compact support modulo Z. There are finitely many pairs (s,t), so their supports have a compact union modulo Z. Consequently (77) has uniformly finite propagation as u varies. Its possible nonzero blocks define a locally finite graph on G/J. There are only finitely many paths of length at most $N = D+1$ starting at the base coordinate.

For each such path $x_0 = J, x_1, \dots, x_r$, with $r \le N$ and $x_r \notin S_*$, impose

$$P(u)_{x_r,x_{r-1}} \cdots P(u)_{x_1,x_0} = 0. \tag{78}$$

The span of the images of all path products of length at most r is exactly the space C_r in (74), starting with the entire base fiber. Thus (78) says precisely that this sequence does not leave the sum supported on S_* during its first N steps. By Theorem 15.2, this is equivalent to finite closure, and hence to $Z(\mathcal{A}(u)) \neq 0$.

Every block in (77) is homogeneous of degree one in u and degree one in $\overline{u}$. The entries of a product of r blocks therefore have bidegree (r,r). Their vanishing is unchanged by nonzero complex rescaling of u, so these equations define a real algebraic closed subset of projective space. The common nonzero scalar c^r can be omitted from each product. No computation of the formal degree is needed to test the vanishing equations themselves.

Write $u_s = a_s + \mathrm{i}b_s$, split each matrix entry into real and imaginary parts, and add $\sum_s (a_s^2 + b_s^2) = 1$. This is a finite system over the stipulated ordered real coefficient field. Quantifier elimination decides whether it has a solution, and effective real algebraic sampling produces one in its real closure when it does. Apply the finite closure algorithm to that embedding and extract a minimal central block as in Theorem 15.2. This gives the claimed inducing data. Exact arithmetic in an unspecified complex field alone would not justify this decision step. □

If one embedding is unsuccessful, the successful locus is proper and has empty interior and measure zero in projective space. Indeed, at least one of its defining real analytic equations is not identically zero. Its zero set has empty interior and measure zero on the connected real analytic manifold $\mathbf{P}^{m-1}(\mathbf{C})$. Thus generic linear mixing is not a method for finding a successful point; the finite equations must be solved.

16.1. **An exact two-dimensional multiplicity space.** Use the supercuspidal representation from Proposition 13.2:

$$G = \mathrm{PGL}_2(\mathbf{F}_2((t))), \qquad K = \mathrm{PGL}_2(\mathbf{F}_2[[t]]), \qquad \pi = \text{c-Ind}_K^G \varepsilon.$$

Normalize $\mathrm{vol}(K) = 1$, so $d_\pi = 1$. Let w be a unit vector in the canonical inducing line, put $a = \mathrm{diag}(t,1)$, and set $w_a = \pi(a)w$. These vectors are orthogonal. Define

$$D_0 = K_1, \qquad J = D_0 \cap aD_0a^{-1}, \qquad \tau = 1_J.$$

The group J fixes both vectors. For a representative $1 + tA$ modulo t^2, membership in J is the condition $A_{12} = 0$. Hence $[D_0 : J] = 2$, $[K : J] = 12$, and $\mathrm{vol}(J) = 1/12$.

In fact $\pi^J = \mathbf{C}w \oplus \mathbf{C}w_a$. The congruence calculation in Proposition 13.2 gives $\pi^{K_2} = W_0 \oplus W_1$. On the three lines of W_1, the characters of K_1/K_2 are

$$(-1)^{A_{12}}, \qquad (-1)^{A_{21}}, \qquad (-1)^{A_{11}+A_{12}+A_{21}+A_{22}}.$$

Only the first is trivial on the kernel of A_{12}, so W_1^J is the line at aK. Since J contains K_2, the assertion follows.

Proposition 16.2. *For the full multiplicity space π^J, the successful locus is exactly*

$$\{[1:0],[0:1]\} \subset \mathbf{P}^1(\mathbf{C}).$$

Every embedding has the same dimension-to-formal-degree ratio 12.

Proof. For $u = (s,t)$ with $|s|^2 + |t|^2 = 1$, take the unit vector $v_u = sw + tw_a$. At $[1:0]$, the scalar matrix coefficient is supported exactly on K, where it is the nonzero character ε. The support graph of its coefficient projection on G/J consequently has the finite connected components gK/J, each of size 12. The identity on one such component is a finite central projection of the block algebra. At $[0:1]$, the same argument uses aKa^{-1} and gives components $gaKa^{-1}/J$, again of size 12.

Suppose now that $st \neq 0$. The vectors $\pi(a^n)w$, $n \in \mathbf{Z}$, are orthonormal because their inducing cosets are distinct. Therefore

$$i_{v_u}^*\pi(a)i_{v_u} = \bar{t}s \neq 0.$$

The scalar support graph contains every edge from $a^n J$ to $a^{n+1}J$. These vertices are distinct, so the component through J is infinite. The graph is G-invariant and vertex transitive; every component is therefore infinite. A central element of its block algebra commutes with all coordinate projections, so it is scalar and diagonal in these one-dimensional fibers. Commutation with a nonzero edge forces its two diagonal entries to agree. Such an element is constant on each infinite component, but has finite support, so it vanishes. Thus no mixed point is successful.

The ratio is $\dim \tau/(d_\pi \operatorname{vol}(J)) = 12$, independently of the embedding. In particular, moving from a mixed point to an endpoint repairs the center obstruction without decreasing this ratio. Extraction at an endpoint then recovers the inducing pair (K, ε) or its conjugate. □

16.2. **Changing the local type by a normalized corner.** Let $K \supset Z$ be open and compact modulo Z, and let $U \subset \pi$ be an actual finite-dimensional K-space of dimension n. Such a space is obtained by taking the K-span of finitely many smooth vectors: a compact open subgroup fixes them, and compactness modulo the scalar center makes their K-span finite-dimensional. Formal-degree orthogonality gives the coefficient projection P_U on $\text{c-Ind}_K^G U$, with diagonal $d_\pi \operatorname{vol}(K/Z)/n$.

Choose one irreducible K-subspace $\sigma \subset U$ of dimension e, and let E be the corresponding G-equivariant diagonal projection. The same orthogonality calculation, applied to an orthonormal basis of σ, gives

$$P_U E P_U = \frac{e}{n} P_U, \qquad P_\sigma = \frac{n}{e} E P_U E. \tag{79}$$

Thus P_σ is an orthogonal projection, its image is isomorphic to π, and its diagonal is $d_\pi \operatorname{vol}(K/Z)/e$. This is an exact change of the occurring local type, and it preserves finite propagation.

There is a useful strict decrease when $J \subset K$ and U is generated by an occurring copy of τ. Then U is a quotient of $\operatorname{Ind}_J^K \tau$, so every irreducible constituent σ that survives in U satisfies

$$\frac{\dim \sigma}{d_\pi \operatorname{vol}(K/Z)} \le \frac{\dim \tau}{d_\pi \operatorname{vol}(J/Z)}. \tag{80}$$

If $\operatorname{Ind}_J^K \tau$ is reducible, the inequality is strict, because each irreducible constituent has dimension strictly less than $[K : J] \dim \tau$. For a general compact span of translated vectors, however, properness of the corner E only gives $e < n$. To decrease the original ratio one needs the additional inequality $e < \dim \tau \operatorname{vol}(K/Z)/\operatorname{vol}(J/Z)$. The existence of that constituent is not supplied by (79).

16.3. **The precise compatibility requirement.** The successful locus can be characterized without the coordinate algebra, but the resulting condition still asks for an imprimitivity decomposition. In the following proposition the multiplicity space is the full space $\operatorname{Hom}_J(\tau, \pi)$.

Proposition 16.3. *The successful locus is nonempty if and only if there exist an open subgroup $L \supset Z$, compact modulo Z, an irreducible representation ρ with $\pi \simeq \text{c-Ind}_L^G \rho$, and an irreducible representation η of $H = J \cap L$, such that*

$$\eta \subset \rho|_H, \qquad \tau \simeq \operatorname{Ind}_H^J \eta. \tag{81}$$

For a prescribed embedding, success is equivalent to the existence of such an orthogonal compact induction decomposition whose inducing-fiber projections preserve the prescribed J-space.

Proof. Suppose first that an embedding is successful, and use its coefficient model to identify π with $P\mathcal{M}$. Extract a minimal central block Q meeting the base fiber, with stabilizer L and image $\rho = PQ\mathcal{M}$. The minimal central blocks meeting the base fiber are its J-translates. Write $H = J \cap L$ and $\eta = QE_{x_0}\mathcal{M}$. Their orthogonal decomposition of V_{x_0} gives $\tau \simeq \operatorname{Ind}_H^J \eta$. Irreducibility of τ forces irreducibility of η.

The map $v \mapsto c^{-1/2} P v$ from the base fiber into π is the original isometric embedding. It is injective because $E_{x_0} P E_{x_0} = c E_{x_0}$, and it maps η into ρ because P commutes with Q. This proves (81). The projections onto the inducing spaces $P(gQg^{-1})\mathcal{M}$ preserve the original J-space, by its displayed decomposition.

Conversely, suppose (81) holds. Choose an isometric copy E of η in the canonical inducing fiber of ρ. The orthogonal sum

$$W = \bigoplus_{jH \in J/H} \pi(j) E$$

is an actual J-space in π, isomorphic to τ. The summands lie in distinct canonical L-inducing fibers. Form the coefficient model of this embedding. In the coordinate at gJ, its fiber decomposes as the sum of the copies of $\pi(gj)E$, labelled by gjL.

Fix a label bL. Its copies can occur only at coordinates in bLJ/J, a finite set of size $[L : J \cap L]$. The sum of all copies with this label is therefore finite-dimensional. Each coordinate projection preserves it, and so does every coefficient block: different canonical L-inducing fibers in π are orthogonal. These finite spaces form an orthogonal decomposition of the whole coordinate model. Every finitely supported vector has finite closure under its block algebra, and Theorem 15.2 proves success. The same argument proves the assertion for a prescribed embedding whose inducing-fiber projections preserve its J-space. □

Call τ primitive if it is not induced from a representation of any proper open subgroup of J. If J is maximal among open subgroups compact modulo Z, Proposition 16.3 has the following consequence:

$$\tau \text{ primitive and successful} \quad\Longrightarrow\quad \pi \simeq \text{c-Ind}_J^G \tau, \qquad \dim \tau = d_\pi \operatorname{vol}(J/Z). \tag{82}$$

Indeed, primitivity forces $H = J$, maximality forces $L = J$, and the irreducible inclusion $\tau = \eta \subset \rho$ is then an equality. Thus a primitive occurring type of a maximal compact subgroup with ratio greater than one would have an empty successful locus, even if the representation had a compact inducing pair elsewhere. This implication does not assert that such a supercuspidal example exists.

For a known inducing pair, projecting τ to a single orbit in the Mackey restriction gives only a constituent of $\operatorname{Ind}_{J\cap gLg^{-1}}^{J}(\rho^g|_{J\cap gLg^{-1}})$. It does not give the full induction identity in (81). In particular, an irreducible constituent need not be preserved by the individual inducing-fiber projections. Neither finite polynomial decidability nor occurrence of τ proves this additional identity. In fact, nonemptiness for every occurring type is false, as the following example shows.

16.4. **An empty locus in an already compactly induced representation.** Keep $G = \mathrm{PGL}_2(\mathbf{F}_2((t)))$, its vertex stabilizer K, and the normalization $\operatorname{vol}(K) = 1$. Use the representation

$$\pi_{\mathrm{aff}} = \text{c-Ind}_{L_e}^G \lambda$$

from Proposition 13.4, where L_e is the full stabilizer of an edge e incident to the vertex o fixed by K. Its restriction to the pointwise edge stabilizer $I = K \cap L_e$ is the affine-generic character θ, and $d(\pi_{\mathrm{aff}}) = 3/2$. Put

$$r = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \in K, \qquad J_0 = \langle r \rangle K_2.$$

The element r has order three and cycles the three edges incident to o.

Proposition 16.4. *The character 1_{J_0} occurs in π_{aff}, but its entire successful-embedding locus is empty. Indeed,* $\dim \pi_{\mathrm{aff}}^{J_0} = 1$.

Proof. Let v be a unit vector in the canonical L_e-inducing line. The vectors $v, \pi(r)v, \pi(r)^2 v$ lie in three different edge fibers and are orthogonal. The group K_2 fixes all three: it fixes v because $\theta|_{K_2} = 1$, and it is normal in K. Hence

$$v + \pi(r)v + \pi(r)^2 v$$

is a nonzero J_0-fixed vector.

For the multiplicity assertion, take e to have endpoints $o, a_{-1}o$, as in Proposition 13.4. The K-orbits of edges are represented by $a_{-n}e$, $n \geq 0$. For $n \geq 1$, the element $u_-(t^{n+1})$ belongs to K_2, and

$$a_{-n}^{-1} u_-(t^{n+1}) a_{-n} = u_-(t).$$

The latter element has λ-value -1. Averaging over K_2 therefore kills the canonical line at $a_{-n}e$, and normality in K kills its entire K-span. Thus $\pi_{\mathrm{aff}}^{K_2}$ is precisely the three-dimensional space on the edges incident to o. The action of r cyclically permutes its indicated basis, giving a one-dimensional fixed space.

The fixed point set of r in the tree is exactly $\{o\}$. If it fixed another point, the initial direction of the segment from o to that point would be fixed, contrary to its three-cycle on the link of o. Consequently every compact subgroup containing J_0 lies in K: its fixed point must be fixed by J_0, and hence must equal o.

If a scalar J_0-embedding were successful, Proposition 16.3 would give $\pi_{\mathrm{aff}} \simeq \text{c-Ind}_L^G \rho$ with $J_0 \subset L$. Thus $L \subset K$. Induction in stages gives

$$\pi_{\mathrm{aff}} \simeq \text{c-Ind}_K^G(\operatorname{Ind}_L^K \rho).$$

The finite representation $\operatorname{Ind}_L^K \rho$ is irreducible: otherwise its decomposition would induce a nontrivial decomposition of π_{aff}. The formal-degree formula would then imply

$$\dim \operatorname{Ind}_L^K \rho = d(\pi_{\mathrm{aff}}) \operatorname{vol}(K) = \frac{3}{2},$$

which is impossible. This proves emptiness of the full locus. □

This example requires changing the local type, not just its embedding. It also supplies a successful finite change. The K-span of the displayed J_0-fixed vector is $\tau_e = \mathrm{Ind}_I^K \theta$, of dimension three. This representation is irreducible: its three edge lines have three distinct K_1/K_2-characters, permuted transitively by K. The original fixed vector has a nonzero component in each line and therefore generates this space. The identity

$$\tau_e = \mathrm{Ind}_{K\cap L_e}^K(\lambda|_{K\cap L_e})$$

satisfies Proposition 16.3, so this new type has a successful embedding. Since $[K : K_2] = 48$ and $[J_0 : K_2] = 3$, the ratios in this explicit change are

$$\frac{1}{d(\pi_{\mathrm{aff}})\,\mathrm{vol}(J_0)} = \frac{32}{3}, \qquad \frac{\dim \tau_e}{d(\pi_{\mathrm{aff}})\,\mathrm{vol}(K)} = 2, \qquad \frac{\dim \lambda}{d(\pi_{\mathrm{aff}})\,\mathrm{vol}(L_e)} = 1.$$

The first move is a change to a larger compact subgroup and an actually occurring constituent; the second extracts its finite inducing block. The proof supplies these moves in this example. It does not prove that every unsuccessful type admits such a move.

17. Extraction from a bound on the ranks of the coefficient blocks

Let G be the group of rational points of a connected reductive group over a nonarchimedean local field, and let Z be its center. Let $J \supset Z$ be open and compact modulo Z, and let τ be an irreducible finite-dimensional smooth unitary representation of J, with central character χ. Put $n = \dim \tau$ and $X = G/J$. Write

$$\mathcal{M} = \text{c-Ind}_J^G \tau = \bigoplus_{x\in X} V_x$$

for the algebraic sum of its coordinate fibers, and let $\mathcal{H}$ be its unitary completion. The coordinate projection onto V_x is denoted by E_x. Suppose that P is a nonzero G-equivariant orthogonal projection on $\mathcal{H}$, that $P(V_x)$ has finite coordinate support for every x, and that $\pi = P\mathcal{M}$ is an irreducible smooth representation. These hypotheses hold for the projection associated with a coefficient embedding of an irreducible unitary supercuspidal representation and one occurring copy of τ.

Schur's lemma and equivariance give $E_xPE_x = cE_x$ for a number $0 < c \le 1$ independent of x. Indeed, E_xPE_x commutes with the irreducible action of the stabilizer of x, and $\|Pv\|^2 = c\|v\|^2$ for $v \in V_x$. If $c = 0$, all columns of P vanish, contrary to $P \neq 0$. Fix $x_0 = J$ and set

$$F_{x_0} = \bigoplus_{y\neq x_0} \mathrm{im}(E_yPE_{x_0}), \qquad R = \dim F_{x_0} = \sum_{y\neq x_0} \mathrm{rank}(E_yPE_{x_0}).$$

This is a finite sum. The analogous number at any other coordinate is R.

Theorem 17.1. *If $c = 1$, then $P = 1$ and $\pi = \text{c-Ind}_J^G \tau$. Suppose that $0 < c < 1$. Then $R \ge n$, and at least one of the following conclusions holds:*

1. *The restriction of π to J contains an irreducible representation of dimension at most $R - n$.*
2. *There are an open subgroup $L \supset Z$, compact modulo Z, and an irreducible finite-dimensional smooth representation ρ of L such that $\pi \simeq \text{c-Ind}_L^G \rho$.*

More precisely, define

$$A = (1 - E_{x_0})P|_{V_{x_0}} : V_{x_0} \longrightarrow F_{x_0}, \qquad D_{x_0} = F_{x_0} \ominus A(V_{x_0}).$$

If $PD_{x_0} \neq 0$, conclusion (i) *holds. If $PD_{x_0} = 0$, conclusion* (ii) *holds. In the latter case c^{-1} is an integer and*

$$R = (c^{-1} - 1)n. \tag{83}$$

Proof. Write $x = x_0$. Since $P^2 = P$,

$$A^*A = E_xP(1 - E_x)PE_x = c(1 - c)1_{V_x}. \tag{84}$$

If $c = 1$, this identity makes every off-diagonal block in the column at x zero. Equivariance gives the same assertion at every coordinate, and the diagonal blocks are identities. Hence $P = 1$.

Suppose that $0 < c < 1$. Equation (84) makes A injective, so $R \ge n$ and $\dim D_x = R - n$. The subgroup J preserves F_x: it permutes the summands indexed by y and fixes the coordinate x. It also preserves $A(V_x)$, since A is J-equivariant. Thus D_x and PD_x are finite-dimensional smooth unitary J-representations. If PD_x

is nonzero, it contains an irreducible constituent η and $\dim \eta \le \dim PD_x \le R - n$. Its inclusion in $P\mathcal{M} = \pi$ proves (i).

Now assume that $PD_x = 0$. For $v \in V_x$, the identity $E_x P v = cv$ gives

$$PAv = P(1 - E_x)Pv = (1 - c)Pv.$$

Every $f \in F_x$ has a unique decomposition $f = Av + d$ with $d \in D_x$. Equation (84) therefore implies

$$Pf = (1 - c)Pv = c^{-1} P E_x A^* f. \tag{85}$$

In particular, $P(F_x) \subset P(V_x)$. By equivariance the same assertion holds with x replaced by any coordinate.

For every $y \in X$, put $W_y = P(V_y)$. The map $B_y = c^{-1/2} P|_{V_y}$ is an isometry onto W_y, so the orthogonal projection onto W_y is

$$Q_y = B_y B_y^* = c^{-1} P E_y P.$$

If $y \neq x$, the range of $E_y P E_x$ is a summand of F_x. Applying (85) to that range gives

$$Q_y(Pv) = c^{-1} P E_y P v \in W_x \qquad (v \in V_x).$$

Thus Q_y preserves W_x. Since Q_y is self-adjoint, it also preserves $W_x^\perp$, and consequently $Q_x Q_y = Q_y Q_x$. Translation gives this commutation relation for every pair of coordinates.

The restrictions of the commuting projections Q_y to any fixed W_x are simultaneously diagonalizable: they generate a commutative star subalgebra of the finite-dimensional algebra $\mathrm{End}(W_x)$. Their simultaneous eigenspaces are simultaneous eigenspaces on $\overline{\pi} = P\mathcal{H}$ as well, because Q_x acts as the identity on W_x. We call the nonzero simultaneous eigenspaces with at least one eigenvalue equal to one the atoms. Each atom is contained in some W_x, and is therefore finite-dimensional and has finite coordinate support. Distinct atoms are orthogonal, since some self-adjoint projection Q_y acts on them with different eigenvalues. Moreover,

$$\pi = P\mathcal{M} = \sum_{x \in X} W_x = \bigoplus_{W \text{ an atom}} W,$$

where both sums on the right are algebraic. Equivariance of the family (Q_x) makes G permute its atoms. Irreducibility of π implies that this action is transitive: the sum of the atoms in any orbit is a nonzero G-invariant subspace.

Choose an atom W and let $L = \{g \in G : gW = W\}$. A basis of W consists of smooth vectors, so a common open subgroup fixes that basis pointwise. Hence L is open, and is consequently closed. The center acts by χ, so $Z \subset L$. Let $S \subset X$ be the finite nonempty coordinate support of W. Every element of L preserves S. For any $x \in S$, the group $L \cap G_x$ therefore has finite index in L. It is closed in G_x, contains Z, and G_x/Z is compact. It follows that L/Z is compact.

Let ρ be the action of L on W. The transitive orthogonal decomposition into atoms defines a G-equivariant isomorphism

$$\text{c-Ind}_L^G \rho \longrightarrow \pi,$$

by sending the canonical inducing fiber to W and its translates to the corresponding atoms. This map is injective by orthogonality and surjective by the displayed algebraic decomposition. If ρ had a nonzero proper L-invariant subspace, the direct sum of its translates in the distinct atoms would be a nonzero proper G-invariant subspace of π. Therefore ρ is irreducible, proving (ii).

It remains to compute R in this case. On $\overline{\pi}$ the coordinate decomposition gives the strongly convergent identity

$$c \sum_{y \in X} Q_y = \sum_{y \in X} P E_y P = 1.$$

On an atom W, each Q_y is either zero or the identity. Its membership set $S_W = \{y : W \subset W_y\}$ is finite: for a nonzero $w \in W$, $\|E_y w\|^2 = c\|w\|^2$ when $y \in S_W$, and is zero otherwise, so S_W is precisely the finite coordinate support of w. The displayed identity gives $|S_W| = c^{-1}$, independent of W.

The identity $B_y^* B_x = c^{-1} E_y P E_x$ shows that

$$\mathrm{rank}(E_y P E_x) = \dim(W_x \cap W_y).$$

Indeed, commuting orthogonal projections make $Q_y|_{W_x}$ the orthogonal projection onto $W_x \cap W_y$, and the isometries B_x, B_y identify this map with $B_y^* B_x$. Summing over $y \neq x$, each atom in W_x contributes its dimension exactly $c^{-1} - 1$ times. This proves (83). $\square$

Corollary 17.2. *If $R < 2n$, then π is compactly induced from an irreducible finite-dimensional representation of an open subgroup compact modulo Z. More precisely, either $c = 1$ and $P = 1$, or $R = n$, $c = 1/2$, and the atoms in the proof of Theorem 17.1 give the inducing pair.*

Consequently, if an irreducible unitary supercuspidal representation is not compactly induced in this way, every coefficient projection attached to any occurring irreducible J-type satisfies $R \geq 2 \dim \tau$.

Proof. For $c = 1$ the theorem already gives the conclusion. For $0 < c < 1$, Theorem 18.1 below proves that $n < R < 2n$ is impossible. Thus $R = n$ and $c = 1/2$. In this case $D_{x_0} = 0$, so the atom construction applies. The final assertion is the contrapositive. □

For a coefficient projection of a unitary supercuspidal representation with formal degree d_π, the usual counting norm on G/J gives $c = d_\pi \operatorname{vol}(J/Z)/n$. At a fixed J, decreasing the dimension therefore also decreases the normalized ratio $n/(d_\pi \operatorname{vol}(J/Z))$. The local complement has dimension $R - n$, but the range $n < R < 2n$ cannot supply a strict decrease, because it does not occur. At $R = 2n$, a reducible complement does give a smaller occurring type, as proved in Corollary 18.3. No argument here puts a model of every supercuspidal in either range.

There is also a criterion that does not require a least-dimensional type. If π is unitary supercuspidal and $R - n < cn$, then $PD_{x_0} = 0$ and the theorem extracts an inducing pair. To verify this, every irreducible J-type η occurring in π satisfies $\dim \eta \geq d_\pi \operatorname{vol}(J/Z) = cn$. In fact, for a unit vector v in a copy of η, compact-group and supercuspidal Schur orthogonality give

$$\frac{\operatorname{vol}(J/Z)}{\dim \eta} = \int_{J/Z} |\langle \pi(j)v, v\rangle|^2 \, dj \leq \int_{G/Z} |\langle \pi(g)v, v\rangle|^2 \, dg = d_\pi^{-1}.$$

Thus a nonzero PD_{x_0} would contradict the dimension bound in the first alternative of the theorem. The rank gap also shows that this numerical criterion is contained in the terminal case $R = n$: it implies $R/n < 1 + c < 2$.

The successful change of type following Proposition 16.4 gives an explicit instance. For the affine-generic representation of $\mathrm{PGL}_2(\mathbf{F}_2((t)))$, take the irreducible three-dimensional vertex type τ_e. Its translated spaces W_x are the spans of the three canonical edge lines incident to the vertex x. Two such spaces share one line exactly when their vertices are adjacent, and otherwise are orthogonal. Thus $c = 1/2$, the three nonzero off-diagonal blocks in each column have rank one, and $R = n = 3$. The atoms are the canonical edge lines, so the theorem recovers the full edge stabilizer and its inducing character.

18. The global complement and a gap in the coefficient ranks

Continue with the coordinate representation $\mathcal{H}$, the equivariant orthogonal projection P, and the irreducible coordinate fibers V_x of dimension n from Section 17. The projection has finite coordinate support in each column, and $E_x P E_x = cE_x$. Assume $0 < c < 1$. Write $P_{yx} = E_y P|_{V_x}$ and put

$$F_x = \bigoplus_{y \neq x} \operatorname{im} P_{yx}, \qquad R = \dim F_x, \qquad r = R/n.$$

The operator identities and rank gap use irreducibility of the coordinate representation of J and apply in every residue characteristic. Irreducibility of $P\mathcal{M}$ is needed only for the conclusions about a single compactly induced representation.

Theorem 18.1. *One has*

$$r \geq 1, \qquad \frac{1}{r+1} \leq c \leq \frac{r}{r+1}. \tag{86}$$

If either bound on c is an equality, then r is a positive integer. Moreover,

$$R = n \quad \text{or} \quad R \geq 2n. \tag{87}$$

In the first case $c = 1/2$, and the orthogonal projections onto the spaces $P(V_x)$ commute. If $P\mathcal{M}$ is irreducible, their atoms give an irreducible compact induction as in Theorem 17.1.

Suppose that $R = 2n$ and $1/3 < c < 2/3$. Then the global complement construction below gives a finite-propagation equivariant unitary isomorphism

$$U : \bigoplus_{x \in X}^{\ell^2} D_x \longrightarrow \mathcal{H}, \qquad D_x = F_x \ominus (1 - E_x)P(V_x),$$

where the summands on the left are distinct abstract copies of the indicated physical subspaces. If $c \neq 1/2$, then

$$\operatorname{pr}_{D_x} P|_{D_x} = (3c-1)1_{D_x}, \qquad (U^*PU)_{xx} = c1_{D_x}. \tag{88}$$

If $c = 1/2$, the unitary U is the sum of the literal inclusions of the physical subspaces D_x.

Proof of Theorem 18.1. Put $t = c(1-c)$ and $A_x = (1-E_x)P|_{V_x}$. The column identity (84) gives $A_x^*A_x = t1_{V_x}$. Thus A_x is injective, $R \geq n$, and $D_x = F_x \ominus A_x(V_x)$ has dimension $R-n$.

The physical subspaces $D_x \subset \mathcal{H}$ may overlap. Form their abstract Hilbert direct sum

$$\mathcal{D} = \bigoplus_{x\in X}^{\ell^2} D_x.$$

Equivariance of the subspaces gives the induced action of G on $\mathcal{D}$. On the algebraic sum define $T : \mathcal{D} \to \mathcal{H}$ by summing the literal inclusions $D_x \hookrightarrow \mathcal{H}$. We now compute its bounded extension.

For $y \neq x$, let Q_{yx} be the orthogonal projection in V_y onto $\operatorname{im} P_{yx}$. At each y, the finite sum $\sum_{x\neq y} Q_{yx}$ commutes with the irreducible action of the stabilizer of y. Its trace is

$$\sum_{x\neq y} \operatorname{rank} P_{yx} = \sum_{x\neq y} \operatorname{rank} P_{xy} = R,$$

since $P_{xy} = P_{yx}^*$. Hence $\sum_{x\neq y} Q_{yx} = r1_{V_y}$, and therefore

$$\sum_x \operatorname{pr}_{F_x} = r1_{\mathcal{H}}.$$

Moreover $A_x = (P - c1)|_{V_x}$, so

$$\operatorname{pr}_{A_x(V_x)} = t^{-1}(P-c1)E_x(P-c1), \qquad \sum_x \operatorname{pr}_{A_x(V_x)} = t^{-1}(P-c1)^2.$$

These identities hold as quadratic forms on finitely supported vectors, where all sums involved are finite. Subtracting gives

$$\sum_x \operatorname{pr}_{D_x} = r1 - \frac{(P-c1)^2}{c(1-c)} = aP + b(1-P), \qquad a = r - \frac{1-c}{c}, \qquad b = r - \frac{c}{1-c}. \tag{89}$$

The left side is positive, and the right side is bounded. Consequently the adjoint map $v \mapsto (\operatorname{pr}_{D_x} v)_x$ is bounded, so T extends to a bounded equivariant operator and

$$TT^* = aP + b(1-P). \tag{90}$$

Both P and $1-P$ are nonzero, so positivity gives $a, b \geq 0$. These inequalities are precisely the two bounds on c in (86).

The local trace calculation is

$$\operatorname{Tr}(P|_{D_x}) = c(R+n) - n, \qquad \operatorname{Tr}((1-P)|_{D_x}) = (1-c)(R+n) - n. \tag{91}$$

Here the notation denotes the trace of a compression; D_x is not assumed invariant under either projection. To check the first identity, the compression to F_x has trace cR, while $PA_x = (1-c)P|_{V_x}$ gives trace $(1-c)n$ on $A_x(V_x)$. Subtract these traces. The second identity follows from $\dim D_x = R - n$.

If both bounds are strict, then $a, b > 0$. Define

$$U = \big(a^{-1/2}P + b^{-1/2}(1-P)\big)T. \tag{92}$$

Equation (90) gives $UU^* = 1$. The maps T and U have finite support in every row and column. Indeed, each D_x is supported inside the finite support of the off-diagonal column of P at x, and rows are finite by self-adjointness of P. The remaining factor in (92) is a linear combination of P and the identity.

Let $C_x = \operatorname{pr}_{D_x} P|_{D_x}$ and put $K = 1 - U^*U$ on the abstract sum $\mathcal{D}$. Since U is a coisometry and differs from T by an invertible left factor, K is the orthogonal projection onto $\ker U = \ker T$. The restriction of T to each abstract fiber is its literal isometric inclusion in $\mathcal{H}$. Consequently

$$K_{xx} = 1 - a^{-1}C_x - b^{-1}(1-C_x), \qquad \operatorname{Tr} K_{xx} = R - 2n. \tag{93}$$

To prove the trace identity, rewrite (91) as

$$\operatorname{Tr} C_x = nca, \qquad \operatorname{Tr}(1-C_x) = n(1-c)b.$$

The compression of U^*U to D_x therefore has trace $a^{-1}nca+b^{-1}n(1-c)b = n$. Subtracting from $\dim D_x = R-n$ gives (93). Positivity now yields $R \geq 2n$. At equality every compression K_{xx} is positive with trace zero. Thus K vanishes on every coordinate fiber and hence is zero. In that case U is unitary.

If $c = 1/(r+1)$, the first trace is zero, and positivity gives $PD_x = 0$. If $c = r/(r+1)$, the second trace is zero, and $(1-P)D_x = 0$. In the latter case the off-diagonal ranges for $1-P$ are the same as for P, and its transverse map is $-A_x$, so the space D_x is unchanged. In either case, write Q for the projection with zero action on D_x, and write $q = 1/(r+1)$ for its diagonal scalar. Equivariance gives $QD_y = 0$ for all y.

The commutation argument in the proof of Theorem 17.1 applies to Q without any irreducibility assumption on its image. We give the relevant steps to make this point explicit. Put $W_x = Q(V_x)$ and let $\mathsf{Q}_x = q^{-1}QE_xQ$ be the orthogonal projection onto W_x. Decomposing F_x into the image of $(1-E_x)Q|_{V_x}$ and D_x gives $QF_x \subset W_x$. Thus Q_y preserves W_x for every y. As it is self-adjoint, it commutes with Q_x. The projections Q_x therefore commute pairwise.

Choose a nonzero simultaneous eigenspace W in some W_x. It is a simultaneous eigenspace in $Q\mathcal{H}$, because Q_x acts as the identity on it. For $0 \neq w \in W$, the eigenvalue of each Q_y is zero or one, and

$$\|E_y w\|^2 = \langle QE_yQw, w\rangle = q\langle \mathsf{Q}_y w, w\rangle.$$

The vector w has finite coordinate support, since it belongs to $Q(V_x)$. Summing the displayed equality over y gives $q|S_W| = 1$, where S_W is the nonempty finite set of coordinates where the eigenvalue is one. Hence $r+1 = q^{-1}$ is an integer, proving the endpoint assertion.

If $R < 2n$, the two bounds on c cannot both be strict by (93). Thus r is an integer, and $1 \leq r < 2$ forces $r = 1$. The bounds on c then both give $c = 1/2$. The preceding commutation argument applies to P itself. When $P\mathcal{M}$ is irreducible, its atoms form a single G-orbit, and the final compact-induction assertion follows from the atom construction in Theorem 17.1.

Finally suppose that $R = 2n$ and $1/3 < c < 2/3$. Then $a, b > 0$, and the preceding kernel calculation makes U unitary. Compressing the identity $U^*U = T^*(a^{-1}P + b^{-1}(1-P))T = 1$ to D_x gives

$$a^{-1}C_x + b^{-1}(1-C_x) = 1.$$

Here $a = (3c-1)/c$ and $b = (2-3c)/(1-c)$. If $c \neq 1/2$, then $a \neq b$, and solving this equation gives $C_x = (3c-1)1_{D_x}$. Furthermore $U^*PU = a^{-1}T^*PT$, whose diagonal compression is $a^{-1}C_x = c1_{D_x}$. This proves (88). If $c = 1/2$, then $a = b = 1$ and (92) gives $U = T$. □

The endpoint construction also extracts the irreducible image of P when the commuting projections come from $1-P$. The following coordinate completion treats both cases.

Proposition 18.2. *Suppose that $\pi = P\mathcal{M}$ is irreducible. If $c = 1/(r+1)$ or $c = r/(r+1)$, then r is a positive integer and there are an open subgroup $L \supset Z$, compact modulo Z, and an irreducible finite-dimensional representation ρ of L such that $\pi \simeq \text{c-Ind}_L^G \rho$.*

More precisely, put $m = r+1$. There is a positive divisor k of n and an orthogonal decomposition of $\mathcal{M}$ into G-permuted coordinate blocks, each supported on exactly m coordinates and of dimension $mk \leq mn$. Each block is preserved by P and every coordinate projection. The inducing fiber has dimension k at the lower endpoint and $(m-1)k = rk$ at the upper endpoint. After choosing a block whose support contains $x_0 = J$, its full stabilizer L satisfies

$$[J : J \cap L] = n/k, \qquad [L : J \cap L] = m. \tag{94}$$

Proof. At the lower endpoint put $Q = P$, and at the upper endpoint put $Q = 1-P$. In both cases Q has diagonal $q = 1/m$, and the projections $\mathsf{Q}_x = q^{-1}QE_xQ$ onto $W_x = Q(V_x)$ commute by the endpoint argument in Theorem 18.1. That argument does not require irreducibility of $Q\mathcal{M}$. Its nonzero simultaneous eigenspaces with at least one eigenvalue equal to one are finite-dimensional orthogonal atoms W. Their algebraic sum is $Q\mathcal{M}$. For each atom, the membership set $S_W = \{x : \mathsf{Q}_x|_W = 1\}$ is its coordinate support and has cardinality m.

Fix such an atom, and put $k_W = \dim W$. If $x \in S_W$, then

$$\langle E_x w, E_x w'\rangle = q\langle w, w'\rangle \qquad (w, w' \in W). \tag{95}$$

Indeed, the left side is $\langle QE_xQw, w'\rangle$, and $QE_xQ = q\mathsf{Q}_x$. Thus E_xW has dimension k_W, and $k_W \leq n$. Define the physical coordinate block

$$B_W = \bigoplus_{x \in S_W} E_x W \subset \mathcal{M}. \tag{96}$$

It has dimension mk_W and is preserved by each E_x.

If $W \neq W'$ are atoms, their images at any common coordinate are orthogonal: for $w \in W$ and $w' \in W'$, one has $\langle E_x w, E_x w' \rangle = q\langle \mathsf{Q}_x w, w' \rangle = 0$. Hence the blocks B_W are pairwise orthogonal. They cover every coordinate fiber. To see this, $q^{-1/2} Q|_{V_x}$ identifies V_x with W_x, and W_x is the orthogonal sum of the atoms W with $x \in S_W$. For $v \in V_x$, decompose Qv into those atoms and use $v = q^{-1} E_x Q v$. This proves the algebraic decomposition

$$\mathcal{M} = \bigoplus_W B_W. \tag{97}$$

The space W is contained in B_W, because every $w \in W$ is the sum of its coordinates in S_W. For $x \in S_W$, the identity $QE_x w = q\mathsf{Q}_x w = qw$ gives $Q(E_x W) \subset W$. Consequently Q preserves B_W and its restriction is the orthogonal projection onto W. Thus

$$P(B_W) = W \quad \text{at the lower endpoint}, \qquad P(B_W) = B_W \ominus W \quad \text{at the upper endpoint}. \tag{98}$$

Every space in (98) is nonzero: its dimension is k_W or $(m-1)k_W$, and $m \geq 2$. Its coordinate support is exactly S_W. This is immediate for W at the lower endpoint. At the upper endpoint, for $x \in S_W$, choose $y \in S_W$ with $y \neq x$. For every $w \in W$, the vector $E_x w - E_y w$ lies in B_W and is killed by Q, so it belongs to $P(B_W)$. Its coordinate at x is $E_x w$. Consequently

$$E_x P(B_W) = E_x W \quad (x \in S_W), \qquad B_W = \bigoplus_x E_x P(B_W) \tag{99}$$

at both endpoints.

Equivariance of Q makes G permute the atoms, the blocks B_W, and their nonzero images under P. Equation (97) therefore decomposes π as an orthogonal algebraic sum of these images. Irreducibility of π implies that G acts transitively on them, hence on the atoms. In particular, all k_W have a common value k.

Let $C_W = P(B_W)$ and let L be its full stabilizer. Formula (99) recovers B_W from C_W, and $Q(B_W) = W$ recovers W from B_W. Conversely, the coordinate completion recovers B_W from W, and P then recovers C_W. All these operations are equivariant, so the three spaces have the same full stabilizer. The subspace C_W has a basis of smooth vectors, so L is open and contains Z. Its coordinate support is the finite nonempty set S_W. The action on this support puts $L \cap G_x$ at finite index in L for any coordinate in the support; this subgroup is closed in G_x, whose quotient by Z is compact. Hence L/Z is compact. Writing ρ for the action on C_W, the transitive orthogonal decomposition gives $\pi = \text{c-Ind}_L^G \rho$. A proper L-invariant subspace of C_W would generate a proper G-invariant subspace of π, so ρ is irreducible. Formula (98) gives its stated dimension.

Choose W with $x_0 \in S_W$. The subgroup J permutes the nonzero mutually orthogonal subspaces $E_{x_0} W'$ with $x_0 \in S_{W'}$. Irreducibility of V_{x_0} makes this action transitive, since the sum over any orbit is J-invariant. Each summand has dimension k, so there are n/k of them. This proves that k divides n and gives the first index in (94). It also proves transitivity of G on the incidences (x, W') with $x \in S_{W'}$. Fixing W then shows that L is transitive on S_W. This set has cardinality m, and its stabilizer at x_0 is $J \cap L$, proving the second index. □

Corollary 18.3. *Suppose that $P\mathcal{M}$ is irreducible, $R = 2n$, and $1/3 < c < 2/3$. If D_{x_0} is reducible as a representation of J, then $P\mathcal{M}$ contains an irreducible J-type of dimension strictly smaller than n. If D_{x_0} is irreducible, the unitary U gives another coordinate model with irreducible fibers of dimension n, and the transported projection $U^* P U$ has diagonal $c1_{D_x}$ at every coordinate, including when $c = 1/2$.*

Proof. The trace identity gives $\mathrm{Tr}(P|_{D_{x_0}}) = (3c-1)n > 0$. Thus P is nonzero on at least one irreducible summand of D_{x_0}. If D_{x_0} is reducible, every such summand has dimension smaller than n. Its nonzero image under P is an isomorphic occurring J-type.

If D_{x_0} is irreducible, the first assertion about the transported model follows from the unitary equivariance of U. Formula (88) gives its diagonal when $c \neq 1/2$. When $c = 1/2$, Schur's lemma makes $\mathrm{pr}_{D_x} P|_{D_x}$ scalar, and its trace from (91) is $n/2$. Since $U = T$ in this case, the transported diagonal is again $c1_{D_x}$. □

The condition $R < 2n$ gives $R = n$, $c = 1/2$, and immediate extraction. Strict dimension descent can first occur at $R = 2n$, through a reducible D_{x_0}. At larger rank, the projection K in (93) records relations among the physical subspaces D_x. Both T and PT vanish on its range, so this construction does not supply occurrence of π there. The preceding identities do not force a smaller rank sum after a change of model or a model satisfying an endpoint condition.

19. Extraction by rotation when the complement has the original type

Let G be the group of rational points of a connected reductive group over a nonarchimedean local field, and let Z be its center. Let $J \supset Z$ be open and compact modulo Z, and let τ be an irreducible smooth unitary representation of J, of dimension n. Put

$$\mathcal{M} = \text{c-Ind}_J^G \tau = \bigoplus_{x \in G/J} V_x,$$

and write $\mathcal{H}$ for its Hilbert completion. Suppose P is the coefficient projection associated with one occurring copy of τ in an irreducible unitary supercuspidal representation π. Thus $P\mathcal{M} \simeq \pi$, the columns of P have finite support, and $E_x P E_x = cE_x$, where $c = d_\pi \operatorname{vol}(J/Z)/n$.

For each coordinate x, define

$$F_x = \bigoplus_{y \neq x} \operatorname{im}(E_y P E_x), \qquad D_x = F_x \ominus (1 - E_x)P(V_x), \qquad R = \dim F_x.$$

Equivariance makes R independent of x. Fix $x_0 = J$, so that D_{x_0} is a finite-dimensional smooth J-representation.

Theorem 19.1. *Suppose that $R = 2n$, $c = 1/2$, and $D_{x_0} \simeq \tau$ as J-representations. Then π is compactly induced from an irreducible finite-dimensional representation of an open subgroup compact modulo Z.*

Proof. Set $A = 2P - 1$. This is a self-adjoint unitary operator. Its diagonal blocks vanish, and

$$A(V_x) = 2(1 - E_x)P(V_x), \qquad F_x = A(V_x) \oplus D_x.$$

The displayed sum is orthogonal. Since $\dim F_x = 2n$ and A is unitary, each D_x has dimension n.

We first prove that the spaces D_x themselves form an orthogonal decomposition of $\mathcal{H}$. Denote the orthogonal projection onto a closed subspace U by e_U. The operator e_{F_x} is diagonal in the coordinate decomposition. On the coordinate V_y, the sum of these operators is

$$E_y \Big(\sum_x e_{F_x} \Big) E_y = \sum_{x \neq y} e_{\operatorname{im}(E_y P E_x)}.$$

The sum is finite and commutes with G_y. Irreducibility of V_y makes it scalar. Its trace is

$$\sum_{x \neq y} \operatorname{rank}(E_y P E_x) = \sum_{x \neq y} \operatorname{rank}(E_x P E_y) = R = 2n,$$

where self-adjointness gives the middle equality. Therefore $\sum_x e_{F_x} = 2\,\mathrm{Id}_{\mathcal{H}}$. On the other hand, $e_{A(V_x)} = AE_xA$, so $\sum_x e_{A(V_x)} = \mathrm{Id}_{\mathcal{H}}$. Consequently,

$$\sum_x e_{D_x} = \mathrm{Id}_{\mathcal{H}}\,. \tag{100}$$

All these sums converge strongly. They can also be checked on the dense space $\mathcal{M}$, where finite support of the columns and rows makes every relevant sum finite.

For $d \in D_y$, equation (100) gives

$$\|d\|^2 = \sum_x \|e_{D_x} d\|^2 = \|d\|^2 + \sum_{x \neq y} \|e_{D_x} d\|^2.$$

Thus D_y is orthogonal to D_x whenever $x \neq y$. Equation (100) also shows that they span the Hilbert space. Their algebraic sum is $\mathcal{M}$: a vector with finite coordinate support is orthogonal to all but finitely many D_x, because $D_x \subset F_x$ and the support graph is locally finite.

Choose a unitary J-isomorphism $V_{x_0} \longrightarrow D_{x_0}$. Translating it defines a G-equivariant unitary operator T on $\mathcal{H}$ with

$$T(V_x) = D_x, \qquad T\mathcal{M} = \mathcal{M}.$$

The choice is independent of coset representatives by J-equivariance. Its columns have finite support. Moreover,

$$E_x T E_x = 0, \qquad E_x P T E_x = 0. \tag{101}$$

The first equality uses $D_x \subset (1 - E_x)\mathcal{M}$. For the second, if $d \in D_x$ and $v \in V_x$, then

$$\langle Pd, v\rangle = \langle d, Pv\rangle = \frac{1}{2}\langle d, v + Av\rangle = 0.$$

Next, PD_{x_0} is a nonzero copy of τ in π, orthogonal to $P(V_{x_0})$. Orthogonality follows from

$$\langle Pd, Pv\rangle = \langle d, Pv\rangle = 0.$$

To prove nonvanishing, take the trace of P on the orthogonal decomposition $F_{x_0} = A(V_{x_0}) \oplus D_{x_0}$. Since F_{x_0} is a coordinate subspace and every diagonal block of P is $1/2$, $\mathrm{Tr}(P|_{F_{x_0}}) = n$. Here and below this notation means the trace of the compression when the subspace is not invariant. Since A commutes with P and is unitary, $\mathrm{Tr}(P|_{A(V_{x_0})}) = n/2$. Hence

$$\mathrm{Tr}(P|_{D_{x_0}}) = n/2 > 0.$$

The J-equivariant map $P|_{D_{x_0}}$ is therefore nonzero and is injective by irreducibility. Thus $m = \dim \mathrm{Hom}_J(\tau, \pi) \geq 2$.

The formal-degree ratio is $r = 1/c = 2$. Proposition 31.1 gives $m \leq 2$, hence $m = 2$ and

$$\mathcal{M} \simeq \pi \oplus \pi, \qquad \mathrm{End}_G(\mathcal{M}) \simeq M_2(\mathbf{C}). \tag{102}$$

The coefficient embeddings in that proposition give an orthogonal identification of the Hilbert completion with two copies of the Hilbert completion of π. Choose it so that $P = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$. For any matrix B in this commutant, its coordinate compression is

$$E_x B E_x = \frac{1}{2}\,\mathrm{Tr}_2(B) E_x. \tag{103}$$

Indeed, for the matrix units supplied by the two orthogonal coefficient embeddings, this compression is $\frac{1}{2}\delta_{ij}E_x$, exactly as in the proof of Proposition 31.1. Equivariance gives the same formula at every coordinate.

Write $T = (t_{ij})$ in this matrix algebra. The two equalities in (101), applied through (103), give $t_{11} + t_{22} = 0$ and $t_{11} = 0$. Thus both diagonal entries vanish. Unitarity implies

$$T = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}, \qquad |a| = |b| = 1.$$

Multiplying T by a scalar of absolute value one, chosen so that its square times ab is one, makes $T^2 = 1$. Since T remains unitary, it then satisfies $T = T^*$. This change does not alter $T(V_x) = D_x$. In this normalization,

$$AT = -TA, \qquad (AT)^* = -AT, \qquad (AT)^2 = -1.$$

Define a new equivariant orthogonal projection

$$P' = \frac{1 + iAT}{2}. \tag{104}$$

It is a rank-one projection in the matrix algebra (102), so $P'\mathcal{M} \simeq \pi$. It has finite columns, since A and T do. Formula (103) also gives $E_x P' E_x = \frac{1}{2}E_x$.

The original finite coordinate space F_x is invariant under P'. Indeed, its orthogonal decomposition is now $F_x = A(V_x) \oplus T(V_x)$, and for $v \in V_x$,

$$AT(Av) = -Tv, \qquad AT(Tv) = Av.$$

Thus AT preserves F_x, and its two eigenspaces there have dimension n each. In particular, $P'F_x$ is nonzero and has dimension n.

Since F_x is a direct sum of coordinate subspaces, every E_y also preserves it. It follows that every block $E_y P' E_z$ preserves F_x. Choose a nonzero vector in one coordinate summand of F_x. It has finite closure under all the coordinate projections and these blocks. The irreducibility of the action on that coordinate fiber makes the whole fiber have finite closure: finitely many translates under its stabilizer span the fiber, and the sum of their finite closures is finite-dimensional. Translating the fiber to V_{x_0} gives the positive outcome of Theorem 15.2 for P'. That theorem extracts an open subgroup compact modulo Z and an irreducible finite-dimensional representation compactly inducing $P'\mathcal{M} \simeq \pi$. □

The theorem uses the isomorphism $D_{x_0} \simeq \tau$, not merely equality of dimensions. Without that isomorphism, the two orthogonal occurring J-spaces need not increase the multiplicity of the same type, and the matrix algebra in (102) has not been obtained.

19.1. **An actual example.** The hypotheses occur for a reductive group. Let $F = \mathbf{Q}_3$, let E/F be unramified quadratic with nontrivial automorphism σ, and form the division quaternion algebra $D = E \oplus E\Pi$, where $\Pi^2 = 3$ and $\Pi a = \sigma(a)\Pi$. The norm $N_{E/F}(a) - 3N_{E/F}(b)$ of $a + b\Pi$ has terms of opposite valuation parity, proving that D is a division algebra. Let $G = \mathrm{PGL}_1(D)(F)$ and let K_1 be the image of $F^\times(1+\mathfrak{p}_D)$ in $D^\times/F^\times$. Reduction of units and conjugation by Π give

$$G/K_1 \simeq (\mathbf{F}_9^\times/\mathbf{F}_3^\times) \rtimes \langle\bar{\Pi}\rangle \simeq D_8.$$

Here $D_8 = \langle r, s : r^4 = s^2 = 1,\ srs = r^{-1}\rangle$ has order eight. The residue quotient is cyclic of order four, and Frobenius acts on it by inversion. The group G is compact, so inflations of irreducible representations of this finite quotient are supercuspidal.

Let J be the inverse image of $\{1, r^2\}$ and let τ send r^2 to -1. Inflate the irreducible representation π given by

$$\pi(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \pi(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Its restriction to J is two copies of τ. With Haar measure normalized by $\mathrm{vol}(G) = 1$, we have $d_\pi = 2$, $\mathrm{vol}(J) = 1/4$, and $c = 1/2$.

Choose the occurring line generated by $v = (2, 1)/\sqrt{5}$. Representatives of G/J in the quotient are $1, r, s, rs$, and their coefficient overlaps with v are

$$\langle\pi(r)v, v\rangle = 0, \qquad \langle\pi(s)v, v\rangle = \frac{3}{5}, \qquad \langle\pi(rs)v, v\rangle = \frac{4}{5}.$$

Thus precisely two off-diagonal blocks in each column of the coefficient projection are nonzero. They have rank one, so $n = 1$ and $R = 2$. Since J acts by τ on the entire induced representation, its one-dimensional space D_{x_0} is isomorphic to τ. All hypotheses of Theorem 19.1 therefore hold. In this example an inducing pair can also be given directly: take the inverse image L of $\langle r\rangle$ and the inflated character $\rho(r) = i$. The two characters ρ and ρ^{-1} of $\langle r\rangle$ are interchanged by s, and

$$\pi \simeq \text{c-Ind}_L^G \rho.$$

20. Extraction from two orbits of coefficient blocks

Let G be a reductive local group with center Z, let $J \supset Z$ be open and compact modulo Z, and let τ be an irreducible unitary representation of J of dimension n. Let P be a coefficient projection on $\mathcal{M} = \text{c-Ind}_J^G \tau$, with irreducible supercuspidal image π, finite columns, and diagonal $P_{x,x} = c1_{V_x}$. Write $R = \sum_{y\neq x} \mathrm{rank}\, P_{y,x}$.

The nonzero off-diagonal blocks define an invariant undirected graph on $X = G/J$. An orbit of ordered edges under G is equivalent to an orbit of active neighbors under G_x. Reversing an ordered edge defines the inverse orbit.

Theorem 20.1. *Suppose $R = 2n$ and the ordered active edges have exactly two G-orbits, exchanged by reversal. Then $c = 1/3$, and π is compactly induced from an irreducible finite-dimensional representation of an open subgroup compact modulo Z. The atoms in the coefficient construction each have coordinate support of size three.*

Proof. Choose one of the two orbits as the forward orientation. At each vertex x, the outgoing column restricted to either neighbor orbit is a nonzero G_x-equivariant map

$$V_x \longrightarrow \bigoplus_{y \text{ in that orbit}} \mathrm{im}\, P_{y,x}.$$

It is injective by irreducibility. Each target has dimension at least n, and their dimensions sum to $R = 2n$, so both dimensions equal n.

Set $E_{x,y} = \mathrm{im}\, P_{x,y} \subset V_x$. For each of the two neighbor orbits at x, Schur's lemma makes $\sum_y P_{x,y}P_{y,x}$ a positive scalar multiple of 1_{V_x}. The sum of the ranks is n. Consequently the spaces $E_{x,y}$ in that orbit are an orthogonal decomposition of V_x, and each summand is the same scalar times its range projection. Indeed, concatenate the positive square roots of the summands into a map from the direct sum of their ranges to V_x. The map is square and its product with its adjoint is scalar, so it is a scaled unitary. Reversing an edge

exchanges the two orbit types and preserves singular values. Thus their scalars are equal, and the diagonal projection equation gives

$$a = \frac{c(1-c)}{2}, \qquad P_{y,x} = \sqrt{a}\, U_{y,x},$$

where $U_{y,x}$ is a unitary map from $E_{x,y}$ onto $E_{y,x}$, extended by zero on the orthogonal complement.

Define an equivariant operator S by

$$S_{y,x} = \begin{cases} a^{-1/2} P_{y,x}, & x \longrightarrow y \text{ is forward,} \\ 0, & \text{otherwise.} \end{cases}$$

It has finite rows and columns. It is unitary. To check this explicitly, at a fixed target y the images of all incoming forward blocks are an orthogonal decomposition of V_y: their source vertices are one neighbor orbit at y. Hence distinct source columns are orthogonal. At a fixed source x, the initial spaces of its outgoing forward blocks are also an orthogonal decomposition of V_x, so the diagonal of S^*S is the identity. Thus $S^*S = 1$. The same argument with sources and targets interchanged gives $SS^* = 1$.

Self-adjointness of P and the two inverse orbits give

$$P = c1 + \sqrt{a}(S + S^*). \tag{105}$$

Thus S commutes with P, preserves its algebraic irreducible image, and acts there as a scalar z of absolute value one. In particular, $PS = zP$. There are no diagonal blocks of S. There are also no diagonal blocks of S^2: a two-step contribution to $(S^2)_{x,x}$ would require both orientations of one edge to be forward, contrary to the disjointness of the inverse orbits. Multiplying (105) by S and compressing to V_x therefore yields

$$\sqrt{a}\, 1_{V_x} = (PS)_{x,x} = zc1_{V_x}.$$

The numbers $\sqrt{a}, c$ are positive, so $z = 1$ and $a = c^2$. Combining with $a = c(1-c)/2$ gives $c = 1/3$.

For the local neighbor complement $D_x = F_x \ominus (1 - E_x)P(V_x)$, the trace identity is now

$$\mathrm{Tr}(E_{D_x} P|_{D_x}) = c(R+n) - n = 0.$$

Positivity gives $PD_x = 0$. The rank-budget atom theorem therefore extracts compact induction. Its membership formula gives coordinate support size $c^{-1} = 3$ for every atom.

The unitary also satisfies $S^3 = 1$. On $P\mathcal{M}$, it is the identity. On $(1-P)\mathcal{M}$, equation (105) gives $S+S^* = -1$, hence $S^2 + S + 1 = 0$. These two assertions prove the claim on all of $\mathcal{M}$. In particular, forward triangles must occur in the support graph: the diagonal of S^3 is the identity, and each term in that diagonal is a product along a directed three-cycle. □

No building adjacency, absence of four-cycles, or identification of the local complement with the original type is needed for this theorem. Irreducibility of the image is essential in the step $S|_\pi = z$.

20.1. **Consequences for adjacency support in projective linear groups.** Let $G = \mathrm{PGL}_m(F)$, let $J = \mathrm{PGL}_m(\mathcal{O}_F)$, and assume nonzero off-diagonal blocks only join adjacent building vertices. The neighbor orbits have type differences $i \in \{1, \dots, m-1\}$, and inversion sends i to $m - i$. Each active orbit has total rank at least n, by the injective outgoing-column map in the proof. Hence $R = 2n$ allows at most two active orbits.

If there are two active orbits, they form an inverse pair $i, m-i$. The theorem gives extraction and $c = 1/3$. Since directed three-cycles occur, three forward type increments must sum to zero modulo m: $3i \equiv 0 \pmod m$. Thus m is divisible by three, and the two types are $m/3, 2m/3$. The atoms have support on the three vertices of an actual two-dimensional facet. Their stabilizers are full stabilizers of those facets, because an atom has exactly that coordinate support and the commuting projection construction gives one atom per nonzero simultaneous membership eigenspace. In particular, for every odd m the rank boundary $R = 2n$ gives actual extraction; if m is odd and not divisible by three, no such adjacency-supported coefficient projection exists.

For even m, the only alternative left is a single self-inverse active orbit of type $m/2$. It has no active triangles, so its off-diagonal projection equation forces $c = 1/2$. This case is not covered by the two-orbit theorem. Nor does finite propagation of a general coefficient projection imply adjacency support.

20.2. **Two orbits each preserved by reversal.**

Theorem 20.2. *Retain $R = 2n$, and suppose the ordered active edges have two G-orbits, each preserved by reversal. Then $c = 1/2$, the local complement is isomorphic to τ, and the rotation theorem extracts compact induction for π.*

Proof. The outgoing-column argument in Theorem 20.1 shows that each orbit has total rank n. The positive-square-root argument in that proof gives, for each orbit $j = 1, 2$, a number $b_j > 0$ such that its blocks are b_j times unitary maps between their ranges. Define S_j by dividing the blocks in that orbit by b_j and setting all other blocks to zero. As before, the decompositions of the initial and final spaces show that S_j is unitary. Since its orbit is preserved by reversal, it is also self-adjoint. Consequently

$$P = c1 + b_1 S_1 + b_2 S_2, \qquad S_1^2 = S_2^2 = 1, \qquad b_1^2 + b_2^2 = c(1-c).$$

Both S_j have zero diagonal. The product $S_1 S_2$ has zero diagonal as well: a two-step return would require the same edge to lie in the two disjoint orbits. Expanding $P^2 = P$ gives

$$(2c-1)(b_1 S_1 + b_2 S_2) + b_1 b_2 (S_1 S_2 + S_2 S_1) = 0. \tag{106}$$

The anticommutator $S_1 S_2 + S_2 S_1$ commutes with S_1. Thus, if $c \neq 1/2$, commuting (106) with S_1 shows that $S_1 S_2 = S_2 S_1$.

In that case each S_j commutes with P and acts on its irreducible image as a scalar $\varepsilon_j \in \{1, -1\}$. Compressing $PS_j = \varepsilon_j P$ to a coordinate, using the vanishing diagonals just established, gives $b_j = \varepsilon_j c$. Positivity forces both signs to be positive and $b_1 = b_2 = c$. Therefore $2c^2 = c(1-c)$, or $c = 1/3$. The commuting involutions then give

$$P = (1 + S_1 + S_2)/3.$$

On their simultaneous eigenspaces, the possible eigenvalues of this operator are $1, 1/3, -1/3$. Since P is a projection, only the first can occur. Hence $P = 1$, contradicting its diagonal $c = 1/3$.

It follows that $c = 1/2$. Equation (106) now gives $S_1 S_2 = -S_2 S_1$. Each orbit space $F_{x,j}$ has dimension n and the outgoing column is a G_x-equivariant isomorphism from V_x onto it. Thus $F_x \simeq 2\tau$ as a G_x-representation. Its outgoing-column subspace is one copy of τ, so $D_x \simeq \tau$. These are exactly the hypotheses of Theorem 19.1. □

For completeness, the rotation can also be seen directly in these operators. The two anticommuting self-adjoint involutions generate a copy of $M_2(\mathbf{C})$ commuting with G. The projection P is a rank-one projection in that algebra. Since its image is irreducible, every rank-one projection in the same matrix algebra has image isomorphic to π. In particular,

$$P' = \frac{1 + \mathrm{i} S_1 S_2}{2}$$

has image π. The finite coordinate space $F_x = S_1(V_x) \oplus S_2(V_x)$ is invariant under P', because $S_1 S_2$ sends $S_1 v$ to $-S_2 v$ and $S_2 v$ to $S_1 v$. Thus the finite-closure extraction applies to P' directly.

Combining the two theorems, every coefficient projection with $R = 2n$ and at least two active neighbor orbits gives actual compact induction. There cannot be more than two such orbits, since each has total rank at least n. Inversion either exchanges the two or preserves each of them, which are precisely the cases proved above. The only case left by this orbit count is a single orbit preserved by reversal.

21. Equality in a rank inequality and central support

Let $\mathcal{M} = \bigoplus_{x \in X} V_x$ be the coordinate bundle of $\text{c-Ind}_J^G \tau$, where τ is irreducible and $n = \dim \tau$. Let P be an equivariant orthogonal coefficient projection with finite columns, diagonal $P_{x,x} = c1$, and irreducible algebraic image π. Suppose $0 < c < 1$ and $R = 2n$. Join distinct vertices x, y when $P_{y,x} \neq 0$, and denote the finite neighbor set of x by $N(x)$. Set

$$E_{x,y} = \operatorname{im} P_{x,y} \subset V_x, \qquad Q_{x,y} = \operatorname{pr}_{E_{x,y}}, \qquad \mathcal{C}_x = C^*(Q_{x,y} : y \in N(x)) \subset \operatorname{End}(V_x).$$

Schur's lemma and the rank sum give

$$\sum_{y \in N(x)} Q_{x,y} = 2\, 1_{V_x}. \tag{107}$$

Indeed the sum commutes with G_x, and its trace is $R = 2n$. The algebra $\mathcal{C}_x$ is unital by this equation.

21.1. **The equality argument.**

Lemma 21.1. *Let V, V' be finite-dimensional Hilbert spaces with families of orthogonal projections (Q_j) and (Q'_k) satisfying $\sum_j Q_j = 2I_V$ and $\sum_k Q'_k = 2I_{V'}$. Suppose a finite family of nonzero maps $T_i : V \to V'$ satisfies $\sum_i T_i = 0$. Suppose its initial spaces $I_i = (\ker T_i)^\perp$ are contained in ranges of distinct Q_{a_i}, and its final spaces $F_i = \operatorname{im} T_i$ are contained in ranges of distinct Q'_{b_i}. Put $W = \sum_i I_i$, $W' = \sum_i F_i$, and $t = \sum_i \operatorname{rank} T_i$. Then*

$$\dim W = \dim W' = t/2, \qquad \sum_i \operatorname{pr}_{I_i} = 2\operatorname{pr}_W, \qquad \sum_i \operatorname{pr}_{F_i} = 2\operatorname{pr}_{W'}.$$

Moreover W reduces every Q_j, with $Q_{a_i} W = I_i$ and $Q_j W = 0$ for omitted indices. The analogous assertions hold for W' and the Q'_k.

Proof. Factor the zero sum through $\bigoplus_i I_i$. The first map sends $v \in W$ to $(\operatorname{pr}_{I_i} v)_i$ and is injective, because the I_i span W. The second sends $(v_i)_i$ to $\sum_i T_i v_i$ and has range W'. The composite is zero. Hence the rank inequality for a product gives

$$\dim W + \dim W' \le \dim\Big(\bigoplus_i I_i\Big) = t.$$

Let $p_i = \operatorname{pr}_{I_i}$. The assumptions give $\sum_i p_i \le \sum_j Q_j = 2I$. Each p_i is supported on W, so $t = \operatorname{Tr}(\sum_i p_i) \le 2\dim W$. Applying the same argument to final spaces gives $t \le 2\dim W'$. All three inequalities must be equalities. The positive operator $2I_W - \sum_i p_i|_W$ has trace zero, so is zero. This proves the asserted identities of projections.

Now

$$\sum_i (Q_{a_i} - p_i) + \sum_{j \notin \{a_i\}} Q_j = 2(I - \operatorname{pr}_W).$$

Every summand on the left is positive. Its compression to W is therefore zero, and positivity implies that it annihilates W. Thus $Q_{a_i}|_W = p_i|_W$ and every omitted Q_j kills W. Since $I_i \subset W$, this proves that all Q_j preserve W, and self-adjointness makes W reducing. It also gives $Q_{a_i} W = I_i$. The proof for W' is identical. □

For distinct coordinates x, z, the off-diagonal projection equation is

$$(2c-1)P_{z,x} + \sum_{y \in N(x) \cap N(z)} P_{z,y} P_{y,x} = 0, \tag{108}$$

where the first term is omitted if it is zero. A nonzero product indexed by y has initial space in $E_{x,y}$ and final space in $E_{z,y}$. If the direct term is present, its initial and final spaces are $E_{x,z}$ and $E_{z,x}$. Its indices z at the source and x at the target are distinct from all common-neighbor indices. Thus Lemma 21.1 applies to every nonempty family of nonzero terms in (108). No claim is needed when all its terms vanish.

21.2. **Full central support.** The central support of a projection $Q \in \mathcal{C}_x$ is the least central projection of $\mathcal{C}_x$ dominating Q. It is full when that central projection is the identity. Equivalently, Q is nonzero in every simple summand of $\mathcal{C}_x$. One consequence used below is that a projection of full central support cannot annihilate a nonzero reducing subspace for $\mathcal{C}_x$. In fact, on a simple summand the algebra acts as $M_a(\mathbf{C}) \otimes I_b$, its reducing subspaces are $\mathbf{C}^a \otimes U$, and a nonzero projection in the matrix factor acts nontrivially whenever $U \neq 0$.

Theorem 21.2. *Suppose every $Q_{x,y}$ has full central support in $\mathcal{C}_x$. Let $d = |N(x)|$, independent of x by equivariance. If $c \neq 1/2$, every connected component of the support graph is K_{d+1}. If $c = 1/2$, every component is $K_{d,d}$. Every component has at most $4n$ vertices. In either case π is compactly induced from an irreducible representation of an open subgroup compact modulo Z. No transitivity on ordered edges is required.*

Proof. Whenever (108) has a nonzero term, its source reducing space W in Lemma 21.1 is nonzero. Full central support therefore prevents omission of any neighbor index at x. The same holds at z for the final reducing space.

Suppose $c \neq 1/2$, and let x, z be adjacent. The direct term in (108) is nonzero. Thus all source and target neighbor indices must occur, giving

$$N(x) \setminus \{z\} = N(z) \setminus \{x\}.$$

Adjacent vertices consequently have the same closed neighborhood. Along any path these closed neighborhoods remain equal, so the whole connected component is one closed neighborhood, a clique on $d+1$ vertices.

Now suppose $c = 1/2$. If a two-step product $P_{z,y}P_{y,x}$ is nonzero, the relation (108) contains only common-neighbor terms. All neighbor indices must occur at both endpoints, so $N(x) = N(z)$. This also shows that

such endpoints cannot be adjacent, since equal open neighborhoods cannot contain one another's distinct base vertices.

At a fixed vertex y, group its neighbors according to equality of their open neighborhoods. If x, z lie in different groups, the preceding implication forces $P_{z,y}P_{y,x} = 0$. Hence $E_{y,x} \perp E_{y,z}$. For each group, take the span of its ranges in V_y. These spans are nonzero, mutually orthogonal, and together span V_y by (107). Each reduces $\mathcal{C}_y$: projections indexed by that group preserve its span and projections from other groups kill it. Full central support rules out two distinct nonzero spans. Thus all neighbors of y have the same open neighborhood.

Put $A = N(y)$, and let $B = N(x)$ for one $x \in A$. Every vertex of A has neighborhood B. For $b \in B$, all neighbors of b have the same neighborhood as x, namely B. Since $y \in B$, any such neighbor also belongs to $N(y) = A$. Conversely every element of A is adjacent to b. Thus $N(b) = A$. The sets A, B are disjoint, each has size d, and they form a whole component isomorphic to $K_{d,d}$.

Every active block has rank at least one, so $d \le R = 2n$. The component sizes are therefore at most $2n+1$ in the first case and $4n$ in the second, both bounded by $4n$.

In either case, the finite coordinate space belonging to a component is invariant under P. Its image is nonzero and has that entire component as coordinate support, since every diagonal block is $c1$. These images form an orthogonal algebraic decomposition of π, permuted transitively by G. The stabilizer L of one image is open, since it contains a common open stabilizer of a finite basis. It contains Z, and it preserves a nonempty finite coordinate set. Its intersection with one coordinate stabilizer has finite index and is compact modulo Z, so the same holds for L. The orthogonal decomposition gives compact induction from that image. Any proper invariant subspace of the inducing image would produce a proper G-subrepresentation of π, so the inducing representation is irreducible. □

21.3. **Three active neighbors.**

Theorem 21.3. *Suppose every vertex has three active neighbors and G is transitive on ordered active edges. Then 3 divides n, every nonbacktracking two-step product has rank $n/3$, and every component of the support graph is one of K_4, $K_{3,3}$, or the ordinary cube graph. In particular, π is compactly induced as above. Full central support is not assumed.*

Proof. At a vertex write the three range projections as Q_1, Q_2, Q_3. Their complementary projections satisfy $(I - Q_1) + (I - Q_2) + (I - Q_3) = I$. Projections whose sum is the identity are pairwise orthogonal: on the range of any one of them, positivity forces all the others to vanish. Thus the Q_i commute. Transitivity on the three neighbors makes their ranks equal, so each has rank $2n/3$ and each complement has rank $n/3$. In particular $3 \mid n$. For $i \ne j$, Q_iQ_j is the third complementary projection and has rank $n/3$.

For a two-step path x, y, z with $x \ne z$, the map $P_{y,x}$ surjects onto $E_{y,x}$, and $P_{z,y}$ is injective on $E_{y,z}$ with kernel its orthogonal complement. Since the two range projections at y commute,

$$\operatorname{rank}(P_{z,y}P_{y,x}) = \dim(E_{y,x} \cap E_{y,z}) = n/3 > 0.$$

Consequently, if x, z are nonadjacent, they cannot have y as their unique common neighbor: otherwise their off-diagonal equation would have one nonzero term and could not vanish.

The permutation action of G_y on its three neighbors is transitive. The graph induced on these neighbors is therefore regular. Its degree is either zero or two, since a regular graph on three vertices cannot have degree one. If that degree is two, the closed neighborhood of y is K_4 and all its vertices already have their three neighbors there. It is a connected component. By vertex transitivity, all components are of this form.

Otherwise the graph is triangle-free. A transitive permutation group on three points is transitive on their three unordered pairs. Hence the number μ of common neighbors of two vertices at distance two is constant throughout the graph. The preceding nonzero-product argument gives $\mu \ge 2$, and the degree bound gives $\mu \le 3$.

If $\mu = 3$, two vertices at distance two have identical neighborhoods. At a vertex y, any two of its neighbors are at distance two and hence have the same neighborhood. The elementary argument in the preceding proof now identifies the component with $K_{3,3}$.

Suppose $\mu = 2$. Let a, b, c be the neighbors of a vertex v. Besides v, the pairs a, b, a, c, and b, c have unique common neighbors u_{ab}, u_{ac}, u_{bc}. These three vertices are distinct. For example, if $u_{ab} = u_{ac}$, that vertex would be adjacent to a, b, c and would share three neighbors with v, contrary to $\mu = 2$.

The vertices u_{ab} and u_{ac} share a, so they have one more common neighbor. It cannot be b, because that would make u_{ac} adjacent to all of a, b, c and give the same contradiction. It cannot be c for the analogous

reason. Therefore it is the third neighbor of each of u_{ab}, u_{ac}; call it w. Applying the same argument to u_{ab}, u_{bc} shows that the third neighbor of u_{bc} is also w. Triangle-freeness and $\mu = 2$ ensure that $v, a, b, c, u_{ab}, u_{ac}, u_{bc}, w$ are eight distinct vertices. Each already has all three neighbors in this set. The edges just listed are exactly those of the cube: v joins a, b, c, each u_{ij} joins i, j, w, and there are no other edges. This is the entire component.

All components in the three alternatives are finite. The last paragraph of Theorem 21.2 gives compact induction. □

22. Central refinement for blocks of rank at most two

Let P be the coefficient projection of a unitary supercuspidal representation π on $\mathcal{M} = \text{c-Ind}_J^G \tau$, where τ is irreducible of dimension n. Write $\mathcal{M} = \bigoplus_{x \in X} V_x$, with $X = G/J$, and suppose that $P_{x,x} = c1$ and $R = 2n$. In particular $0 < c < 1$. For an active edge $\{x, y\}$, let $Q_{x,y}$ be the orthogonal projection onto $\operatorname{im} P_{x,y}$ in V_x. Set

$$\mathcal{C}_x = C^*(Q_{x,y} : y \sim x) \subset \operatorname{End}(V_x).$$

These are finite-dimensional unital algebras with adjoint. Indeed, the sum of the projections is invariant under G_x, and its trace is $R = 2n$. Schur's lemma therefore gives

$$\sum_{y \sim x} Q_{x,y} = 2\, 1_{V_x}. \tag{109}$$

The arguments below use Lemma 21.1 and Theorem 21.2. Full central support of a projection means that its component in every simple summand of the stated algebra is nonzero.

Lemma 22.1. *Suppose each coordinate has an orthogonal decomposition*

$$V_x = \bigoplus_{a \in A_x} V_{x,a}.$$

Assume that these decompositions are transported by G and that G_x acts transitively on its summands. Let their common dimension be k. Suppose also that, for every old coefficient block, its refined blocks satisfy

$$\sum_{a \in A_x,\, b \in A_y} \operatorname{rank}(P_{y,x})_{b,a} = \operatorname{rank} P_{y,x}. \tag{110}$$

Then the refined coordinates form a compact-induction model with irreducible fibers of dimension k, diagonal $c1$, and rank budget $2k$.

If each active range projection has full central support in the algebra generated by the refined range projections at its coordinate, then every refined support component is finite, with at most $4k$ coordinates. The original coefficient closure of any V_x is supported on at most $4n$ original coordinates. In particular, π is actually compactly induced.

Proof. Fix a summand $V_{x,a}$, and let J_a be its stabilizer in $G_x = J$. It contains Z and the open kernel of the smooth finite-dimensional representation τ, and has index n/k in J. If it had a proper nonzero J_a-invariant subspace, the orthogonal sum of its J-translates would be a proper nonzero J-invariant subspace of V_x. Thus $V_{x,a}$ is irreducible under J_a, and

$$\tau = \operatorname{Ind}_{J_a}^J V_{x,a}, \qquad \mathcal{M} = \text{c-Ind}_{J_a}^G V_{x,a}.$$

These identifications use the given literal orthogonal summands. In particular, the refined coordinate set is transitive under G. Its diagonal blocks are $c1$; the blocks between distinct summands of the same old coordinate are zero. Summing (110) over the target coordinates shows that the sum of the new rank budgets over the n/k summands at x is $2n$. Transitivity makes the budgets equal, so each is $2k$.

Theorem 21.2 makes each refined component a clique if $c \neq 1/2$, or a complete bipartite graph if $c = 1/2$. If its degree is d, then $d \leq 2k$, since every active block has positive rank. Transitivity makes the graph regular. Thus its components have $d + 1 \leq 2k + 1 \leq 4k$ vertices in the first case and $2d \leq 4k$ in the second.

The old fiber V_x is the sum of n/k refined fibers. The union of their components has at most $4n$ refined coordinates, hence at most $4n$ original coordinates. Its full coordinate space is preserved by P and by every old coordinate projection. It therefore contains the coefficient closure of V_x under all the old coefficient blocks.

For completeness, choose a refined component C and its setwise stabilizer L. The stabilizer of one refined coordinate is an open subgroup compact modulo Z; it lies in L with finite index $|C|$. Hence L is also open and

compact modulo Z. The nonzero space $W = P(\bigoplus_{(x,a)\in C} V_{x,a})$ is finite dimensional and L-stable. The image of P is the orthogonal sum of its translates over G/L. Irreducibility of that image forces W to be irreducible under L, and gives $\pi \simeq \text{c-Ind}_L^G W$. □

We will use two elementary facts about the local algebras. Write their representation on V_x as

$$V_x = \bigoplus_{a\in A_x} (H_a \otimes M_a), \qquad \mathcal{C}_x = \bigoplus_{a\in A_x} (\mathrm{End}(H_a) \otimes 1_{M_a}). \tag{111}$$

The spaces $V_{x,a} = H_a \otimes M_a$ are the minimal central blocks. The group G_x permutes them, and irreducibility of V_x makes this action transitive. Its stabilizer $G_{x,a}$ acts irreducibly on $V_{x,a}$, by the subspace argument in Lemma 22.1. If the component of a generator $Q_{x,y}$ in one such block has rank one, then $\dim M_a = 1$. This follows because the rank of any nonzero element $T \otimes 1_{M_a}$ is a positive multiple of $\dim M_a$.

Theorem 22.2. *If all the active coefficient blocks have rank one, then the original coefficient closure of each V_x is supported on at most $4n$ original coordinates, and π is actually compactly induced.*

Proof. Each $Q_{x,y}$ is supported in exactly one minimal central block of (111). Every block is encountered, by (109), and therefore every multiplicity space M_a has dimension one. Equivalently, these blocks are the spans of the components of the graph whose vertices are the range lines and whose edges join nonorthogonal lines. Different components span orthogonal spaces. Products of the rank-one projections along a chain of nonorthogonal lines give the rank-one operators between its endpoint lines, so the generated algebra on each component span is its full endomorphism algebra.

Refine to these central blocks. A coefficient block of rank one has its initial line in one source block and its final line in one target block. Thus it gives exactly one nonzero refined block, and (110) holds. The refined local range algebra is $\mathrm{End}(V_{x,a})$, so every nonzero refined range projection has full central support. Apply Lemma 22.1. □

Lemma 22.3. *Suppose that $\mathcal{C}_x = \bigoplus_a \mathrm{End}(V_{x,a})$ and every range projection has rank at most two. Call a generator internal at a if its range is contained in $V_{x,a}$. Call a generator a link if it has one range line in each of two distinct central blocks; these blocks are partners for that generator. If a has an internal generator or more than one distinct partner, then every link line at a is nonorthogonal either to an internal range at a or to a link line with a different partner.*

Proof. Call a link line at a unprobeable if it is orthogonal to every internal range at a and to every link line at a with a different partner. The collection of unprobeable lines is invariant under $G_{x,a}$. If it is nonempty, its span is $V_{x,a}$, because $G_{x,a}$ acts irreducibly there. An internal range would be orthogonal to this span and hence zero. Thus there are no internal generators at a in this case.

For each partner b, let U_b be the span of the unprobeable lines whose partner is b. The nonzero U_b are pairwise orthogonal and their sum is $V_{x,a}$. Every link line with partner b is orthogonal to $U_{b'}$ for $b' \neq b$, by the definition of an unprobeable line. It therefore lies in U_b. In particular, $U_b \neq 0$ for every partner that occurs. All generators restricted to $V_{x,a}$ preserve this orthogonal decomposition. More than one partner would contradict $\mathcal{C}_x|_{V_{x,a}} = \mathrm{End}(V_{x,a})$. Consequently an unprobeable line is possible only when there is no internal generator and there is exactly one partner, as asserted. □

Theorem 22.4. *Suppose every active coefficient block has rank at most two. Then the original coefficient closure of each V_x is supported on at most $4n$ original coordinates, and π is actually compactly induced. No transitivity on ordered edges is required.*

Proof. Suppose first that every generator $Q_{x,y}$ is supported in a single minimal central block. Refine each old coordinate into those blocks. Every old coefficient block has its entire initial space in one refined source fiber and its entire final space in one refined target fiber. It therefore produces exactly one nonzero refined coefficient block, with the same rank. The local range algebra on a refined fiber is a factor, possibly represented with multiplicity, so every nonzero range projection has full central support. Lemma 22.1 applies.

Otherwise some generator has nonzero components in two central blocks. Both components have rank one. Transitivity of G_x on the central blocks implies that every block receives a rank-one component of a generator. Hence all multiplicity spaces in (111) are one dimensional, and $\mathcal{C}_x = \bigoplus_a \mathrm{End}(V_{x,a})$. Each generator is now internal in one central block or is a link between two blocks in the sense of Lemma 22.3. Every central block has at least one partner. Transitivity on the central blocks makes their number of distinct partners constant and makes the existence of an internal generator independent of the block chosen.

Suppose first that internal generators exist, or that each block has more than one distinct partner. Fix a coefficient block $B = P_{y,x}$ whose final space is a link, and write its two incidence lines as $\ell_a \subset V_{y,a}$ and $\ell_b \subset V_{y,b}$. By Lemma 22.3, there is another neighbor z of y whose range projection $Q_{y,z}$ has no component in $V_{y,b}$ and has nonzero overlap with ℓ_a. This generator is either internal at a or links a to a partner different from b. In particular $z \neq x$. The product

$$T = P_{z,y}P_{y,x}$$

has rank one: projection of the final space of B onto $\operatorname{im} Q_{y,z}$ has rank one, and $P_{z,y}$ is injective on its initial space. Moreover, the initial line of T is $B^*\ell_a$.

Apply Lemma 21.1 to the nonzero terms in the (z,x)-block of $P^2 = P$:

$$\sum_{u\neq x,z} P_{z,u}P_{u,x} + (2c-1)P_{z,x} = 0. \tag{112}$$

When the final term is zero it is omitted. The initial spaces of these terms lie under distinct projections at x, and the final spaces lie under distinct projections at z. The lemma supplies a subspace $W \subset V_x$ reducing $\mathcal{C}_x$ such that

$$B^*\ell_a = Q_{x,y}W.$$

Since $\mathcal{C}_x$ is a sum of full endomorphism algebras with multiplicity one, W is a sum of minimal central blocks. If the rank-two projection $Q_{x,y}$ were internal in one central block, its image on W would have rank zero or two. Therefore $Q_{x,y}$ is a link, and the rank-one space $B^*\ell_a$ is one of its two source incidence lines.

The same argument probes ℓ_b. The two lines $B^*\ell_a$ and $B^*\ell_b$ are distinct because B^* is injective on its final space. Consequently B^*, and therefore B, maps the two incidence lines bijectively to the corresponding lines at the other endpoint. This also proves that an old coefficient block cannot have a link at one endpoint and an internal rank-two range at the other: apply the argument to the block or its adjoint.

Refine all coordinates into their minimal central blocks. An internal block of rank one or two produces one refined block of the same rank. A block with link ranges produces exactly two refined blocks of rank one. Thus (110) holds. The refined range projections are exactly the nonzero restrictions of the old generators to the central blocks, so their local algebra is $\operatorname{End}(V_{x,a})$. Lemma 22.1 proves the assertion.

It remains to consider the case in which no internal generators occur and every central block has exactly one distinct partner. The local graph of partners is a disjoint union of pairs, each possibly joined by several link generators. Merge the two blocks in every such pair. These merged spaces are intrinsic to the local range algebra and its generators, so they form a G-equivariant orthogonal refinement with a transitive stabilizer action. Every coefficient block has its entire initial space in one merged source fiber and its entire final space in one merged target fiber. Hence (110) holds.

On a merged fiber $V_{x,a}\oplus V_{x,b}$ the local range algebra is $\operatorname{End}(V_{x,a})\oplus\operatorname{End}(V_{x,b})$. Every active range projection has rank one in each summand, hence full central support. One final application of Lemma 22.1 completes the proof. □

Corollary 22.5. *If each $\mathcal{C}_x$ is a maximal abelian subalgebra of $\operatorname{End}(V_x)$, then the original coefficient closure of each V_x is supported on at most $4n$ original coordinates, and π is actually compactly induced. No bound on the individual block ranks is required.*

Proof. The joint eigenspaces of $\mathcal{C}_x$ are lines. Each belongs to exactly two of the range projections $Q_{x,y}$, by (109). Distinct joint lines have distinct pairs of neighbors: otherwise every generator would act identically on their two-dimensional span, contrary to maximal abelianness.

Fix $B = P_{y,x}$ and a joint eigenline ℓ in its final range. There is a unique other neighbor $z \neq x$ of y whose range contains ℓ. The intersection of the ranges of $Q_{y,x}$ and $Q_{y,z}$ is exactly ℓ, because another joint eigenline in that intersection would have the same neighbor pair. Thus $P_{z,y}B$ has rank one and initial line $B^*\ell$. Lemma 21.1, applied to (112), gives $B^*\ell = Q_{x,y}W$ for a reducing subspace W of $\mathcal{C}_x$. Such a subspace is a sum of joint eigenlines. Since $Q_{x,y}W$ has dimension one, $B^*\ell$ is one of the source joint eigenlines.

Doing this for every final joint line makes B^*, and hence B, a bijection between its initial and final joint lines, with a nonzero scalar on each. Refinement to the joint eigenlines therefore preserves every block's rank sum. Its fibers have dimension one and its local algebras are scalar, so Lemma 22.1 applies. □

Corollary 22.6. *If n is prime, then the original coefficient closure of each V_x is supported on at most $4n$ original coordinates, and π is actually compactly induced. No transitivity on ordered edges is required.*

Proof. The stabilizer G_x acts transitively on the minimal central blocks of $\mathcal{C}_x$. If there are q such blocks, each has dimension k with $qk = n$. Primality gives either $q = 1$ or $k = 1$. In the first case the local range algebra is a factor, and every nonzero range projection has full central support. Theorem 21.2 applies. In the second case $\mathcal{C}_x$ is a maximal abelian algebra, and Corollary 22.5 applies. □

No restriction on the residue characteristic occurs in these arguments. The rank bound in Theorem 22.4 concerns the ranks of the individual coefficient blocks. The common assumption $R = 2n$ remains in force throughout this section.

23. Commuting range projections with arbitrary multiplicity

Let P be an equivariant orthogonal coefficient projection with finite columns on $\mathcal{M} = \bigoplus_{x\in X} V_x = \text{c-Ind}_J^G \tau$, where τ is irreducible of dimension n, and suppose that the algebraic image of P is an irreducible representation π. Assume that $P_{x,x} = c1$, where $0 < c < 1$, and that the sum of the ranks of the off-diagonal blocks in every column is $R = 2n$. For distinct coordinates put

$$E_{x,y} = \operatorname{im} P_{x,y}, \qquad Q_{x,y} = \operatorname{pr}_{E_{x,y}}, \qquad \mathcal{C}_x = C^*(Q_{x,y} : y \neq x) \subset \operatorname{End}(V_x).$$

An inactive block has $E_{x,y} = 0$ and $Q_{x,y} = 0$. As before, Schur's lemma gives $\sum_{y\neq x} Q_{x,y} = 2\,1_{V_x}$. We assume throughout this section that every $\mathcal{C}_x$ is commutative. Its joint eigenspaces may have arbitrary dimension.

Theorem 23.1. *Under these assumptions, the coefficient closure of each original fiber V_x is supported on at most $4n$ original coordinates. There is an equivariant orthogonal refinement of the coordinates whose support components are triangles when $c \neq 1/2$ and four-cycles when $c = 1/2$. The refined fibers are irreducible under their stabilizers, and π is compactly induced from an irreducible representation of an open subgroup compact modulo Z.*

The proof uses Lemma 21.1. We give the required refinement and all the transport identities, including the comparison of the two routes around a four-cycle.

23.1. **Joint eigenspaces and the Gram operators.** On a joint eigenspace, every $Q_{x,y}$ acts by zero or one. Their sum is two, so exactly two distinct neighbors occur. For distinct $y, w \neq x$, define

$$V_x^{y,w} = E_{x,y} \cap E_{x,w}.$$

Zero spaces are omitted. These are exactly the joint eigenspaces, and

$$V_x = \bigoplus_{\{y,w\}} V_x^{y,w}, \qquad E_{x,y} = \bigoplus_{w\neq x,y} V_x^{y,w}. \tag{113}$$

Both sums are orthogonal. There cannot be two different joint eigenspaces with the same pair $\{y, w\}$, since all the generators would act identically on their sum. The group transports these decompositions. Irreducibility of V_x under G_x makes its action on the nonzero summands transitive, because the sum of any orbit of summands is an invariant subspace. Thus all the nonzero spaces in (113), at all coordinates, have one common dimension.

Lemma 23.2. *For every active block $B = P_{y,x}$, the operator B^*B preserves every joint eigenspace in V_x, and BB^* preserves every joint eigenspace in V_y.*

Proof. The diagonal equation for $P^2 = P$ is

$$\sum_{u\neq x} P_{x,u}P_{u,x} = c(1-c)1_{V_x}. \tag{114}$$

The positive summand indexed by u is supported on $E_{x,u}$. Let A, A' be different summands of (113). Their two neighbor pairs have at most one element in common. If they have no common element, every summand of (114) has zero block from A to A'. If their unique common element is u, only the summand indexed by u can have a nonzero such block. That block is nevertheless zero, because the right side of (114) has zero block from A to A'. Thus every summand in (114) preserves every joint eigenspace. This proves the assertion for B^*B. The same argument at y proves it for BB^*. □

The restriction of B is an isomorphism $E_{x,y} \to E_{y,x}$, and its adjoint is an isomorphism in the reverse direction. For a target joint eigenspace $V_y^{x,z} \subset E_{y,x}$, put

$$I_{x;y,z} = P_{x,y}V_y^{x,z} \subset E_{x,y}. \tag{115}$$

This notation is used only when x, y, z are distinct; it denotes zero if the target joint eigenspace is zero.

Lemma 23.3. *For a fixed active edge $\{x, y\}$, the spaces $I_{x;y,z}$ form an orthogonal decomposition of $E_{x,y}$. Each is a sum of its intersections with the joint eigenspaces in V_x. Consequently, for every nonzero $V_x^{y,w}$,*

$$V_x^{y,w} = \bigoplus_{z \neq x,y} \big(V_x^{y,w} \cap I_{x;y,z}\big). \tag{116}$$

Moreover $BI_{x;y,z} = V_y^{x,z}$ and $I_{x;y,z} = B^{-1}(V_y^{x,z})$, where the inverse image is taken inside $E_{x,y}$.

Proof. Consider a nonzero $V_y^{x,z}$. The initial space of the nonzero product $P_{z,y}P_{y,x}$ is

$$\operatorname{im}(P_{x,y}P_{y,z}) = P_{x,y}E_{y,z} = P_{x,y}(E_{y,x} \cap E_{y,z}) = I_{x;y,z}.$$

Here $P_{x,y}$ kills $E_{y,x}^{\perp}$, and the projections onto $E_{y,x}$ and $E_{y,z}$ commute. Apply Lemma 21.1 to the nonzero terms of the (z, x) equation for $P^2 = P$. It gives a subspace $W \subset V_x$ reducing $\mathcal{C}_x$ such that $I_{x;y,z} = Q_{x,y}W$. A reducing subspace for this commutative algebra is the sum of its intersections with the joint eigenspaces. The same is therefore true of $I_{x;y,z}$.

The target joint eigenspaces $V_y^{x,z}$ are mutually orthogonal and sum to $E_{y,x}$. Their images under B^* sum to $E_{x,y}$. They are orthogonal as well: if u, v belong to two different target joint eigenspaces, then $\langle B^*u, B^*v\rangle = \langle u, BB^*v\rangle = 0$ by Lemma 23.2. This proves the orthogonal decomposition of $E_{x,y}$. Its summands reduce every joint-eigenspace projection, so restricting it to $V_x^{y,w}$ gives (116).

Finally, BB^* is positive and invertible on $E_{y,x}$ and preserves each of its joint eigenspaces. Hence $BB^*V_y^{x,z} = V_y^{x,z}$. It follows that $BI_{x;y,z} = V_y^{x,z}$. Since B is an isomorphism between the two edge ranges, this also proves the inverse-image assertion. □

23.2. **Triangles and four-cycles.** Suppose first that $c \neq 1/2$. Fix a nonzero joint eigenspace $A = V_x^{y,w}$. Only the blocks indexed by y, w can act nontrivially on A. Restricting the (y, x) equation for $P^2 = P$ to A therefore gives

$$(2c - 1)P_{y,x}|_A + P_{y,w}P_{w,x}|_A = 0. \tag{117}$$

The first map is injective on A. Its image lies in both $E_{y,x}$ and $E_{y,w}$, by (117), so $P_{y,x}A \subset V_y^{x,w}$. The two joint eigenspaces have the same dimension, and thus equality holds. Interchanging y, w gives $P_{w,x}A = V_w^{x,y}$. Applying the same statement at these other coordinates shows that the three joint eigenspaces $V_x^{y,w}$, $V_y^{x,w}$, and $V_w^{x,y}$ form an entire refined support component. It is a triangle.

Now suppose that $c = 1/2$. For distinct x, y, w with $V_x^{y,w} \neq 0$, define the slice obtained by the route through y by

$$S_{x;y,w;z}^{(y)} = V_x^{y,w} \cap I_{x;y,z} \quad (z \neq x, y).$$

Equation (116) is an orthogonal decomposition into these slices. We next compare it with the decomposition obtained by the route through w.

Fix $z \notin \{x, y, w\}$. Restricting the (z, x) equation for $P^2 = P$ to $A = V_x^{y,w}$ leaves exactly the two possible terms

$$P_{z,y}P_{y,x}|_A + P_{z,w}P_{w,x}|_A = 0. \tag{118}$$

Either term may be zero. The initial space of the first term on V_x is $I_{x;y,z}$, which reduces the projection onto A by Lemma 23.3. Its restriction to A therefore has initial space $A \cap I_{x;y,z}$. The second term likewise has initial space $A \cap I_{x;w,z}$. The equality of the two maps up to sign in (118) makes their kernels on A equal, and hence

$$S_{x;y,w;z}^{(y)} = S_{x;y,w;z}^{(w)}. \tag{119}$$

There are no missing slices at the other two possible endpoints. At $z = w$, the (w, x) equation restricted to A has only the term $P_{w,y}P_{y,x}|_A$, and hence this term is zero. Its initial space is $A \cap I_{x;y,w}$, so $S_{x;y,w;w}^{(y)} = 0$. The equation at $z = y$ similarly gives $S_{x;y,w;y}^{(w)} = 0$.

Write the common space in (119) as $S_x^{y,w;z}$; the pair $\{y, w\}$ is unordered. We have the canonical orthogonal decomposition

$$V_x = \bigoplus_{\{y,w\}} \bigoplus_{z \notin \{x,y,w\}} S_x^{y,w;z}. \tag{120}$$

To check transport, let $B = P_{y,x}$. The inverse-image assertion in Lemma 23.3 gives

$$\begin{aligned} BS_x^{y,w;z} &= B\big(V_x^{y,w} \cap B^{-1}V_y^{x,z}\big) \\ &= BV_x^{y,w} \cap V_y^{x,z} \\ &= I_{y;x,w} \cap V_y^{x,z} = S_y^{x,z;w}. \end{aligned} \tag{121}$$

In the last equality, the route through x can be used to define the slice, by (119). The map in (121) is an isomorphism, because B is injective on $E_{x,y}$. The same argument applies to every side of the square. Thus a nonzero slice $S_x^{y,w;z}$ has exactly the following four refined coordinates in its support component:

$$S_x^{y,w;z}, \qquad S_y^{x,z;w}, \qquad S_z^{y,w;x}, \qquad S_w^{x,z;y}.$$

Their only active connections are the four sides in the order displayed. Indeed, each slice is contained in a joint eigenspace with precisely its two listed neighbor indices, and (121) sends it onto the single listed slice at each such neighbor. The four original coordinates are distinct by construction. The component is therefore exactly a four-cycle.

23.3. **Closure in the original coordinates.** In the first case use the refinement (113); in the second use (120). Each is an intrinsic orthogonal decomposition transported by G. At a fixed original coordinate the summands are permuted transitively by G_x, since otherwise the sum over one orbit would be a proper nonzero invariant subspace of the irreducible V_x. If J' stabilizes one summand $V' \subset V_x$, then V' is irreducible under J': a proper invariant subspace would give, by summing its translates over G_x/J', a proper invariant subspace of V_x. In particular the refinement is itself a compact-induction coordinate model.

Every original fiber has at most n nonzero refined summands. Each is in a component with at most four refined coordinates. The union of the components meeting V_x therefore contains at most $4n$ refined coordinates, and at most $4n$ original coordinates. The sum of these refined coordinate spaces is preserved by P and by every original coordinate projection: P preserves each component, and an original coordinate projection either fixes or kills each refined fiber. It contains V_x. Consequently it contains the closure of V_x under all the original coefficient blocks, proving the asserted bound in the original model.

Finally choose one refined support component C and put $W = P(\bigoplus_{v\in C} V'_v)$. This finite-dimensional space is nonzero and has full refined coordinate support C, since every refined diagonal block is $c1$. Its translates from different components are orthogonal, and their algebraic sum is π. The stabilizer L of W equals the setwise stabilizer of C. It contains the stabilizer of a refined coordinate with finite index, so it is open and compact modulo Z. The resulting orthogonal decomposition identifies π with $\mathrm{c\text{-}Ind}_L^G W$. Irreducibility of π forces W to be irreducible under L, completing the proof.

24. Central refinement using projections that meet one central factor

Retain the coefficient-projection setup with irreducible coordinate fibers of dimension n, diagonal $c1$, and rank budget $R = 2n$. For each coordinate x, let $Q_{x,y}$ be the projection onto $\operatorname{im} P_{x,y}$ and let $\mathcal{C}_x = C^*(Q_{x,y} : y \sim x)$. Write its representation as

$$V_x = \bigoplus_{a\in A_x} V_{x,a}, \qquad V_{x,a} = H_a \otimes M_a, \qquad \mathcal{C}_x = \bigoplus_{a\in A_x} (\mathrm{End}(H_a) \otimes 1_{M_a}).$$

For a local generator Q_i, write $S_i = \{a : Q_i|_{V_{x,a}} \neq 0\}$ and $E_{i,a} = Q_i V_{x,a}$. These definitions are made at a fixed coordinate when no coordinate index is displayed. Consider the condition

$$E_{i,a} = \sum_{\substack{j\neq i \\ S_i\cap S_j=\{a\}}} Q_i \operatorname{im} Q_j \qquad (i,\ a \in S_i). \tag{122}$$

Each summand on the right is contained in the one indicated central block.

Lemma 24.1. *Suppose* (122) *holds at every coordinate. Refinement into the minimal central blocks preserves the sum of the ranks of every old coefficient block. The refined coordinate fibers are irreducible, have a common dimension k, and have rank budget $2k$. If the old local algebra has more than one central block, then $k < n$.*

At a refined coordinate $V_{x,a}$, the new local range algebra contains the old factor $\mathrm{End}(H_a) \otimes 1_{M_a}$. *These assertions do not require* $\dim M_a = 1$. *They do not assert that the new local range algebra is a factor or that* (122) *persists.*

Proof. Fix an old edge block $B = P_{y,x}$ and a target incidence space $F_b = Q_{y,x}V_{y,b}$. For every generator $Q_{y,z}$ used in (122), put $L_z = Q_{y,x}\operatorname{im} Q_{y,z} \subset F_b$. If $L_z = 0$, omit this generator. Otherwise $P_{z,y}B$ is nonzero and has initial space B^*L_z. Lemma 21.1, applied to its off-diagonal projection equation, expresses this as $Q_{x,y}W_z$, where W_z reduces $\mathcal{C}_x$. Such a space is preserved by every central projection of $\mathcal{C}_x$. Thus B^*L_z is a direct sum of its components in the source central blocks. Taking spans in (122) gives the same assertion for B^*F_b.

Let e_a and f_b denote the source and target central projections. Since B^* is injective on F_b, the preceding direct sum gives

$$\dim F_b = \dim B^*F_b = \sum_a \dim e_aB^*F_b = \sum_a \operatorname{rank}(e_aB^*f_b).$$

The last equality holds because B^* vanishes on the orthogonal complement of F_b inside $V_{y,b}$. Sum this equality over b and take adjoints to obtain

$$\sum_{a,b} \operatorname{rank}(f_bBe_a) = \operatorname{rank} B.$$

Lemma 22.1 supplies irreducibility of the refined fibers and their rank budget $2k$. Transitivity of the stabilizer action on the old central blocks gives $k < n$ when there is more than one such block.

Finally, fix an old range projection $Q_{x,y}$ and restrict it to $V_{x,a}$. The ranges of the new blocks from coordinates over y span $Q_{x,y}V_{x,a}$: their sum is the projection to $V_{x,a}$ of the entire old range. Therefore $Q_{x,y}|_{V_{x,a}}$ is the support projection of the sum of those new range projections. In a finite-dimensional algebra this support projection is a polynomial in that positive sum, obtained by assigning value zero at zero and value one at every positive eigenvalue. It belongs to the new local range algebra. The old restricted generators are all present, so their algebra $\operatorname{End}(H_a) \otimes 1_{M_a}$ is contained there. □

Theorem 24.2. *Suppose* $\dim M_a = 1$ *for every central block and* (122) *holds at every coordinate. Then each coefficient block maps its nonzero source central incidence spaces bijectively onto its target central incidence spaces. Refinement into the minimal central blocks preserves the sum of ranks of each old block. The original coefficient closure of each* V_x *is supported on at most* $4n$ *original coordinates, and* π *is compactly induced from an irreducible finite-dimensional representation of an open subgroup compact modulo* Z*.*

Proof. Fix $B = P_{y,x}$. Its initial and final spaces decompose as

$$E = \operatorname{im} B^* = \bigoplus_a E_a, \qquad F = \operatorname{im} B = \bigoplus_b F_b,$$

where $E_a = Q_{x,y}V_{x,a}$ and $F_b = Q_{y,x}V_{y,b}$, and zero summands are omitted. The maps $B : E \to F$ and $B^* : F \to E$ are invertible.

Fix a target component F_b. A generator $Q_{y,z}$ occurring in (122) for $Q_{y,x}$ meets the central support of $Q_{y,x}$ only in b, and $z \neq x$. Put $L_z = Q_{y,x}\operatorname{im} Q_{y,z} \subset F_b$. If $L_z \neq 0$, the two-step product $P_{z,y}B$ is nonzero and has initial space

$$\operatorname{im}(B^*P_{y,z}) = B^*L_z.$$

Apply Lemma 21.1 to the nonzero terms of the (z, x) block of $P^2 = P$, including the direct term when it is nonzero. The lemma gives a reducing subspace W_z of $\mathcal{C}_x$ such that

$$B^*L_z = Q_{x,y}W_z.$$

Multiplicity one makes W_z a sum of complete central blocks. Hence B^*L_z is a sum of some of the entire source incidence spaces E_a. Taking spans over the generators in (122) shows that B^*F_b is a sum of such E_a.

The spaces F_b are a direct sum and B^* is invertible. Their images are therefore a direct sum as well. Since each image is a sum of the orthogonal spaces E_a, these sums use disjoint index sets and partition all the source indices. There is consequently a function f from source incidences to target incidences such that

$$B^*F_b = \bigoplus_{a:f(a)=b} E_a.$$

For a with $f(a) = b$, adjointness implies $BE_a \subset F_b$: its inner product with any other $F_{b'}$ is zero.

Apply the same argument to the reversed block B^*. It says that BE_a is a sum of entire target incidence spaces. Since it is nonzero and contained in $F_{f(a)}$, it must equal $F_{f(a)}$. Two distinct source incidences cannot have the same value of f, because B is injective and their images would then have nonzero intersection. Thus f is a bijection, and B restricts to an isomorphism $E_a \longrightarrow F_{f(a)}$ for every a.

Refine every coordinate into its minimal central blocks. Each old coefficient block now has one nonzero refined block for each of its source incidence spaces, and the ranks sum to $\sum_a \dim E_a = \operatorname{rank} B$. Thus rank is preserved. At a refined coordinate, the new range projections are exactly the nonzero restrictions of the old $Q_{x,y}$ to that central block. They generate its full endomorphism algebra and have full central support. The stabilizer action is transitive on central blocks by irreducibility of the original fiber, and its action on each refined fiber is irreducible. Lemma 22.1 therefore gives the stated $4n$ support bound and compact induction. □

Corollary 24.3. *Suppose that the central supports of any two distinct local range projections have at most one common central factor. Then the descent conclusions of Lemma 24.1 hold. If all multiplicities are one, the compact-induction conclusions of Theorem 24.2 hold as well. There is no bound on the individual coefficient-block ranks and no transitivity assumption on ordered edges.*

Proof. Fix Q_i and $a \in S_i$. Restrict the identity $\sum_j Q_j = 2I$ to $V_{x,a}$ and compress to $E_{i,a}$. Since Q_i is the identity on this space, it gives

$$\sum_{j \neq i} Q_{i,a} Q_{j,a} Q_{i,a} = 1_{E_{i,a}},$$

where $Q_{i,a} = Q_i|_{V_{x,a}}$. The ranges of the positive summands therefore span $E_{i,a}$. The range of $Q_{i,a}Q_{j,a}Q_{i,a}$ equals $Q_{i,a} \operatorname{im} Q_{j,a}$: this is the finite-dimensional identity $\operatorname{im}(AA^*) = \operatorname{im} A$, applied to $A = Q_{i,a}Q_{j,a}$.

Every nonzero summand has $a \in S_i \cap S_j$. By hypothesis this intersection is exactly $\{a\}$, so its range is the full space $Q_i \operatorname{im} Q_j$ appearing in (122). Thus that equation holds for every incidence. Apply Lemma 24.1, and in multiplicity one apply Theorem 24.2. □

The spanning condition is stronger than mere pairwise nonzero overlap. Its proof uses the sum of all probes at an incidence, so individual probes may have any positive rank. The unresolved configurations include central supports sharing several factors for which those factors cannot be separated by the stated spans. No matching of central components is asserted in that situation.

Corollary 24.4. *Suppose some local range projection is supported in a single minimal central block. Then* (122) *holds, so central refinement preserves the rank budget as in Lemma 24.1. If all multiplicities are one, the original closure has support at most* $4n$ *and* π *is compactly induced.*

Proof. Call a generator internal at a if its range lies in $V_{x,a}$. Transitivity on central blocks makes the family of internal generators at every block nonempty. Its range span is invariant under the stabilizer of that block. This stabilizer acts irreducibly on $V_{x,a}$, so the internal ranges span all of $V_{x,a}$.

If Q_i has more than one central factor in its support, the internal generators at $a \in S_i$ are distinct from Q_i and meet its support only in a. Their compressed ranges therefore span $Q_i V_{x,a} = E_{i,a}$. If Q_i is itself internal at a, every other generator meeting its range meets its central support only in a. Compressing $\sum_j Q_j = 2I$ to its range proves the required spanning condition just as in Corollary 24.3. □

Corollary 24.5. *Suppose each local range algebra has exactly two minimal central blocks. If every active range projection meets both blocks, full central support already gives compact induction. Otherwise central refinement preserves* $R = 2n$ *and lowers the fiber dimension to* $n/2$*. If all multiplicities are one, compact induction and the* $4n$ *support bound hold in either case.*

Proof. A nonzero projection that does not meet both blocks is internal in one of them. Apply Corollary 24.4; in the other case apply Theorem 21.2. □

Under the hypothesis of Corollary 24.3, there is also a useful orthogonality statement with arbitrary multiplicities. For every old block $B = P_{y,x}$, the operator B^*B preserves the source central blocks and BB^* preserves the target central blocks. Indeed, fix two different source blocks. At most one generator can meet both: otherwise two distinct generator supports would have two common factors. The corresponding off-diagonal central component of $\sum_y P_{x,y}P_{y,x} = c(1-c)I$ therefore has at most one nonzero term, which must be zero. Reversing the edge proves the assertion at the target. Consequently the subspaces B^*F_b for distinct target incidences are orthogonal, not merely a direct sum. This still does not assert commutation of slices arising from different old coefficient blocks.

Theorem 24.6. *Suppose any two distinct central-support sets occurring among the local range projections meet in at most one factor; several generators may have the same support set. If every central factor belongs to only one such support set, merging factors with that support gives compact induction and the* $4n$ *support bound, with arbitrary multiplicities. Otherwise* (122) *holds and gives the descent of Lemma 24.1. In multiplicity one, compact induction and the* $4n$ *bound hold in either case.*

Proof. Fix a central block V_a and, for each distinct support set S containing a, let U_S be the span in V_a of the ranges of all generators with support exactly S. These spaces are nonzero and together span V_a. Define

$$Z_S = U_S \cap \Big(\sum_{T \neq S} U_T\Big)^{\perp}.$$

Their sum is invariant under the stabilizer of V_a, which permutes the support sets containing a. If some Z_S is nonzero, irreducibility makes that sum all of V_a. The spaces Z_S are pairwise orthogonal. Every U_S is orthogonal to every Z_T with $T \neq S$, so $U_S \subseteq Z_S$; the reverse inclusion is part of the definition. Thus $V_a = \bigoplus_S U_S$, and every restricted generator preserves each summand.

This is impossible if more than one support set occurs at a. To include multiplicities explicitly, every restricted generator has the form $q \otimes 1_{M_a}$, so $U_S = H_S \otimes M_a$ for a subspace $H_S \subset H_a$. A nontrivial orthogonal decomposition into these spaces preserved by all generators would give a nontrivial invariant subspace for $\mathrm{End}(H_a)$ on H_a, a contradiction. Hence, when there is more than one support set at a, every Z_S is zero.

For a generator Q_i with support S and $a \in S$, its incidence space $E_{i,a}$ lies in U_S. The equality $Z_S = 0$ implies

$$\sum_{T \neq S} Q_i U_T = E_{i,a}.$$

Indeed the orthogonal complement of the left side inside $E_{i,a}$ is $E_{i,a} \cap (\sum_{T \neq S} U_T)^{\perp}$, which is contained in Z_S. By the hypothesis on distinct support sets, every $T \neq S$ containing a meets S only in a. Expanding the definitions of U_T proves (122).

Transitivity on central blocks makes the number of distinct support sets containing a block independent of the block. Thus either the preceding argument applies at all blocks, or every block belongs to exactly one support set. In the latter case these support sets partition the central blocks. Merge the blocks in each part. Every old coefficient block has its entire initial space in one merged source fiber and its entire final space in one merged target fiber, so rank is preserved. Each active range projection meets every central factor of its merged fiber, hence has full central support there. The merging is intrinsic and is permuted transitively by the coordinate stabilizer. Lemma 22.1 gives compact induction and the stated support bound. In the other case use Lemma 24.1, and in multiplicity one use Theorem 24.2. □

In particular, if the local algebras have multiplicity one and every range projection meets at most two minimal central blocks, compact induction follows with no bound on its rank. Distinct supports of size at most two satisfy the intersection hypothesis of Theorem 24.6.

Theorem 24.7. *For each generator* Q_i*, put* $E_i = \operatorname{im} Q_i$ *and consider the subspaces* $Q_i \operatorname{im} Q_j \subset E_i$ *for* $j \neq i$*. Suppose every central incidence space* $E_{i,a}$ *can be obtained from these subspaces by finitely many sums and intersections. Then the rank-preserving central descent and old-factor inclusion of Lemma 24.1 hold. If all multiplicities are one, the bijection of central incidences, compact induction, and the* $4n$ *support bound of Theorem 24.2 hold.*

Proof. For an edge block $B = P_{y,x}$ and any other neighbor z of y, the initial space of $P_{z,y}B$ is $B^*(Q_{y,x} \operatorname{im} Q_{y,z})$. When it is nonzero, saturation writes it as $Q_{x,y}W$ for a reducing subspace W of $\mathcal{C}_x$. It is therefore preserved by every source central projection. Sums and intersections of subspaces with this property retain it. The map B^* is invertible from the final edge range to the initial one and hence preserves both sums and intersections. The assumed finite constructions therefore make every B^*F_b preserved by source central projections. The dimension calculation in Lemma 24.1 gives rank preservation and its remaining conclusions.

In multiplicity one, each initial space supplied by saturation is a sum of complete source incidence spaces. These coordinate sums are closed under sums and intersections. Thus every B^*F_b is a sum of complete source incidences. Applying the same statement to B^* gives the reverse assertion. The bijection argument in Theorem 24.2 applies without change, and gives compact induction and the support bound. □

Corollary 24.8. *Suppose every nonzero component of a local range projection in a minimal central block has rank one. Suppose also that at each central block the range lines of any two nonzero restricted generators are nonorthogonal, and that different central blocks occur in different sets of generators. Then the original coefficient closure of every V_x is supported on at most $4n$ coordinates, and π is compactly induced. Individual edge blocks may have arbitrarily large rank and their central supports may share several factors.*

Proof. A rank-one component forces the multiplicity of its central block to be one. Every block receives such a component, so all multiplicities are one. For a central block a, let $I_a = \{j : a \in S_j\}$. Its restricted projections have rank one and sum to twice the identity. Thus $|I_a| = 2 \dim V_a$. All the central blocks have the same dimension, so the sets I_a have equal cardinality. By hypothesis they are distinct; therefore none contains another.

Fix i and $a \in S_i$. Nonorthogonality at every common central factor gives

$$Q_i \operatorname{im} Q_j = \bigoplus_{b \in S_i \cap S_j} E_{i,b} \qquad (j \neq i).$$

Consequently

$$E_{i,a} = \bigcap_{j \in I_a \setminus \{i\}} Q_i \operatorname{im} Q_j.$$

The intersection has at least one index because $|I_a| \geq 2$. It retains the line $E_{i,a}$. For any other $b \in S_i$, incomparability supplies $j \in I_a \setminus I_b$, and this j cannot equal i; its probe removes $E_{i,b}$. This proves the displayed identity. Theorem 24.7 now applies. □

25. A descent measured by the multiplicity of the local factors

Retain the coefficient-projection setup with irreducible fibers of dimension n, diagonal $c1$, and rank budget $R = 2n$. Write the local range algebra and its representation as

$$V_x = \bigoplus_{\alpha=1}^{q} (H_{x,\alpha} \otimes M_{x,\alpha}), \qquad \mathcal{C}_x = \bigoplus_{\alpha=1}^{q} (\operatorname{End}(H_{x,\alpha}) \otimes 1_{M_{x,\alpha}}).$$

The coordinate stabilizer acts transitively on the minimal central blocks. Consequently $a = \dim H_{x,\alpha}$ and $b = \dim M_{x,\alpha}$ are independent of x, α, and $n = qab$. We call b the multiplicity of the local factors.

Proposition 25.1. *Suppose every active coefficient block has rank at most $2b$. Then either the original coefficient closure of each V_x has support at most $4n$ and π is compactly induced, or refinement into the minimal central blocks gives a new coefficient model with irreducible fibers of dimension $ab < n$, the same diagonal $c1$, and rank budget $2ab$. This refinement preserves the formal-degree ratio.*

More generally, the conclusion holds when each range projection is supported in one minimal central block or is supported in exactly two such blocks and has matrix-factor rank one in each. There is no rank bound on the projections of the first kind.

Proof. Every nonzero component of a range projection has rank a positive multiple of b. The numerical hypothesis therefore implies the more general stated condition. We prove the assertion under that condition.

If $q = 1$, all range projections have full central support, so Theorem 21.2 applies. Suppose that $q > 1$. Call a generator internal if it is supported in one central block. If any internal generator exists, Corollary 24.4 gives the required refinement. We may therefore assume that every generator meets two central blocks and has range $\ell_\alpha \otimes M_{x,\alpha}$ in each, where ℓ_α is a line in $H_{x,\alpha}$. Say that these two central blocks are partners for that generator. Transitivity on the central blocks makes their number of distinct partners constant.

If each central block has exactly one partner, the pairs of partners partition the central blocks. Merge each pair. Every old coefficient block has its entire initial space in one merged source fiber and its entire final space in one merged target fiber. Thus the sum of the ranks of its refined blocks is its old rank. On a merged fiber every active range projection meets both original central blocks, so it has full central support. Lemma 22.1 gives finite closure with the stated bound and compact induction.

Suppose instead that each central block has at least two partners. Fix a central block $H \otimes M$ and a line $\ell \subset H$ belonging to a generator whose other central block is β. Call this line unprobeable if it is orthogonal to every incidence line at H whose partner differs from β. The family of subspaces $\ell \otimes M$ obtained from unprobeable lines is invariant under the stabilizer of the central block. If this family is nonempty, its span is the whole $H \otimes M$, by irreducibility of that stabilizer representation.

For each partner β, let $U_\beta \subset H$ be the span of the unprobeable lines with partner β. These spaces are mutually orthogonal and their sum is H. Every incidence line with partner β is orthogonal to U_γ for $\gamma \neq \beta$, and thus lies in U_β. In particular $U_\beta \neq 0$ for every partner. All the matrix-factor projections of the local generators preserve the resulting orthogonal decomposition of H. They generate $\mathrm{End}(H)$, so there can be only one nonzero summand. This contradicts the existence of two partners. There are therefore no unprobeable lines.

For any incidence $\ell_\alpha \otimes M_{x,\alpha}$ of a generator Q_i, choose a generator Q_j with a different partner and a nonorthogonal incidence line at α. Their central supports meet only at α, and

$$Q_i \operatorname{im} Q_j = \ell_\alpha \otimes M_{x,\alpha}.$$

Thus condition (122) holds at every incidence. Lemma 24.1 gives the refinement and the rank budget $2ab$. Since $q > 1$, its fiber dimension is strictly smaller than n.

If J' stabilizes a central block, then $[J : J'] = q$. Hence the new fiber dimension is n/q and $\mathrm{vol}(J'/Z) = \mathrm{vol}(J/Z)/q$. Their quotient, and therefore the formal-degree ratio, is unchanged. □

Lemma 25.2. *After any refinement supplied by Lemma 24.1, let $\mathcal{A}$ be the new local range algebra on an old central block $H \otimes M$, where $\dim H = a$ and $\dim M = b$. There is a unital algebra with adjoint $\mathcal{B} \subset \mathrm{End}(M)$ such that*

$$\mathcal{A} = \mathrm{End}(H) \otimes \mathcal{B}.$$

If $\mathcal{A}$ is a factor, the new coefficient model has finite support components and gives compact induction. Otherwise its common factor multiplicity b' satisfies $b' \leq b/2$.

Proof. Lemma 24.1 shows that $\mathrm{End}(H) \otimes 1$ is contained in $\mathcal{A}$. Choose matrix units (e_{ij}) of $\mathrm{End}(H)$ and write $T = \sum_{i,j} e_{ij} \otimes T_{ij}$ for $T \in \mathcal{A}$. Multiplication by the old matrix units gives

$$(e_{ki} \otimes 1)T(e_{j\ell} \otimes 1) = e_{k\ell} \otimes T_{ij} \in \mathcal{A}.$$

Summing the diagonal cases gives $1 \otimes T_{ij} \in \mathcal{A}$. Therefore, with $\mathcal{B} = \{S : 1 \otimes S \in \mathcal{A}\}$, every matrix coefficient of T belongs to $\mathcal{B}$ and $\mathcal{A} = \mathrm{End}(H) \otimes \mathcal{B}$.

The new coordinate stabilizer acts irreducibly on $H \otimes M$ and normalizes $\mathcal{A}$. It consequently permutes its minimal central blocks transitively. All the simple factors of $\mathcal{B}$, in their representation on M, have the same matrix size a' and multiplicity b'. If there are q' such factors, then

$$\mathcal{B} \simeq \bigoplus_{\nu=1}^{q'} (M_{a'}(\mathbf{C}) \otimes 1_{b'}), \qquad b = q'a'b'.$$

When $q' = 1$, the algebra $\mathcal{A}$ is a factor. Every nonzero range projection then has full central support, and Theorem 21.2 applies to the refined model. When $q' > 1$, the displayed dimension identity gives $b' \leq b/2$. □

Lemma 25.3. *Under the hypotheses of Lemma 24.1, fix one old central block $H \otimes M$. For an old edge range projection write its restriction as $q_i \otimes 1_M$. Each new range projection obtained by refining that edge has the form $q_i \otimes p_{i,\beta}$, where $p_{i,\beta}$ is the orthogonal projection onto a subspace $L_{i,\beta} \subset M$. For fixed i, the nonzero $L_{i,\beta}$ form an algebraic direct decomposition of M. If the new multiplicity algebra $\mathcal{B}$ is commutative, this decomposition is orthogonal and is a partition of its minimal central blocks.*

Proof. Use the notation of the proof of Lemma 24.1. The pullback of a target central incidence F_β is spanned by spaces $Q_i W_z$, with W_z reducing the old local algebra. On $H \otimes M$, each such reducing space is $H \otimes U_z$ for some $U_z \subset M$. Thus the component of $B^* F_\beta$ in this old central block is $\operatorname{im} q_i \otimes L_{i,\beta}$, where $L_{i,\beta} = \sum_z U_z$. This is the range of the adjoint of the corresponding refined coefficient block, so its orthogonal range projection is $q_i \otimes p_{i,\beta}$.

The target incidences are a direct sum, and B^* is injective on their sum. Their pullbacks therefore form a direct sum. Those pullbacks are graded by the source central blocks, so their components in $H \otimes M$ form a direct decomposition of $\operatorname{im} q_i \otimes M$. This gives the stated direct decomposition of M. The projection $p_{i,\beta}$ belongs to $\mathcal{B}$, by the matrix-unit argument in Lemma 25.2. If $\mathcal{B}$ is commutative, all these projections commute. Projections onto disjoint subspaces that commute are orthogonal, so they form an orthogonal decomposition. Each is a sum of minimal central projections of $\mathcal{B}$, giving the asserted partition. □

Corollary 25.4. *Suppose the multiplicity b of the local factors is at most three and condition* (122) *holds. Then the original coefficient closure of each V_x is supported on at most $4n$ original coordinates, and π is compactly induced. The same conclusion holds under the hypotheses of Proposition 25.1 when $b \leq 3$. In particular it holds when $b = 2$ and every active coefficient block has rank at most four, or when $b = 3$ and every active coefficient block has rank at most six.*

Proof. For $b = 1$, apply Theorem 24.2. For $b = 2$ or $b = 3$, refine into the old minimal central blocks, as in Lemma 24.1. Their common dimension is $k = ab$. By Lemma 25.2, the new local algebra is $\mathrm{End}(H) \otimes \mathcal{B}$, where $\mathcal{B} \subset M_b(\mathbf{C})$. If $\mathcal{B}$ is a factor, Theorem 21.2 gives finite closure of every new fiber on at most $4k$ new coordinates. Otherwise the transitivity used in that lemma gives $b = q'a'b'$ with $q' > 1$. Since b is prime, $q' = b$ and $a' = b' = 1$. Thus $\mathcal{B}$ is the diagonal algebra $\mathbf{C}^b$, and the new local factors have multiplicity one.

For $b = 2$, Corollary 24.5 gives finite closure with bound $4k$. For $b = 3$, use Lemma 25.3. For every old edge its new multiplicity projections form a partition of the three minimal central blocks of $\mathcal{B}$. At least one of these partitions is nontrivial: otherwise all the new range projections would belong to $\mathrm{End}(H) \otimes 1$, contrary to $\mathcal{B} = \mathbf{C}^3$. Every nontrivial partition of three points has a singleton part. The corresponding new range projection is therefore internal in one of the new minimal central blocks. Apply Corollary 24.4; the new factors have multiplicity one, so it gives compact induction and the same closure bound $4k$.

An old fiber is the sum of n/k new fibers, so the union of their finite closures has at most $4n$ new coordinates and hence at most $4n$ old coordinates. It is preserved by P and by the old coordinate projections, and contains the closure of the old fiber. The associated finite refined components give compact induction as in Lemma 22.1.

Finally, Proposition 25.1 either already supplies finite closure with the required bound or obtains its refinement by condition (122). The preceding argument applies in the latter case as well. □

Corollary 25.5. *Suppose condition* (122) *holds and the multiplicity b of the old local factors is a prime or is four. Then π is compactly induced. The same conclusion holds under the hypotheses of Proposition 25.1 for these values of b.*

Proof. Refine into the old minimal central blocks. This preserves the rank budget $R = 2 \dim V'$ for the new irreducible fiber V'. If the new support has at least two orbits of ordered active edges, Theorems 20.1 and 20.2 give compact induction. Otherwise it has a single orbit. All new range projections then have one common number t of minimal central blocks in their support. This number is intrinsic to the new local algebra and is preserved by the new coordinate stabilizer.

If the new algebra is a factor, apply Theorem 21.2. Suppose it is not. Write its multiplicity algebra as in Lemma 25.2, so $b = q'a'b'$ with $q' > 1$. If b is prime, then $q' = b$ and $a' = b' = 1$. The multiplicity algebra is therefore $\mathbf{C}^b$. By Lemma 25.3, every old edge gives a partition of these b central blocks. Each nonzero part is the central support of an actual new range projection and hence has size t. Therefore t divides b. Primality gives $t = 1$ or $t = b$.

When $t = b$, every new range projection has full central support, so Theorem 21.2 applies. When $t = 1$, every new coefficient block has its entire initial range in one minimal central source block and its entire final range in one minimal central target block. Refining into these central blocks preserves every coefficient rank. The restricted range projections generate the old factor on each refined fiber, so every nonzero range projection has full central support there. Lemma 22.1 again gives compact induction.

Now let $b = 4$. If $q' = 2$, then $a'b' = 2$. For $b' = 1$, the new local factors have multiplicity one and Corollary 24.5 applies. For $b' = 2$, that corollary either gives compact induction directly or gives a further central refinement by the exclusive-probe condition. Corollary 25.4 applies to that refinement, since its old multiplicity is two. The only other possibility is $q' = 4$ and $a' = b' = 1$. The multiplicity algebra is then $\mathbf{C}^4$, and the partition argument gives $t \mid 4$. The cases $t = 1, 4$ were handled above. If $t = 2$, every new range projection meets exactly two central factors, all with multiplicity one. Distinct support sets of size two meet in at most one factor, so Theorem 24.6 gives compact induction.

Finally, Proposition 25.1 either gives compact induction directly or obtains its descent from the exclusive-probe condition, to which the preceding argument applies. □

These statements give an actual occurrence-preserving descent with rank budget equal to twice the new fiber dimension. They do not prove that the hypothesis of Proposition 25.1, or the spanning condition (122), persists after a descent. The new algebra $\mathcal{B}$ may be generated by noncommuting projections coming from different old edges. Thus the multiplicity drop bounds any sequence for which the stated refinement remains available, but does not by itself establish exhaustion of all local algebras with $R = 2n$.

26. The rank boundary in fiber dimensions at most eleven

Retain the coefficient-projection setup with irreducible fibers of dimension n, irreducible image π, diagonal $c1$, and rank budget $R = 2n$.

Lemma 26.1. *Suppose $n = 10$, each local range algebra is $\bigoplus_{a=1}^{5} M_2(\mathbf{C})$ with multiplicity one, and there are five active neighbors. Suppose each range projection has rank one in four central factors and is zero in the remaining factor. Then the original coefficient closure of each fiber has support at most $4n$, and π is actually compactly induced.*

Proof. Each central block is omitted by the same number of generators, because its stabilizer action is transitive on the five blocks. There are five omissions in total, so each block is omitted once. Label the generators Q_i so that Q_i omits block i. This identifies their permutation action with the action on the five central blocks. Let $K \subset S_5$ be this transitive permutation image of the coordinate stabilizer J.

The kernel of $J \to K$ preserves each central block and commutes with every restricted range projection. It therefore acts by scalars on that block, since the restricted projections generate its full $M_2(\mathbf{C})$. The stabilizer J_a of a central block acts irreducibly on its two-dimensional space. Consequently the point stabilizer K_a cannot be cyclic: a lift of one generator would otherwise generate the image of J_a up to scalars, and an eigenline of that unitary lift would be invariant under J_a.

It follows that K contains A_5. Here is an elementary verification. A Sylow five subgroup C of K is generated by a five-cycle. If C is normal, K is contained in its normalizer, the affine group of the five points $\mathbf{F}_5$, whose point stabilizers are cyclic. Otherwise K has six Sylow five subgroups, and its order is 30, 60, or 120, since the normalizer of a five-cycle in S_5 has order 20. Order 30 is impossible: its six Sylow five subgroups already contain 24 nonidentity elements; the number of Sylow three subgroups must then be one, rather than ten. That normal subgroup of order three is centralized by a subgroup of order five, producing an element of order fifteen in S_5. The remaining orders give $K = A_5$ or $K = S_5$.

For fixed a, the group K_a acts doubly transitively on the four labels different from a. Let q_i be the rank-one restriction of Q_i to this central block, for $i \neq a$. All values $\operatorname{tr}(q_i q_j)$ with $i \neq j$ are equal. Since $\sum_{i \neq a} q_i = 2I$, compression to one line gives $1 + 3\operatorname{tr}(q_i q_j) = 2$. Thus every pair of these four lines has nonzero overlap, with squared absolute inner product $1/3$.

Fix an edge block $B = P_{y,x}$, and label its target range projection as Q_i at y. Its final space is the sum of four incidence lines F_a indexed by $a \neq i$. For every other generator Q_j at y, the preceding nonzero overlaps give

$$Q_i \operatorname{im} Q_j = \bigoplus_{a \notin \{i,j\}} F_a.$$

This is the coordinate hyperplane obtained by omitting F_j. The initial space of the corresponding two-step product is its image under B^*. Lemma 21.1 expresses that image as $Q_{x,y}W$, where W reduces the source local algebra. Multiplicity one makes W a sum of central blocks. Hence B^* sends each of the four target coordinate hyperplanes to a source coordinate hyperplane.

The map B^* is invertible between the four-dimensional edge ranges. It therefore sends distinct hyperplanes to distinct hyperplanes and preserves their intersections. Intersecting three of them isolates each target incidence line, so B^* maps the four target lines bijectively to the source lines. The same is true of B. Refinement into the central blocks consequently preserves every block rank sum. Its local range algebras are $M_2(\mathbf{C})$, and Lemma 22.1 applies. □

Corollary 26.2. *Let P be a coefficient projection with irreducible coordinate fibers of dimension n, irreducible image π, and rank budget $R = 2n$. If $n \leq 11$, then π is compactly induced from an irreducible representation of an open subgroup compact modulo Z.*

Proof. Each active neighbor orbit contributes at least n to the rank budget: its sum of column Gram operators is a nonzero positive scalar on the irreducible fiber, so the sum of the block ranks in that orbit is at least n. There are therefore at most two such orbits. With two orbits, Theorems 20.1 and 20.2 give the conclusion. Assume there is just one. Write d for its degree and s for its common block rank, so $ds = 2n$ and $s \leq n$.

Degree two gives invertible edge blocks. A cycle of length at least five or an infinite path would have nonadjacent endpoints of a two-step path with only one common neighbor, contradicting $P^2 = P$. Thus its components are triangles or squares, and finite-component extraction applies. Degree three is covered by

Theorem 21.3, and $s \leq 2$ by Theorem 22.4. Dimension one is covered by Theorem 21.2, and prime dimensions by Corollary 22.6. For composite $n \leq 11$, the factorization $ds = 2n$ therefore leaves only

$$(n, d, s) = (6, 4, 3),\ (8, 4, 4),\ (9, 6, 3),\ (10, 4, 5),\ (10, 5, 4).$$

Write the local range algebra as a sum of q factors on equal-dimensional central blocks of dimension k, so $qk = n$. If $q = 1$, apply Theorem 21.2. If the algebra is commutative, apply Theorem 23.1. We may exclude these two cases. Transitivity on neighbors makes the unordered list of ranks of a generator on the central blocks independent of that generator. Transitivity on central blocks makes the number of occurrences of each rank independent of the block.

For $n = 6$, the remaining possibilities are $(q, k) = (2, 3)$ and $(3, 2)$. In the first case a generator of rank three either meets both blocks, giving full central support, or is the identity on one block and zero on the other. The latter alternative for all generators makes the algebra commutative. For $(q, k) = (3, 2)$, the possible rank lists are $(1, 1, 1)$ and $(2, 1, 0)$. The first has full central support. The second would give four occurrences of rank two equally distributed among three blocks, which is impossible.

For $n = 8$, the remaining possibilities are $(q, k) = (2, 4)$ and $(4, 2)$. The first is treated exactly as the preceding two-block case, since $s = k$. For $(q, k) = (4, 2)$, the possible lists are $(2, 2, 0, 0)$, $(2, 1, 1, 0)$, and $(1, 1, 1, 1)$. In the first case the generators on each block are zero or its identity, so they commute. In the second case each block receives one identity and two rank-one projections. Their sum is $2I$, so the two rank-one projections are complementary and again commute. The last list gives full central support.

For $n = 9$, only $(q, k) = (3, 3)$ remains. Since the algebra is not commutative, its factor on a three-dimensional central block is $M_3(\mathbf{C})$ with multiplicity one. The possible rank lists are $(3, 0, 0)$, $(2, 1, 0)$, and $(1, 1, 1)$. The first makes the algebra commutative and the last gives full central support. In the middle case each block receives two rank-two projections and two rank-one projections. Write the first two as $I - p_1$ and $I - p_2$, and the latter as q_1, q_2, with all four p_i, q_i rank one. The tight sum of the range projections gives

$$p_1 + p_2 = q_1 + q_2.$$

Both positive sums have the same range U, of dimension at most two. Every generator on the block preserves U and $U^{\perp}$. Since U is nonzero and proper, this contradicts generation of $M_3(\mathbf{C})$.

For $n = 10$, the remaining possibilities are $(q, k) = (2, 5)$ and $(5, 2)$. The two-block algebra has multiplicity one, since it is not commutative and five is prime. Corollary 24.5 applies. For $(q, k) = (5, 2)$ and $(d, s) = (4, 5)$, a generator can have zero, one, or two rank-two components. The latter counts would give four or eight occurrences equally distributed among five blocks, which is impossible. All five components therefore have rank one, giving full central support. For $(d, s) = (5, 4)$, the possible rank lists are $(2, 2, 0, 0, 0)$, $(2, 1, 1, 0, 0)$, and $(1, 1, 1, 1, 0)$. In the first two cases the same tight-sum arguments as for $n = 8$ make the local algebra commutative. The last case is Lemma 26.1. This completes the proof. □

This is a consequence at the rank boundary $R = 2n$. It gives no bound on R for a general occurring type, and hence does not establish exhaustion for all supercuspidal representations.

27. Descent along individual coefficient orbits

Let G be a reductive local group with center Z, and let π be an irreducible unitary supercuspidal representation of formal degree d_π. Fix a Haar measure on G/Z. For an occurring irreducible representation τ of an open subgroup $J \supset Z$ compact modulo Z, put $n = \dim \tau$ and define

$$\mathfrak{a}(J, \tau) = \frac{n}{d_\pi \operatorname{vol}(J/Z)}.$$

Choose a coefficient projection P with image π on $\mathcal{M} = \text{c-Ind}_J^G \tau = \bigoplus_{x \in G/J} V_x$. Write $\mathcal{H}$ for its Hilbert completion and E_x for the coordinate projections. The diagonal of P is $c1$, where $c = \mathfrak{a}(J, \tau)^{-1}$, and its rows and columns have finite support.

Every occurring pair satisfies $\mathfrak{a}(J, \tau) \geq 1$. For a unit vector v in a copy of τ, compact-group and supercuspidal Schur orthogonality give

$$\frac{\operatorname{vol}(J/Z)}{n} = \int_{J/Z} |\langle \pi(j)v, v \rangle|^2 \, dj \leq \int_{G/Z} |\langle \pi(g)v, v \rangle|^2 \, dg = d_\pi^{-1}.$$

If equality gives $\mathfrak{a}(J, \tau) = 1$, the coefficient projection has diagonal one. The positive operator $1 - P$ then vanishes on every coordinate fiber, so $P = 1$ and $\pi \simeq \text{c-Ind}_J^G \tau$.

An active orbital means a G-orbit of ordered pairs (x,y) with $x \neq y$ and $P_{y,x} := E_y P|_{V_x} \neq 0$. Fix one such orbital $\mathcal{O}$, choose $x_0 = J$ and a neighbor y_0 in it, and put

$$H = G_{x_0} \cap G_{y_0}, \qquad d = [J : H], \qquad s = \operatorname{rank} P_{y_0,x_0}, \qquad B = ds.$$

Thus d is the number of neighbors of x_0 in this orbital, and B is its contribution to the rank sum. The orbital is self-paired if reversal preserves it.

Proposition 27.1. *For every active orbital, $B \geq n$. If the orbital is self-paired, let L be the setwise stabilizer of $\{x_0, y_0\}$. There is an L-representation of dimension s all of whose irreducible summands occur in π. For every such summand ρ, with $t = \dim \rho$, one has*

$$\mathfrak{a}(L,\rho) = \frac{dt}{2n}\mathfrak{a}(J,\tau) \leq \frac{B}{2n}\mathfrak{a}(J,\tau). \tag{123}$$

Consequently $B < 2n$ gives strict decrease of the ratio. If $B = 2n$ and $d > 2$, the pair $(\mathfrak{a}, \dim)$ decreases strictly in lexicographic order.

For any active orbital, there is an H-representation of dimension s all of whose irreducible summands occur in π. For every such summand η, with $t = \dim \eta$, one has

$$\mathfrak{a}(H,\eta) = \frac{dt}{n}\mathfrak{a}(J,\tau) \leq \frac{B}{n}\mathfrak{a}(J,\tau). \tag{124}$$

In particular, for an orbital that is not self-paired, $B = n$ and $d > 1$ give strict lexicographic decrease.

For every self-paired orbital, $B \geq 2cn$. If $B = 2cn$, the representation of L just constructed is irreducible and compactly induces π.

Proof. The outgoing column restricted to the chosen neighbor orbit is a nonzero J-map from V_{x_0} to the direct sum of its block ranges. Irreducibility of τ makes this map injective. Its target has dimension $ds = B$, proving $B \geq n$.

Put $e = \{x_0, y_0\}$ and $M = P_{y_0,x_0}$, let $V_{x,e} = \operatorname{im} M^* \subset V_{x_0}$ and $V_{y,e} = \operatorname{im} M \subset V_{y_0}$, and write $M = UA$ for its polar decomposition on these ranges. Thus A is positive definite on $V_{x,e}$ and $U : V_{x,e} \to V_{y,e}$ is unitary.

Suppose first that the orbital is self-paired. Then $[L : H] = 2$, so L is open and compact modulo Z, and

$$\operatorname{vol}(H/Z) = \frac{\operatorname{vol}(J/Z)}{d}, \qquad \operatorname{vol}(L/Z) = \frac{2\operatorname{vol}(J/Z)}{d}. \tag{125}$$

The positive polar graph is

$$W_e = \{2^{-1/2}(v + Uv) : v \in V_{x,e}\} \subset V_{x_0} \oplus V_{y_0}. \tag{126}$$

It has dimension s and is preserved by L. To verify the latter assertion, the off-diagonal operator $\left(\begin{smallmatrix} 0 & M^* \\ M & 0 \end{smallmatrix}\right)$ on $V_{x,e} \oplus V_{y,e}$ is invertible, self-adjoint, and L-equivariant. Its polar involution has W_e as its positive eigenspace.

Under the isometric graph parametrization in (126), the compression of P to W_e is $c1 + A$, which is positive definite. Since P is an orthogonal projection, $\|Pw\|^2 = \langle Pw, w\rangle \geq c\|w\|^2$ for $w \in W_e$. Hence P injects this entire L-representation into π. Every irreducible summand ρ therefore occurs, and $t \leq s$. The second volume identity in (125) proves (123). If $B = 2n$ and $d > 2$, then $t \leq s = 2n/d < n$, so either the ratio decreases or it stays equal and the dimension decreases.

The intersection move uses $V_{y,e}$ itself as an H-representation. The projection P is injective on V_{y_0} because its diagonal there is $c1$; thus every irreducible summand η of $V_{y,e}$ occurs in π. The first volume identity gives (124). When $B = n$ and $d > 1$, one has $t \leq s = n/d < n$, proving the stated decrease.

Finally choose an irreducible summand ρ of W_e. Its occurrence and $\mathfrak{a}(L,\rho) \geq 1$ imply

$$1 \leq \frac{dt}{2n}\mathfrak{a}(J,\tau) \leq \frac{B}{2cn}.$$

This proves $B \geq 2cn$. If equality holds, every irreducible summand of W_e has dimension at least $2cn/d = s$. Since $\dim W_e = s$, this representation is irreducible. Its ratio is one, so it compactly induces π by the preceding coefficient-projection criterion. □

The new coefficient model can be written directly from the finite data used in these occurrence maps. This also specifies the model to which the next orbital test applies. Write $\operatorname{Ind}_{J'}^{G,\ell^2} \rho$ for the Hilbert completion of compact induction.

Lemma 27.2. *Let J' be L or H in Proposition 27.1, and let ρ be a chosen irreducible summand of the corresponding physical graph or range space. Choose an isometric equivariant inclusion $\iota : V_\rho \hookrightarrow \mathcal{M}$. Write $t = \dim \rho$ and define*

$$\alpha 1 = \iota^* P \iota, \qquad c' = \frac{d_\pi \operatorname{vol}(J'/Z)}{t} = \frac{cn \operatorname{vol}(J'/Z)}{t \operatorname{vol}(J/Z)}.$$

Then $\alpha > 0$. Let $T : \operatorname{Ind}_{J'}^{G,\ell^2} \rho \to \mathcal{H}$ be the synthesis of the translated inclusions ι. It is bounded, equivariant, and has finite rows and columns. The new coefficient projection is

$$P' = \frac{c'}{\alpha} T^* P T. \tag{127}$$

Its image is isomorphic to π, its diagonal is $c'1$, and its rows and columns have finite support. For the intersection move, $\alpha = c$.

Proof. Schur's lemma makes $\iota^* P \iota$ scalar, and the injection established in Proposition 27.1 makes that scalar positive. The translated inclusions are supported on one vertex for the intersection move and on the two endpoints of an edge for the graph move. Every vertex is incident to finitely many such translated supports, with a uniform bound by equivariance. Hence T is bounded and has finite rows and columns.

The map $V = \alpha^{-1/2} P \iota : V_\rho \to \pi$ is isometric. For an orthonormal basis $w_1, \dots, w_t$ of its image and $v \in \pi$, Schur orthogonality gives

$$\int_{G/Z} \sum_{i=1}^{t} |\langle v, \pi(g) w_i \rangle|^2 \, dg = \frac{t}{d_\pi} \|v\|^2.$$

The integrand is invariant under right translation by J', because this subgroup acts unitarily on the span of the w_i. Dividing by $\operatorname{vol}(J'/Z)$ therefore identifies the sum over G/J' with $c'^{-1}\|v\|^2$. Equivalently, $PTT^*P = (\alpha/c')P$. Thus $(c'/\alpha)^{1/2} PT$ is a coisometry onto $P\mathcal{H}$, and (127) is its initial projection. Its diagonal is $(c'/\alpha)\iota^* P \iota = c'1$. Finite support follows by composition, and the adjoint coisometry identifies its smooth algebraic image with π. In the intersection move, ι is contained in one original coordinate fiber, so $\iota^* P \iota = c1$. □

The formula for c' uses only the old diagonal c, the two fiber dimensions, and the subgroup indices in (125).

Theorem 27.3. *Fix an initial occurring pair (J_0, τ_0), with $n_0 = \dim \tau_0$ and $a_0 = \mathfrak{a}(J_0, \tau_0)$. Let $K_1, \dots, K_h$ represent the full setwise stabilizers of the finitely many G-orbits of facets in the reduced Bruhat–Tits building. Put*

$$B_i = \left\lfloor \frac{n_0 \operatorname{vol}(K_i/Z)}{\operatorname{vol}(J_0/Z)} \right\rfloor, \qquad B_{\max} = \max_i B_i.$$

Every sequence of the preceding moves in which $(\mathfrak{a}(J, \tau), \dim \tau)$ decreases strictly in lexicographic order has at most $B_{\max} \sum_i B_i - 1$ moves. The bound applies to arbitrary choices of the occurring summands and subsequent coefficient embeddings. It does not require knowledge of d_π.

Proof. An open subgroup $J \supset Z$ compact modulo Z has a fixed point in the reduced building. It preserves the unique open facet containing that point and is therefore contained in its full setwise stabilizer. After conjugation we may assume $J \subset K_i$ for some i. These subgroups are open and compact modulo Z, so $[K_i : J]$ is finite. Unimodularity makes the volume unchanged by conjugation. Consequently

$$\mathfrak{a}(J, \tau) = \frac{[K_i : J] \dim \tau}{d_\pi \operatorname{vol}(K_i/Z)}. \tag{128}$$

The numerator $m = [K_i : J] \dim \tau$ is a positive integer. If the ratio is at most a_0, then

$$m \le \lfloor a_0 d_\pi \operatorname{vol}(K_i/Z) \rfloor = B_i.$$

Thus all such ratios belong to the finite union of the sets $\{m/(d_\pi \operatorname{vol}(K_i/Z)) : 1 \le m \le B_i\}$, of total cardinality at most $\sum_i B_i$. Moreover $\dim \tau \le m \le B_{\max}$. There are therefore at most $B_{\max} \sum_i B_i$ possible pairs of ratio and dimension below the initial ratio. Strict lexicographic decrease never repeats a pair, proving the bound. Finally, $a_0 d_\pi = n_0 / \operatorname{vol}(J_0/Z)$, which gives the stated expression for the integers B_i without a formal-degree input. □

One can test the exact dimensions in (123) and (124), beyond the sufficient bounds involving B alone. At a terminal model with no successful inducing pair, one has $c < 1$, and the following restrictions hold if every eligible constituent move has been tested: for each self-paired orbital and each irreducible summand of its positive graph, either $dt > 2n$ or $(d, t) = (2, n)$; for each orbital that is not self-paired and each irreducible summand of its range representation, either $dt > n$ or $(d, t) = (1, n)$. Indeed, the opposite strict inequalities lower the ratio, while equality lowers the dimension unless the displayed exceptional pair occurs. The bound $t \le s \le n$ rules out any other equality case.

In particular, every remaining self-paired orbital has $B \ge 2n$, with $B = 2n$ possible only for $d = 2$ and $s = n$. Every remaining orbital that is not self-paired has $B > n$ or $d = 1$ and $s = n$. Equality $B = 2cn$ at a self-paired orbital has already yielded compact induction. These residual conditions give a further extraction at the rank boundary.

Corollary 27.4. *Starting with any occurring pair, repeatedly apply a strictly decreasing constituent move whenever one is available. After at most the bound in Theorem 27.3, the procedure reaches either an inducing pair or a terminal coefficient model. If the latter has total rank sum $R \le 2n$, its finite coordinate components give an inducing pair. Thus every remaining terminal model has $R > 2n$.*

Proof. Termination has been proved. A model with no active orbital has $P = c1$, hence $c = 1$ and gives compact induction. Otherwise use the terminal restrictions above. If $R \le 2n$ and a self-paired orbital is active, there can only be one orbital, and it has $B = 2n$, $d = 2$, and $s = n$. If no self-paired orbital is active, the orbitals occur in distinct inverse pairs. Each has budget at least n. Thus there are exactly two, and the terminal condition forces $d = 1$ and $s = n$ for each.

In both cases the active support graph is a simple graph of degree two, and every edge block is an invertible map between the coordinate fibers of dimension n. Each connected component is a finite cycle or a doubly infinite path. In a path or a cycle of length at least five, choose distinct nonadjacent vertices x, y with exactly one common neighbor z. The (y, x) block of $P^2 = P$ then reads

$$0 = P_{y,x} = (P^2)_{y,x} = P_{y,z}P_{z,x},$$

contradicting invertibility. Consequently every component is a triangle or a cycle of length four.

For a component S, the finite coordinate space $\mathcal{M}_S = \bigoplus_{x \in S} V_x$ is preserved by P, and $W_S = P\mathcal{M}_S$ is nonzero. Its coordinate support is all of S, since $P_{x,x} = c1$ with $c > 0$. The spaces W_S are orthogonal, and their algebraic sum is π. The group acts transitively on the support components because it acts transitively on the coordinate vertices. Let L be the full stabilizer of W_S, equivalently of S. It is open and contains Z. The subgroup $L \cap G_x$ has finite index in L for $x \in S$, and is compact modulo Z, so L/Z is compact. The transitive orthogonal decomposition identifies π with the compact induction of the L-representation on W_S. This representation is irreducible, since a proper invariant subspace would generate a proper invariant subspace of π. □

Corollary 27.5. *Among all occurring pairs, the ratio $\mathfrak{a}$ attains a global minimum. Choose a pair of least dimension among those attaining that minimum. Every coefficient model for this pair admits no strictly decreasing orbital move. If one such model has $R \le 2n$, then π is compactly induced. Otherwise every such model has $R > 2n$.*

Proof. Choose any initial occurring pair. Formula (128) puts all ratios at most its ratio in a nonempty finite set. This set has a smallest attained value, which is the global minimum because every omitted ratio is larger than the initial one. Among the pairs attaining this minimum, the positive integer dimension has a least value. A strictly decreasing orbital move would contradict one of these two choices. Corollary 27.4 therefore applies. □

The minimum-pair statement is an existence result. The finite descent need not reach a global minimum, and the residual case $R > 2n$ has not been excluded. These results therefore give a bounded reduction of the local model rather than compact-induction exhaustion.

28. Transverse extraction with the radical retained

Fintzen's extraction theorem assumes tame splitting and $p \nmid |W|$ [10, Assumption 2.1 and Theorem 6.1]. The prime restriction enters before the finite Heisenberg representation is constructed. Lemma 3.4 identifies the centralizer of an almost stable covector with a twisted Levi subgroup. Proposition 3.12 replaces an almost

strongly stable covector by a generic one in the required coset; its proof uses saturation of coroot lattices and integral projections. The definition of the residual depth function in Theorem 6.1 also uses the existence of an almost stable representative in every coadjoint lattice coset. Finally, Lemma 5.1 and Corollary 5.3 correct transverse errors while preserving actions already extracted at previous stages. The following local argument replaces transverse correction by a calculation with its radical. It does not supply the other steps of that theorem.

Let $\mathcal{P}$ be a smooth affine parahoric group scheme over $\mathcal{O}$, with connected reductive generic fiber G. Put $K = \mathcal{P}(\mathcal{O})$, $L = \mathrm{Lie}(\mathcal{P})(\mathcal{O})$, and $K_b = \ker(\mathcal{P}(\mathcal{O}) \to \mathcal{P}(\mathcal{O}/\varpi^b\mathcal{O}))$ for $b \geq 1$. The coadjoint convention is $(g\gamma)(X) = \gamma(\mathrm{Ad}(g^{-1})X)$. For $\beta \in L^\vee$, define

$$D_\beta : L \longrightarrow L^\vee, \qquad D_\beta(Y)(X) = -\beta([Y, X]).$$

Write $R = \ker(D_\beta \otimes F)$ and $W = \mathrm{im}(D_\beta \otimes F)$. Since the commutator form is alternating, $W = \mathrm{ann}(R)$. The lattices $R_L = R \cap L$ and $W_L = W \cap L^\vee$ are saturated. Choose $\mathcal{O}$-linear complements

$$L = R_L \oplus N, \qquad L^\vee = W_L \oplus Q. \tag{129}$$

Restriction identifies Q with $R_L^\vee$. Let $\delta \geq 0$ be the largest Smith exponent of $D_\beta : N \to W_L$; equivalently, δ is the smallest nonnegative integer with $\varpi^\delta W_L \subset D_\beta(N)$. If $D_\beta = 0$, set $\delta = 0$ and $N = W_L = 0$.

Proposition 28.1. *For every integer $m \geq 2\delta + 1$ and every $C \in \varpi^m L^\vee$, there exist $g \in K_{m-\delta}$ and $Z \in \varpi^m Q$ such that*

$$\beta + C = g(\beta + Z), \qquad Z - \mathrm{pr}_Q(C) \in \varpi^{2m-2\delta} Q. \tag{130}$$

In particular, $g\beta - \beta \in \varpi^m L^\vee$. No smoothness or reductivity assumption on the stabilizer of β is required.

Proof. Smoothness of $\mathcal{P}$ gives integral formal coordinates at the identity. After identifying their tangent lattice with L, they define an analytic map $q : \varpi L \to K_1$ with $q(0) = 1$, differential the identity, and $q(\varpi^b L) = K_b$ for $b \geq 1$. The coadjoint action in these coordinates has integral power series coefficients. For $Y \in N \cap \varpi L$ and $Z \in Q \cap \varpi L^\vee$, write

$$q(Y)(\beta + Z) - \beta = \mathcal{A}(Y, Z) + \mathcal{R}(Y, Z), \qquad \mathcal{A}(Y, Z) = D_\beta(Y) + Z.$$

Every monomial in $\mathcal{R}$ has total degree at least two. The map $\mathcal{A} : (N \oplus Q) \otimes F \to L^\vee \otimes F$ is an isomorphism, and its inverse loses at most δ powers of ϖ. More precisely,

$$\mathcal{A}^{-1}(\varpi^s L^\vee) \subset \varpi^{s-\delta} N \oplus \varpi^s Q.$$

Put $\mathcal{T}(U) = C - \mathcal{R}(\mathcal{A}^{-1}U)$ on $\varpi^m L^\vee$. The integral coefficients and the absence of constant and linear terms give

$$\mathcal{R}(\mathcal{A}^{-1}U) \in \varpi^{2m-2\delta} L^\vee.$$

If $U - V \in \varpi^s L^\vee$ with $s \geq m$, the difference of two monomials contains one factor from $\mathcal{A}^{-1}(U - V)$ and at least one other factor of valuation at least $m - \delta$. Hence

$$\mathcal{T}(U) - \mathcal{T}(V) \in \varpi^{s+m-2\delta} L^\vee. \tag{131}$$

Since $m - 2\delta \geq 1$, the map $\mathcal{T}$ preserves the complete ball $\varpi^m L^\vee$ and strictly contracts it. Starting at $U_0 = C$ and setting $U_{j+1} = \mathcal{T}(U_j)$ therefore gives a convergent sequence, whose limit U satisfies $U + \mathcal{R}(\mathcal{A}^{-1}U) = C$. Write $(Y, Z) = \mathcal{A}^{-1}U$ and set $g = q(Y)$. Then $Y \in \varpi^{m-\delta} N$, $Z \in \varpi^m Q$, and the first equality in (130) holds. Since $Z = \mathrm{pr}_Q(U)$ and $U - C \in \varpi^{2m-2\delta} L^\vee$, the second assertion follows. Finally, g preserves $L^\vee$, so $g\beta - \beta = C - gZ \in \varpi^m L^\vee$. □

Fix an additive character $\psi : F \to \mathbf{C}^\times$ whose annihilator of $\mathcal{O}$ is $\mathcal{O}$. For integers $b < n \leq 2b$ and $\gamma \in L^\vee$, write

$$\theta_{\gamma;b,n}(h) = \psi\big(\varpi^{-n}\gamma(\ell_{b,n}(h))\big), \qquad \ell_{b,n} : K_b/K_n \simeq \varpi^b L/\varpi^n L.$$

The map $\ell_{b,n}$ is the canonical square-zero linearization and is K-equivariant. Distinct characters are parametrized by $L^\vee/\varpi^{n-b}L^\vee$.

Corollary 28.2. *Suppose $\theta_{\beta+C;a,n}$ occurs in a smooth representation π of $G(F)$, where $a < n \leq 2a$, $C \in \varpi^m L^\vee$, and $m \geq 2\delta + 1$. Choose g, Z as in Proposition 28.1. Then $\theta_{\beta+Z;a,n}$ occurs in π, and*

$$I_{G(F)}(\theta_{\beta+Z;a,n}) = g^{-1} I_{G(F)}(\theta_{\beta+C;a,n}) g. \tag{132}$$

Conjugation also gives an isomorphism of their Hecke algebras that sends support in $K_a x K_a$ to support in $K_a g^{-1} x g K_a$.

Proof. The element g lies in K and normalizes K_a and K_n. Equivariance of $\ell_{a,n}$ and (130) give

$$\theta_{\beta+C;a,n}(h) = \theta_{\beta+Z;a,n}(g^{-1}hg).$$

Thus $\pi(g^{-1})$ sends a nonzero vector of the first character to a vector of the second. Conjugating both subgroups in the definition of an intertwiner proves (132). Pullback under $x \mapsto gxg^{-1}$ gives the Hecke algebra isomorphism. It preserves convolution because inner conjugation by $g \in K$ preserves Haar measure. □

There is also a refinement statement in which the finer character is selected from the representation before the correction is made.

Proposition 28.3. *Let a, A, n be positive integers with $a \le A < n \le 2a$, and put $m = n - A$. Suppose $m \ge 2\delta + 1$. Every smooth representation containing $\theta_{\beta;A,n}$ contains a character $\theta_{\beta+Z;a,n}$ for some $Z \in \varpi^m Q$.*

More precisely, let $\mathcal{E}$ be the finite set of extensions of $\theta_{\beta;A,n}$ to K_a. Write each extension as $\theta_{\lambda;a,n}$ with $\lambda \in \beta + \varpi^m L^\vee$, and choose g_λ, Z_λ by Proposition 28.1. Then there is an isomorphism of smooth $G(F)$-representations

$$\text{c-Ind}_{K_A}^{G(F)} \theta_{\beta;A,n} \simeq \bigoplus_{\lambda \in \mathcal{E}} \text{c-Ind}_{K_a}^{G(F)} \theta_{\beta+Z_\lambda;a,n}. \tag{133}$$

The summands retain their labels even if some are isomorphic.

Proof. The congruence commutator estimate gives $[K_a, K_A] \subseteq K_{a+A} \subseteq K_n$, since $a + A \ge 2a \ge n$. Consequently K_a preserves the $\theta_{\beta;A,n}$-isotypic subspace. The group K_n acts trivially there, and K_a/K_n is finite abelian because $n \le 2a$. Any nonzero such subspace therefore contains a character of K_a/K_n extending $\theta_{\beta;A,n}$. Under square-zero linearization, extension is exactly the condition $\lambda - \beta \in \varpi^{n-A} L^\vee = \varpi^m L^\vee$. Corollary 28.2 now gives the first assertion.

Finite abelian induction gives

$$\text{Ind}_{K_A}^{K_a} \theta_{\beta;A,n} = \bigoplus_{\lambda \in \mathcal{E}} \theta_{\lambda;a,n},$$

with every extension occurring once. Transitivity of compact induction and conjugation by each g_λ give (133). □

The hypothesis $m \ge 2\delta + 1$ is a condition on the character already under consideration. Passing to a finer character does not increase this prescribed coarse precision. The propositions do not justify replacing an occurring character by an arbitrary chosen refinement. They retain an actual constituent and then conjugate it. Moreover, Q need not be preserved by the stabilizer of β. Thus (133) does not identify its summands with representations of that stabilizer or with a smaller reductive group. In particular, it does not replace the assertion about previously extracted subgroup actions in [10, Lemma 5.1(b)(iii)].

The retained coordinates have a precise Lie algebra interpretation. The space R is a Lie subalgebra: for $r_1, r_2 \in R$ and $X \in \text{Lie}(G)$, Jacobi gives

$$\beta([[r_1, r_2], X]) = \beta([r_1, [r_2, X]]) - \beta([r_2, [r_1, X]]) = 0.$$

Moreover, $\beta([R, R]) = 0$, and hence

$$(\beta + Z)([r_1, r_2]) = Z([r_1, r_2]) \qquad (r_1, r_2 \in R). \tag{134}$$

Since R_L is closed under brackets and $Z \in \varpi^m Q$, the functional $\varpi^{-m} Z|_{R_L}$ induces a functional and an alternating commutator form on the finite Lie algebra $R_L/\varpi^s R_L$, for every $s \ge 1$. This is residual finite data, but no group integrating that finite Lie algebra or comparison with a subgroup commutator quotient follows from (134) alone. If $D_\beta = 0$, then $R = L \otimes F$, $Q = L^\vee$, and the normal form is simply $g = 1$, $Z = C$; there is no reduction.

The exact infinitesimal stabilizer requires a further correction to this residual form. The correction can be computed without an exponential map.

Lemma 28.4. *Let $Z \in \varpi^m Q$ with $m > \delta$, and put $\gamma = \beta + Z$. Relative to $L = N \oplus R_L$, write the alternating commutator form $B_\gamma(X, Y) = \gamma([X, Y])$ in blocks*

$$B_\gamma = \begin{pmatrix} A_\gamma & C_\gamma \\ E_\gamma & D_\gamma \end{pmatrix}.$$

The block A_γ is invertible over F, and its inverse loses at most δ powers of ϖ. The form

$$S_\gamma = D_\gamma - E_\gamma A_\gamma^{-1} C_\gamma \tag{135}$$

is alternating in every characteristic. Its radical identifies with that of B_γ by the graph map

$$y \longmapsto -A_\gamma^{-1} C_\gamma y + y, \qquad \ker B_\gamma \simeq \ker S_\gamma. \tag{136}$$

Moreover,

$$\operatorname{rank}_F B_\gamma = \dim_F (N \otimes F) + \operatorname{rank}_F S_\gamma, \tag{137}$$

$$S_\gamma(r_1, r_2) - Z([r_1, r_2]) \in \varpi^{2m-\delta}\mathcal{O} \qquad (r_1, r_2 \in R_L). \tag{138}$$

When $N = 0$, these statements mean $S_\gamma = D_\gamma$ and the graph map is the identity.

Proof. The form B_β is nondegenerate on $N \otimes F$ because its radical is R. Restriction identifies W_L with $N^\vee$, so A_β^{-1} loses at most δ powers of ϖ. Since $A_\gamma - A_\beta \in \varpi^m \operatorname{Hom}_{\mathcal{O}}(N, N^\vee)$, the operator $A_\beta^{-1}(A_\gamma - A_\beta)$ has entries in $\varpi^{m-\delta}\mathcal{O}$. Its geometric series gives the claimed inverse bound for A_γ.

Solving the first block equation $A_\gamma x + C_\gamma y = 0$ gives $x = -A_\gamma^{-1} C_\gamma y$. The second block equation is then $S_\gamma y = 0$, proving (136). The change of variables $x = x' - A_\gamma^{-1} C_\gamma y$ changes the bilinear form by congruence to $A_\gamma \oplus S_\gamma$. This proves alternation and (137); no division by two is involved. Finally, the off-diagonal blocks of B_β vanish and its $R \times R$ block is zero. Thus $C_\gamma, E_\gamma, D_\gamma$ have entries in $\varpi^m\mathcal{O}$, and $D_\gamma(r_1, r_2) = Z([r_1, r_2])$. The inverse bound gives (138). □

Thus the Lie bracket on R gives the leading residual commutator form, but the exact form controlling the next infinitesimal stabilizer is S_γ. Its correction term matters at finer precision. This calculation does not identify the full stabilizer group or construct a representation of it.

Proposition 28.5. *Let $F = \mathbf{F}_2((t))$ and $G = \mathrm{PGL}_2$. The trace pairing identifies $\mathrm{Lie}(G)^*$ with $\mathfrak{sl}_2$. Its covectors with geometrically closed coadjoint orbit are exactly the scalar matrices. For*

$$A = \begin{pmatrix} 0 & 1 \\ t & 0 \end{pmatrix}, \qquad E = F[s]/(s^2 - t),$$

every coset $A + t^m\mathfrak{sl}_2(\mathcal{O})$ with $m \geq 1$ misses this locus. The stabilizer of A is the smooth group $(\operatorname{Res}_{E/F} \mathbf{G}_m)/\mathbf{G}_m$. It becomes $\mathbf{G}_a$ over an algebraic closure and has compact F-points. The covector A is rationally elliptic. If $m \geq 2$, its entire coset $A + t^m\mathfrak{sl}_2(\mathcal{O})$ consists of rationally elliptic covectors.

Proof. The annihilator of FI under the matrix trace pairing is $\mathfrak{sl}_2$, which gives the stated coadjoint identification. Over an algebraic closure, a trace-zero matrix has characteristic polynomial $T^2 + d = (T + z)^2$. It is therefore $zI + N$ with $N^2 = 0$. If $N \neq 0$, conjugate N to E_{12}; conjugation by $\operatorname{diag}(u, 1)$ then has limit zI as $u \to 0$. This scalar matrix is outside the original orbit, which is therefore not closed. Scalar orbits are points. The upper-right entry in the displayed coset is a unit, so the coset contains no scalar matrix.

The cyclic matrix A has matrix centralizer $F[A] = E$. Its group centralizer in GL_2 is $\operatorname{Res}_{E/F} \mathbf{G}_m$, so its coadjoint stabilizer in PGL_2 is the stated quotient. It is the open subvariety

$$\{[u : v] \in \mathbf{P}^1_F : u^2 + tv^2 \neq 0\}$$

and hence is smooth. Over an algebraic closure, $E \otimes_F \overline{F} \simeq \overline{F}[\epsilon]/(\epsilon^2)$; its units modulo scalars form $\mathbf{G}_a$. The F-points are $E^\times/F^\times$. The quotient of valuation groups has order two, and the quotient of unit groups is compact. Hence $E^\times/F^\times$ is compact.

A proper F-parabolic of PGL_2 stabilizes an F-line in F^2. In a basis adapted to that line, the trace pairing of a matrix B with the Lie algebra of its unipotent radical vanishes exactly when the lower-left entry of B is zero, or equivalently when B preserves the line. Since $T^2 - t$ is irreducible, A has no such line. For $B \in A + t^m\mathfrak{sl}_2(\mathcal{O})$ with $m \geq 2$, write

$$B = \begin{pmatrix} t^m z & 1 + t^m v \\ t + t^m w & t^m z \end{pmatrix}.$$

Then

$$\det B = t + t^m w + t^{m+1} v + t^{2m}(z^2 + vw)$$

has valuation one and is not a square in F. Thus B has no F-eigenline either, proving the last assertion. □

In this example the radical correction has an explicit form. Put $H = \operatorname{diag}(1,0) \bmod FI$, $e = E_{12}$, and $f = E_{21}$ in $\mathfrak{pgl}_2$. Then $D_A(H) = A$, $D_A(f) = I$, and $D_A(e) = tI$. Thus one can take $N = \mathcal{O}H \oplus \mathcal{O}f$, $R_L = \mathcal{O}(e + tf)$, $W_L = \mathcal{O}I \oplus \mathcal{O}A$, and $Q = \mathcal{O}E_{21}$ in the coadjoint lattice. Here $\delta = 0$. For $B = \left(\begin{smallmatrix} a & b \\ c & a \end{smallmatrix}\right)$ close to A, the entry b is a unit and direct multiplication gives

$$B = PA_dP^{-1}, \qquad P = \begin{pmatrix} b & 0 \\ a & 1 \end{pmatrix}, \qquad A_d = \begin{pmatrix} 0 & 1 \\ d & 0 \end{pmatrix}, \qquad d = a^2 + bc. \tag{139}$$

If $B - A \in t^m\mathfrak{sl}_2(\mathcal{O})$, then $P \in K_m$ and $A_d - A \in t^mQ$. The nonlinear term a^2 is retained in the radical coordinate. The stabilizer varies with d.

There is no invariant complementary line that would turn this chart into descent through the fixed stabilizer of A. Indeed, for $g = uI + vA$, $D = u^2 + tv^2 \neq 0$, and $B = \left(\begin{smallmatrix} a & b \\ c & a \end{smallmatrix}\right)$, one has

$$gBg^{-1} = B + (c + tb)\frac{uvI + v^2A}{D}.$$

The stabilizer acts trivially on $\mathfrak{sl}_2/(FI + FA)$, whereas its fixed subspace in $\mathfrak{sl}_2$ is exactly $FI + FA$. An invariant complement would map isomorphically to this trivial quotient and hence consist of fixed vectors, which is impossible. This is also an obstruction to an invariant smooth transverse slice, by taking tangent spaces.

Proposition 28.5 identifies an exact failure in the definition of the residual depth function in [10, proof of Theorem 6.1]: the asserted almost stable representative need not exist in the prescribed coset. Rational ellipticity still applies to these covectors and gives compact transporter sectors by the preceding results. For noncompact stabilizers, the transverse argument above supplies neither a parahoric filtration on the stabilizer nor the overlap identities needed for a further induction step. Allowing smooth nonreductive stabilizers therefore requires additional representation theory. The finite Clifford enlargement of [2, Theorem 4.6.8] assumes tame splitting and a residue field of cardinality greater than three, and retains the first genericity condition. Likewise, [3, Lemma 3.4 and Theorem 3.5] uses supergenericity. Neither result provides these missing statements for the stabilizers considered here.

29. Rational cocharacter closure and Levi subgroups

Let $\mathbf{G}$ be a connected reductive group over F, put $G = \mathbf{G}(F)$, and set $\mathfrak{g} = \operatorname{Lie}(\mathbf{G})(F)$. A G-orbit in $\mathfrak{g}^*$ is *cocharacter-closed over F* if, whenever an F-defined cocharacter $\lambda : \mathbf{G}_m \to \mathbf{G}$ has a limit $\lim_{z\to 0} \lambda(z)\beta$, that limit belongs to $G\beta$. This is a condition on rational cocharacters and rational conjugacy.

For an F-split torus $S \subset \mathbf{G}$, put $\mathbf{L} = C_{\mathbf{G}}(S)$ and $\mathfrak{l} = \operatorname{Lie}(\mathbf{L})(F)$. The S-weight decomposition gives a canonical identification

$$(\mathfrak{g}^*)^S = \iota_{\mathbf{L}}(\mathfrak{l}^*), \qquad \iota_{\mathbf{L}}(\eta)|_{\mathfrak{l}} = \eta, \qquad \iota_{\mathbf{L}}(\eta)|_{\mathfrak{g}_\chi} = 0 \quad (\chi \neq 0). \tag{140}$$

Here $\mathbf{L}$ is an F-Levi subgroup of $\mathbf{G}$. Write $\mathcal{E}_{\mathbf{L}}$ for the rationally elliptic covectors in $\mathfrak{l}^*$, defined by (38) with $\mathbf{G}$ replaced by $\mathbf{L}$.

Proposition 29.1. *Suppose that $G\beta$ is cocharacter-closed over F. Choose an F-split torus S of maximal dimension among the F-split tori that fix β, and put $\mathbf{L} = C_{\mathbf{G}}(S)$. Then $\beta = \iota_{\mathbf{L}}(\beta_{\mathbf{L}})$ for a covector $\beta_{\mathbf{L}} \in \mathcal{E}_{\mathbf{L}}$.*

Conversely, for any F-Levi subgroup $\mathbf{L}$ and any $\eta \in \mathcal{E}_{\mathbf{L}}$, the orbit $G\iota_{\mathbf{L}}(\eta)$ is cocharacter-closed over F. No smoothness assumption on the scheme stabilizer of β is required.

Proof. Such a torus S exists: the trivial torus fixes β, and the dimensions of F-split tori in $\mathbf{G}$ are bounded. Since S fixes β, (140) gives $\beta_{\mathbf{L}}$. By [15, Theorem 5.4(i)], the orbit $\mathbf{L}(F)\beta$ is cocharacter-closed over F.

Suppose that $\beta_{\mathbf{L}}$ vanishes on $\operatorname{Lie}(U_P)(F)$ for a proper F-parabolic $P \subset \mathbf{L}$. Choose an F-defined cocharacter λ of $\mathbf{L}$ with $P = P_\lambda(\mathbf{L})$. The positive λ-weight spaces in $\mathfrak{l}$ are $\operatorname{Lie}(U_P)(F)$, so the negative weight components of $\beta_{\mathbf{L}}$ vanish. Therefore

$$\beta' = \lim_{z\to 0} \lambda(z)\beta$$

exists. The equivariance of $\iota_{\mathbf{L}}$ justifies this limit in $\mathfrak{g}^*$ as well as in $\mathfrak{l}^*$. By [15, Corollary 5.1], applied to $\mathbf{L}$, there is $u \in U_P(F)$ with $\beta' = u\beta$. Both S and λ fix β'. Moreover, u centralizes S, since $S \subset Z(\mathbf{L})$. Consequently S and $u^{-1}\lambda u$ fix β. They commute and generate an F-split torus. This torus strictly contains S: otherwise $u^{-1}\lambda u$ would take values in S, hence λ would take values in the central torus S, contrary to $P \neq \mathbf{L}$. This contradicts the choice of S and proves $\beta_{\mathbf{L}} \in \mathcal{E}_{\mathbf{L}}$.

For the converse, choose an F-split torus S with $\mathbf{L} = C_{\mathbf{G}}(S)$ and set $\beta = \iota_{\mathbf{L}}(\eta)$. If a noncentral F-defined cocharacter λ of $\mathbf{L}$ had a limit on η, the same weight calculation would give $\eta|_{\mathrm{Lie}(U_{P_\lambda(\mathbf{L})})(F)} = 0$. The parabolic $P_\lambda(\mathbf{L})$ would be proper, contradicting rational ellipticity. Central cocharacters fix every covector in $\mathfrak{l}^*$. Thus $\mathbf{L}(F)\eta$ is cocharacter-closed over F, and [15, Theorem 5.4(ii)] gives the same property for $G\beta$. □

In particular, the covectors with cocharacter-closed rational orbit form exactly the set

$$\bigcup_{\mathbf{L}} G\iota_{\mathbf{L}}(\mathcal{E}_{\mathbf{L}}), \tag{141}$$

where $\mathbf{L}$ ranges over the F-Levi subgroups. This equality does not assert that the union is dense in $\mathfrak{g}^*$. The rational Hilbert–Mumford theorem [15, Theorem 1.3] produces a cocharacter-closed orbit by a limit; it does not put that orbit inside a prescribed open lattice coset.

The Levi subgroup $\mathbf{L}$ in Proposition 29.1 is the centralizer of S, not the stabilizer $\mathbf{G}_\beta$. For example, when $\beta = 0$, one may take S maximal F-split in $\mathbf{G}$. Then $\mathbf{L}$ is a minimal F-Levi, whereas $\mathbf{G}_\beta = \mathbf{G}$; these differ whenever $\mathbf{G}$ has positive relative semisimple rank. The proposition therefore supplies an algebraic description of the covector. It does not replace the full stabilizer by $\mathbf{L}$ in an intertwining calculation or give an occurrence-preserving reduction of a representation to $\mathbf{L}(F)$.

Proposition 29.2. *Let $F = \mathbf{F}_2((t))$, $E = F[s]/(s^2 - t)$, and $\mathbf{G} = \mathrm{PGL}_4$. Under the trace-pairing identification $\mathfrak{g}^* = \mathfrak{sl}_4(F)$, put*

$$B = \begin{pmatrix} 0 & 1 \\ t & 0 \end{pmatrix}, \qquad A = \begin{pmatrix} B & 0 \\ 0 & B \end{pmatrix}.$$

The orbit GA is cocharacter-closed over F. Its scheme stabilizer is the smooth connected group

$$\mathbf{H} = (\mathrm{Res}_{E/F}\,\mathrm{GL}_2)/\mathbf{G}_m. \tag{142}$$

It contains the central smooth connected F-unipotent subgroup

$$\mathbf{U} = (\mathrm{Res}_{E/F}\,\mathbf{G}_m)/\mathbf{G}_m, \qquad \mathbf{U}_{\overline{F}} \simeq \mathbf{G}_{a,\overline{F}}. \tag{143}$$

In particular, $\mathbf{H}$ is not pseudo-reductive. On rational points,

$$\mathbf{U}(F) = E^\times/F^\times = Z(\mathbf{H}(F)), \qquad \mathbf{H}(F)/Z(\mathbf{H}(F)) \simeq \mathrm{PGL}_2(E). \tag{144}$$

The first group is compact and the last group is noncompact.

Proof. Since t is not a square in F, the polynomial $T^2 - t$ is irreducible. The minimal polynomial of A is $T^2 - t$, and F^4 with T acting by A is the semisimple $F[T]$-module $E \oplus E$. Thus [15, Proposition 10.1] shows that the $\mathrm{GL}_4(F)$-orbit of A is cocharacter-closed over F.

The trace of A is zero. The nondegenerate trace pairing on $\mathfrak{gl}_4(F)$ identifies the dual of $\mathfrak{gl}_4(F)/FI_4 = \mathfrak{g}$ with the annihilator of FI_4, namely $\mathfrak{sl}_4(F)$, and carries the coadjoint action to matrix conjugation. The map $\mathrm{GL}_4(F) \to \mathrm{PGL}_4(F)$ is surjective, so their orbits on this space agree. An F-cocharacter of PGL_4 is rationally conjugate into its diagonal split torus. Cocharacters of that torus lift to the diagonal torus of GL_4, and the conjugating element also lifts on rational points. Hence every cocharacter limit for the PGL_4 action is a cocharacter limit for the GL_4 action. This proves the first assertion.

For every F-algebra R, the endomorphisms of R^4 commuting with A are precisely the endomorphisms of the free rank-two $(E \otimes_F R)$-module $E^2 \otimes_F R$. Their units form $\mathrm{GL}_2(E \otimes_F R)$. Thus the scheme centralizer of A in GL_4 is $\mathrm{Res}_{E/F}\,\mathrm{GL}_2$. Taking its quotient by the scalar $\mathbf{G}_m$ proves (142); this is the stabilizer of the matrix A itself, as required by the coadjoint identification. Weil restriction along the finite field extension preserves smoothness, and the quotient by the smooth central torus is smooth. These groups are geometrically connected.

Scalar multiplication by $E^\times$ gives the central subgroup (143). Over $\overline{F}$ one has $E \otimes_F \overline{F} \simeq \overline{F}[\epsilon]/(\epsilon^2)$. Every unit of this algebra has a unique expression $a(1 + b\epsilon)$, with $a \neq 0$. Modulo the constant units, multiplication becomes addition of b. This proves $\mathbf{U}_{\overline{F}} \simeq \mathbf{G}_a$ and shows that $\mathbf{U}$ is a nontrivial smooth connected F-unipotent normal subgroup of $\mathbf{H}$.

Hilbert's theorem 90 for $\mathbf{G}_m$ gives $\mathbf{H}(F) = \mathrm{GL}_2(E)/F^\times$ and $\mathbf{U}(F) = E^\times/F^\times$. The latter is compact: normalized valuations give $v_E(F^\times) = 2\mathbf{Z}$, and the unit quotient $\mathcal{O}_E^\times/\mathcal{O}_F^\times$ is compact. The natural surjection

$$\mathrm{GL}_2(E)/F^\times \longrightarrow \mathrm{PGL}_2(E)$$

has kernel $E^\times/F^\times$. This kernel is central, and $\mathrm{PGL}_2(E)$ has trivial center, so it is the full center of $\mathbf{H}(F)$. The quotient is noncompact, as is seen from the diagonal classes $\mathrm{diag}(s^r,1)$ as r tends to infinity. This proves (144) and the remaining assertions. □

This example has a noncompact residual stabilizer even though its unipotent central subgroup has compact rational points. Consequently cocharacter closure alone does not place stabilizers in the class of reductive or pseudo-reductive groups to which a further descent theorem could be applied.

30. Finite bounds and comparison of compact inducing data

Let F be a nonarchimedean local field, let G be a connected reductive F-group, and put $Z = Z(G)(F)$. All representations have complex coefficients. The statements in this section impose no restriction on the characteristic, the residue characteristic, or the ramification of G. We first give a quantitative version of coadjoint properness. We then compare arbitrary supplied compact inducing pairs, including pairs obtained from finite cuspidal quotients or from different central extensions and twists.

Choose a maximal F-split torus S, a positive relative root system, and its set Δ of simple roots. Put $M = Z_G(S)$ and let A_G be the maximal F-split central torus. Fix a uniformizer ϖ. The special-parahoric Cartan decomposition [4] gives $G(F) = K_0M(F)K_0$ for a suitable compact open subgroup K_0. The quotient $M(F)/S(F)$ is compact: the group M/S is anisotropic, and the split kernel S has trivial first Galois cohomology. Consequently a finite set of elements of $M(F)$, together with $S(F)$ and $M(F)\cap K_0$, covers $M(F)$. Absorbing the compact subgroup $S(\mathcal{O}_F)$ and choosing finitely many representatives of the relative Weyl group gives fixed compact sets C_1, C_2, each a finite union of translates of K_0, and a finite set $D \subset M(F)$ such that

$$G(F) = \bigcup_{d\in D} C_1\mathcal{A}^+dC_2, \qquad \mathcal{A}^+ = \{\lambda(\varpi) : \lambda \in X_*(S), \langle\alpha,\lambda\rangle \geq 0 \text{ for all } \alpha\in\Delta\}. \tag{145}$$

The use of C_1, C_2 does not require the chosen Weyl representatives to belong to the original parahoric. The common kernel of the relative roots on $X_*(S)$ is $X_*(A_G)$. Thus a bounded set of simple-root values contains only finitely many cocharacter classes modulo $X_*(A_G)$.

30.1. **A bound for mixed coadjoint transporters.** Write $\mathfrak{g} = \mathrm{Lie}(G)(F)$. A covector is rationally elliptic if it does not annihilate the Lie algebra of the unipotent radical of any proper F-parabolic. Denote this locus by $\mathcal{E}$. For every positive relative root γ, choose an $\mathcal{O}_F$-basis $X_{\gamma,1},\dots,X_{\gamma,d_\gamma}$ of a lattice in $\mathfrak{g}_\gamma$. For $\alpha\in\Delta$, let U_α be the unipotent radical of the standard maximal parabolic whose Levi omits α. Its Lie algebra is the sum of the positive root spaces for which the α-coefficient is positive. Let $\mathcal{X}_\alpha$ be the resulting set of basis vectors.

Proposition 30.1. *Let Ω_1,Ω_2 be compact subsets of $\mathcal{E}$, and put*

$$\mathcal{A} = C_1^{-1}\Omega_1, \qquad \mathcal{B} = \bigcup_{d\in D} dC_2\Omega_2.$$

There are nonnegative integers c_1, c_2 such that

$$v_F(\gamma(X_{\eta,j})) \geq -c_1 \qquad (\gamma\in\mathcal{A},\ \eta>0), \tag{146}$$

$$\min_{X\in\mathcal{X}_\alpha} v_F(\delta(X)) \leq c_2 \qquad (\delta\in\mathcal{B},\ \alpha\in\Delta). \tag{147}$$

If $g = c_1'adc_2'$ is a decomposition in (145)*, with $a=\lambda(\varpi)$, and $\Omega_1\cap g\Omega_2\neq\varnothing$, then*

$$0 \leq \langle\alpha,\lambda\rangle \leq c_1+c_2 \qquad (\alpha\in\Delta). \tag{148}$$

In particular, these mixed transporters lie in a bounded finite union of Cartan sets modulo Z.

Proof. The evaluation of a fixed vector on a compact set of covectors is bounded, which gives (146). For each α, restriction to $\mathrm{Lie}(U_\alpha)$ is nowhere zero on the compact set $\mathcal{B}\subset\mathcal{E}$. Its norm therefore has a positive minimum. There are finitely many α, giving (147) with a common c_2. We use the convention $v_F(0)=+\infty$.

Choose $\beta_i\in\Omega_i$ with $\beta_1 = g\beta_2$, and set $\gamma = (c_1')^{-1}\beta_1$ and $\delta = dc_2'\beta_2$. Then $\gamma = a\delta$. For $X\in\mathfrak{g}_\eta$, the coadjoint convention $(a\delta)(X) = \delta(\mathrm{Ad}(a^{-1})X)$ gives

$$\delta(X) = \eta(a)\gamma(X). \tag{149}$$

Suppose $\langle\alpha,\lambda\rangle > c_1+c_2$. Every positive root η occurring in $\mathrm{Lie}(U_\alpha)$ has a positive integral α-coefficient and nonnegative coefficients at the other simple roots. Dominance and (149) imply $v_F(\delta(X)) > c_2$ for every

$X \in \mathcal{X}_\alpha$, contrary to (147). This proves (148). The final assertion follows from the finiteness of the cocharacter classes in this box modulo $X_*(A_G)$. The argument also treats nonreduced relative root systems: it uses the actual root characters and their integral coefficients, not their differentials. □

When the Ω_i are cosets of given lattices, the bound has a finite computational form. A lattice containing $\mathcal{A}$ supplies c_1. To find c_2, increase an integer q and compute, for every α, the finite image of the restrictions of $\mathcal{B}$ to $\mathrm{Lie}(U_\alpha)$ modulo ϖ^{q+1}. A common initial negative power of ϖ accommodates nonintegral values. Stop when zero is absent from each image; then $c_2 = q$ works. Compactness and rational ellipticity prove termination. The images are computed by reducing the input cosets and the compact group action modulo sufficiently deep lattices and congruence kernels, with the finitely many fixed conjugations included. This is a terminating procedure on supplied elliptic data, assuming effective local group and lattice operations; it is not a decision procedure for arbitrary nonelliptic covectors. In relative semisimple rank zero there are no root tests, and $G(F)/Z$ itself is compact.

30.2. **Finite comparison of actual inducing pairs.** Let (J_i, ρ_i), for $i = 1, 2$, be pairs in which $J_i \supset Z$ is open and compact modulo Z, ρ_i is finite-dimensional and smooth with central character ω_i, and

$$\Pi_i = \text{c-Ind}_{J_i}^{G(F)} \rho_i$$

is irreducible supercuspidal. The next argument does not require either ρ_i to contain a linear character with elliptic coadjoint coset. The inclusion $Z \subset J_i$ is automatic for an irreducible compact induction with a central character: a central element acts by a scalar, whereas translating a nonzero function supported on J_i moves its support to a central translate of J_i. Equality of these supports forces the central element to belong to J_i.

Choose finite sets $\mathcal{R} \subset C_1$ and $\mathcal{S} \subset C_2$ such that $C_1 \subset \bigcup_{r\in\mathcal{R}} J_1 r$ and $C_2 \subset \bigcup_{s\in\mathcal{S}} s J_2$. Thus every double coset in $J_1 \backslash G(F)/J_2$ has a representative $rads$ with $r \in \mathcal{R}$, $a \in \mathcal{A}^+$, $d \in D$, and $s \in \mathcal{S}$. For $\alpha \in \Delta$, let U_α^- be the opposite maximal unipotent radical. Put

$$D_{r,\alpha} = r^{-1} J_1 r \cap U_\alpha^-(F), \qquad \rho_{1,r}(u) = \rho_1(rur^{-1}).$$

These groups are compact: a unipotent radical embeds as a closed subgroup of $G(F)/Z$.

Lemma 30.2. *For every $r \in \mathcal{R}$ and $\alpha \in \Delta$, there are at most $\dim \rho_1$ elements $u_{r,\alpha,l} \in D_{r,\alpha}$ such that*

$$\sum_l \mathrm{im}(\rho_{1,r}(u_{r,\alpha,l}) - 1) = V_{\rho_1}. \tag{150}$$

Proof. Mackey restriction to the corresponding parabolic, followed by unnormalized Jacquet coinvariants, has a summand obtained by compact induction of $(\rho_{1,r})_{D_{r,\alpha}}$ to the Levi subgroup. To check this assertion, write $Q = r^{-1} J_1 r \cap P$ and let $\overline{Q}$ be its image in the Levi. Compact-induction adjunction and the defining adjunction of coinvariants give

$$(\text{c-Ind}_Q^P \rho_{1,r})_{U_\alpha^-} \simeq \text{c-Ind}_{\overline{Q}}^{M_P} (\rho_{1,r})_{Q\cap U_\alpha^-}.$$

The parabolic double-coset set is finite because $G(F)/P(F)$ is compact and J_1 is open. Since Π_1 is cuspidal, its Jacquet module vanishes, so $(\rho_{1,r})_{D_{r,\alpha}} = 0$. Equivalently, the images of $\rho_{1,r}(u) - 1$ span V_{ρ_1}. Successively choosing an element whose image enlarges the span proves (150) in at most $\dim \rho_1$ steps. □

Choose a compact open subgroup V contained in

$$\bigcap_{d\in D,\ s\in\mathcal{S}} ds \ker(\rho_2) s^{-1} d^{-1}.$$

For every negative indivisible relative root b, choose a filtration level q_b such that $U_{b,q_b} \subset V$. Write the finitely many elements in (150) in ordered relative root coordinates, $u_{r,\alpha,l} = \prod_b u_{r,\alpha,l,b}$. Only roots whose negatives have positive α-coefficient occur. For multipliable roots use their actual relative root groups and Bruhat–Tits valuations φ_b. Set

$$B = 1 + \max\big(\{0\} \cup \{q_b - \varphi_b(u_{r,\alpha,l,b}) : u_{r,\alpha,l,b} \neq 1\}\big). \tag{151}$$

Choose a finite set $\mathcal{A}_B \subset \mathcal{A}^+$ representing the cocharacter classes modulo $X_*(A_G)$ with $0 \le \langle \alpha, \lambda \rangle < B$ for every α, and put

$$\mathcal{D}_B = \{rads : r \in \mathcal{R},\ a \in \mathcal{A}_B,\ d \in D,\ s \in \mathcal{S}\}.$$

Theorem 30.3. *The representations Π_1 and Π_2 are isomorphic if and only if $\omega_1 = \omega_2$ and, for at least one $g \in \mathcal{D}_B$, the finite system*

$$T\rho_1(h) = \rho_2(g^{-1}hg)T \qquad (h \in J_1 \cap gJ_2g^{-1}) \tag{152}$$

has a nonzero solution. The bound and all the systems are obtained by finite operations from the inducing data and the rank certificates.

Proof. Different central characters exclude a nonzero intertwiner. Suppose they agree, and consider a representative $g = rads$ with $a = \lambda(\varpi) \in \mathcal{A}^+$. If $\langle \alpha, \lambda \rangle \geq B$, every root coordinate used in $u_{r,\alpha,l}$ satisfies

$$\varphi_b(a^{-1}u_{r,\alpha,l,b}a) = \varphi_b(u_{r,\alpha,l,b}) - \langle b, \lambda \rangle \geq \varphi_b(u_{r,\alpha,l,b}) + \langle \alpha, \lambda \rangle > q_b.$$

Thus $a^{-1}u_{r,\alpha,l}a \in V$. By the choice of V, the element $h = ru_{r,\alpha,l}r^{-1}$ belongs to $J_1 \cap gJ_2g^{-1}$ and satisfies $\rho_2(g^{-1}hg) = 1$. Every solution of (152) therefore annihilates the left side of (150), and is zero. All nonzero Mackey terms are consequently represented in $\mathcal{D}_B$. Central cocharacters do not change the double coset since $Z \subset J_i$.

For such a representative, put $H_g = J_1 \cap gJ_2g^{-1}$. Choose a compact open subgroup N of H_g killed by both actions in (152), and replace it by its core under J_1. Only finitely many conjugates occur because J_1/Z is compact and the normalizer of N is open. The resulting subgroup is still compact open, is contained in H_g, and is killed by both actions. Equal central characters make the action on the Hom space trivial on Z as well. Hence the equations factor through the finite group $H_g/(ZN)$, and Gaussian elimination computes their solution space E_g.

Finally, Mackey decomposition and compact-induction adjunction give

$$\mathrm{Hom}_{G(F)}(\Pi_1, \Pi_2) \simeq \bigoplus_{g \in J_1 \backslash G(F)/J_2} E_g.$$

A map from the finite-dimensional inducing space has finitely many nonzero Mackey components, and the preceding bound specifies all possible components. Since the two representations are irreducible, this Hom space is nonzero exactly when they are isomorphic. Duplicate candidates in $\mathcal{D}_B$ do not affect the test for existence of a nonzero solution. In relative semisimple rank zero the root certificates are empty and the Cartan list is already finite modulo the center, so the same conclusion holds. □

For an algorithmic interpretation, the field, root filtrations, compact quotients, intersections, and supplied representation matrices must admit exact effective operations. Each $D_{r,\alpha}$ acts through a finite group, so its rank certificate is a finite matrix calculation. Alternatively, enumerating an effective dense subset finds the certificate in finitely many steps because cuspidality guarantees (150) and the action is locally constant. The finite intersection quotients in (152) are computed inside $J_1/(ZN)$. Central characters are compared on finitely many generators modulo a common open kernel. These hypotheses concern the presentation of the input data; the mathematical equivalence criterion does not restrict G or F.

In particular, one may first twist and descend each constructed inducing pair to $G(F)$ and then apply Theorem 30.3. No common central extension and no comparison of the auxiliary choices is required. The theorem also compares the finite cuspidal base data with the deep coadjoint constructions. It gives a complete equivalence test for the representations supplied by these constructions. An assertion that every supercuspidal representation is supplied by them remains a separate exhaustion assertion.

31. Occurrence multiplicity and formal degree

Let G be the group of rational points of a connected reductive group over a nonarchimedean local field, and let Z be its center. Fix a Haar measure on G/Z. Let π be an irreducible admissible unitary supercuspidal representation of G with central character χ and formal degree d_π. Write V for its space of smooth vectors and $\mathcal{V}$ for its Hilbert completion. Thus the normalization of the formal degree gives

$$\int_{G/Z} |\langle \pi(g)v, w \rangle|^2 \, dg = d_\pi^{-1} \|v\|^2 \|w\|^2.$$

Let $J \supseteq Z$ be open and compact modulo Z, and let τ be an irreducible smooth representation of J occurring in V. Its central character is χ. Equip its finite-dimensional space V_τ with a J-invariant Hermitian inner product, and put

$$v_J = \mathrm{vol}(J/Z), \qquad m = \dim \mathrm{Hom}_J(\tau, V), \qquad r = \frac{\dim \tau}{d_\pi v_J}.$$

The integer m is finite: a compact open subgroup acting trivially on τ embeds its entire τ-isotypic space into a finite-dimensional fixed-vector space of the admissible representation V.

Proposition 31.1. *With this notation, $1 \le m \le r$. Equality $m = r$ holds if and only if*

$$\text{c-Ind}_J^G \tau \simeq \pi^{\oplus m}.$$

In particular, $r = 1$ if and only if $\text{c-Ind}_J^G \tau \simeq \pi$.

Proof. Use the convention $f(jg) = \tau(j)f(g)$ and right translation in compact induction. Its unitary completion $\mathcal{H}$ has norm

$$\|f\|_{\mathcal{H}}^2 = \sum_{Jg \in J\backslash G} \|f(g)\|^2.$$

The summands are independent of the representatives because τ is unitary. In particular, the canonical map $\iota : V_\tau \to \mathcal{H}$, whose image consists of functions supported on J and whose value at 1 is the supplied vector, is an isometry. For a left J-invariant integrable function F on G/Z, the Haar normalization is

$$\int_{G/Z} F(g)\, dg = v_J \sum_{Jg \in J\backslash G} F(g).$$

Choose J-equivariant isometric embeddings $T_1, \dots, T_m : V_\tau \to V$ with mutually orthogonal ranges. They give all the copies of τ in V. Define

$$(\Phi_i v)(g) = r^{-1/2} T_i^* \pi(g) v \qquad (v \in V).$$

The functions satisfy the required J-equivariance. Each coordinate is a smooth matrix coefficient of π, so it has compact support modulo Z. Because $Z \subseteq J$, the function $\Phi_i v$ belongs to $\text{c-Ind}_J^G \tau$.

Let $e_1, \dots, e_{\dim \tau}$ be an orthonormal basis of V_τ. Schur orthogonality, including its polarized form, and the preceding integration formula give

$$\begin{aligned}\langle \Phi_i v, \Phi_j w\rangle_{\mathcal{H}} &= \frac{1}{r v_J} \sum_b \int_{G/Z} \langle \pi(g)v, T_i e_b\rangle \overline{\langle \pi(g)w, T_j e_b\rangle}\, dg \\ &= \delta_{ij} \frac{\dim \tau}{r v_J d_\pi} \langle v, w\rangle = \delta_{ij}\langle v, w\rangle.\end{aligned}$$

Here the inner products are linear in the first variable. Thus the Φ_i extend to isometric G-equivariant maps $\mathcal{V} \to \mathcal{H}$ with mutually orthogonal closed ranges.

These ranges exhaust the closed π-isotypic subspace of $\mathcal{H}$. Indeed, any bounded G-equivariant map $A : \mathcal{V} \to \mathcal{H}$ is determined by evaluation at 1, since $(Av)(g) = (A\pi(g)v)(1)$. The adjoint of this evaluation map is a J-equivariant map $V_\tau \to \mathcal{V}$. Its image consists of smooth vectors, since V_τ has an open kernel. It is therefore a linear combination of the T_i. The same description applies to an algebraic G-equivariant map $A : V \to \text{c-Ind}_J^G \tau$. To see the boundedness needed here, its evaluation map $E : V \to V_\tau$ satisfies $E = Ee_\tau$, where e_τ is the orthogonal projection onto the finite-dimensional τ-isotypic space in V. This follows either from compact-group decomposition or from the formula

$$e_\tau = \frac{\dim \tau}{v_J} \int_{J/Z} \overline{\operatorname{tr} \tau(j)}\, \pi(j)\, dj.$$

The integrand is independent of the lift from J/Z, because the two central characters cancel. Thus E is bounded and its adjoint is a linear combination of the T_i. Evaluation then expresses A as a linear combination of the displayed coefficient maps, which are bounded by the orthogonality calculation.

Let $P = \sum_i \Phi_i \Phi_i^*$ be this isotypic orthogonal projection. The support of ιe is the single coset J, so

$$\Phi_i^* \iota e = r^{-1/2} T_i e, \qquad \iota^* P \iota = \frac{m}{r} \operatorname{Id}_{V_\tau}.$$

Since $0 \le P \le 1$, this proves $m \le r$. If $m = r$, then

$$\|(1 - P)\iota e\|^2 = \langle (1 - P)\iota e, \iota e\rangle = 0.$$

The projection commutes with G, and the translates of the canonical fiber span the dense subspace $\text{c-Ind}_J^G \tau$. Hence $P = 1$.

This equality also gives the assertion for smooth compact induction, without introducing additional vectors from Hilbert completion. For a compactly supported function f, the vector $\Phi_i^* f$ is a finite sum of translates of vectors in $T_i(V_\tau)$, and therefore belongs to V. Consequently

$$f = \sum_i \Phi_i \Phi_i^* f \in \bigoplus_i \Phi_i(V).$$

The summands have zero intersection by orthogonality, and each is isomorphic to π. Conversely, if compact induction is algebraically a direct sum of m copies of π, its dense smooth subspace is contained in the range of P by the description of algebraic coefficient maps above. Thus $P = 1$, and its canonical-fiber diagonal gives $m = r$.

Finally, occurrence implies $m \geq 1$. If $r = 1$, the proved inequality and equality case give $m = 1$ and compact induction equal to π. If compact induction equals π, compact Frobenius reciprocity and Schur's lemma give $m = 1$, and the converse equality case gives $r = 1$. □

When $m = r > 1$, the conclusion is a homogeneous compact induction with

$$\mathrm{End}_G(\text{c-Ind}_J^G \tau) \simeq M_m(\mathbf{C}).$$

A primitive idempotent of this matrix algebra selects one copy of π. The proposition does not identify that idempotent with induction from a subgroup, nor does it make its translates have disjoint support. Thus multiplicity saturation alone does not provide a compact inducing pair for an individual copy when $m > 1$.

32. Extraction from a special vertex and a minimal parabolic

Let $G = \mathbf{G}(F)$ for a connected reductive group over a nonarchimedean local field F, let Z be its center, and let K be the full stabilizer of a special vertex o in the reduced Bruhat–Tits building X. Fix a minimal F-parabolic subgroup P_0 of G. We use the same notation for an algebraic subgroup and its group of F-points when no confusion is possible. There is no restriction on the characteristic, the residue characteristic, or the ramification of a splitting field.

We first give the geometric statement used for extraction. A finite-dimensional unitary representation τ of K defines a bundle with fibers V_x on the vertex orbit $G \cdot o = G/K$. Write $P_{y,x} : V_x \to V_y$ for the blocks of an operator P on $\bigoplus_{x \in G/K} V_x$. Finite propagation means that a single bound on $d(x, y)$ contains every pair for which $P_{y,x} \neq 0$.

Lemma 32.1. *Suppose that $\tau|_{K \cap P_0}$ is irreducible. Every G-equivariant orthogonal projection of finite propagation on the Hilbert direct sum of the fibers V_x is either zero or the identity.*

Proof. The reduced building of a group of relative semisimple rank zero is a point, and the assertion follows from irreducibility. Assume positive rank, and let $P \neq 0$. A nonzero column of P is a nonzero finitely supported vector v with $Pv = v$. This uses local finiteness of the building and finite propagation.

There is a regular Busemann function b with a unique minimum on $\mathrm{supp}(v)$. Here regular means that its endpoint lies in the interior of a chamber at infinity, with a nonzero component in each positive-rank factor. To verify the assertion, choose a point p and a support point x farthest from p, with $r = d(p, x) > 0$. Extend $[p, x]$ to a ray in an apartment, and normalize its Busemann function by $b(p) = 0$. Then $b(x) = -r$ and $b(y) \geq -d(p, y) \geq -r$ for every support point y. If equality held at a distinct y, convexity would give $b(m) \leq -r$ at the midpoint of x, y, whereas the strict CAT(0) midpoint inequality gives $d(p, m) < r$. This contradicts $b(m) \geq -d(p, m)$. Finally perturb the direction into the interior of a chamber at infinity containing the original direction. Under the retraction centered at this chamber, the finitely many Busemann values are linear in the direction, so the strict inequalities persist. Singleton support causes no difficulty.

Let P_ξ be the minimal parabolic stabilizing the regular endpoint of b. Iwasawa decomposition $G = P_\xi K$, as in [4, Proposition 9.1.1], makes P_ξ transitive on G/K, and $b(hx) - b(x)$ is independent of x for $h \in P_\xi$. Thus the finite set

$$\{b(x) - b(y) : P_{y,x} \neq 0\}$$

is independent of x. Let R be its maximum. Self-adjointness makes the global set of jumps symmetric, so $R \geq 0$.

If an off-diagonal block is nonzero, then $R > 0$. To see this, use equivariance to put one endpoint at o. The group K acts transitively on the apartments containing o, so move the nonzero displacement into an apartment containing o and the endpoint of b. Since o is special, the apartment normalizer inside K realizes

the full relative finite Weyl group W. The span of the W-orbit of a nonzero displacement is the sum of the irreducible factors in which that displacement is nonzero. A regular Busemann functional is nonzero on each such factor. Some Weyl translate therefore has nonzero Busemann displacement. Equivariance preserves the nonzero block, and self-adjointness supplies a positive jump. This reasoning applies equally to reducible buildings and to ramified groups.

Suppose there is an off-diagonal block, and let x be the unique minimum of b on $\operatorname{supp}(v)$. For every y with $b(y) = b(x) - R$, the point y is outside that support. Every other support point z satisfies $b(z) - b(y) > R$, and hence $P_{y,z} = 0$. Consequently

$$0 = v_y = (Pv)_y = P_{y,x} v_x.$$

The subspace

$$N_\xi(x) = \bigcap_{b(y)=b(x)-R} \ker P_{y,x}$$

is nonzero, since it contains v_x. It is proper, since the definition and uniformity of R give a nonzero maximal descending block from x. It is stable under $G_x \cap P_\xi$: such an element fixes x, preserves the Busemann values, and permutes the blocks in this intersection.

Every minimal parabolic is K-conjugate to P_0. Indeed, write a conjugating element as kp using $G = KP_0$. Therefore the restriction of V_x to $G_x \cap P_\xi$ is irreducible. The subspace just constructed is a contradiction. All off-diagonal blocks of P vanish. Its block at o is a K-equivariant projection on the irreducible representation τ, and is therefore zero or the identity. Equivariance and $P \neq 0$ give the identity at every vertex. □

Theorem 32.2. *Let π be an irreducible unitary supercuspidal representation of G, and let τ be an irreducible representation of K occurring in π. If $\tau|_{K\cap P_0}$ is irreducible, then*

$$\pi \simeq \text{c-Ind}_K^G \tau.$$

In particular, this holds whenever an occurring τ is one-dimensional.

Proof. Let $T : V_\tau \to \pi$ be an isometric K-equivariant embedding. With counting norm on $K\backslash G$, formal-degree orthogonality makes

$$v \longmapsto \left(g \longmapsto \left(\frac{d_\pi \operatorname{vol}(K/Z)}{\dim \tau} \right)^{1/2} T^* \pi(g) v \right)$$

an isometric G-equivariant coefficient embedding into unitary induction. Its orthogonal range projection has finite propagation: there are finitely many coefficient functions from the finite-dimensional space $T(V_\tau)$, and each has compact support modulo Z. Identifying $K\backslash G$ with G/K by inversion gives the bundle in Lemma 32.1. That lemma makes the range projection the identity. The coefficient embedding takes smooth vectors into compact induction, and its adjoint takes each compactly supported function into a finite sum of smooth vectors. Thus it identifies the smooth compact induction with π. □

There is also a sufficient condition that uses only the supplied compact representation. Write U_Q for the unipotent radical of a parabolic Q.

Corollary 32.3. *Let τ be an irreducible finite-dimensional smooth representation of K with unitary central character. Suppose that*

$$\tau|_{K\cap P_0} \text{ is irreducible}, \qquad \tau^{K\cap U_Q} = 0 \quad \text{for every proper maximal } F\text{-parabolic } Q.$$

Then $\text{c-Ind}_K^G \tau$ *is irreducible supercuspidal.*

Proof. Put $\mathcal{I} = \text{c-Ind}_K^G \tau$. The stated invariant vanishing holds for every proper parabolic: its radical contains the radical of a maximal parabolic containing it. Let $Q = LU$ be such a parabolic. Iwasawa decomposition $G = KQ$ permits representatives in L for $K\backslash G/U$. Mackey decomposition on restriction to U, followed by compact-induction coinvariants, identifies each summand of $\mathcal{I}_U$ with

$$\tau_{K\cap lUl^{-1}} = \tau_{K\cap U} = 0 \qquad (l \in L).$$

The last equality follows from compact averaging: $K \cap U$ is compact, so invariants and coinvariants of its finite-dimensional complex representation are isomorphic. Hence every proper Jacquet module of $\mathcal{I}$ is zero.

We record why $\mathcal{I}$ is admissible. Choose an arbitrarily small compact open subgroup D normal in K. In the Cartan decomposition, the dominant representatives have the form

$$a = zbm \prod_i a_i^{n_i}, \qquad z \in Z, \quad m \in M \cap K, \quad b \in B, \quad n_i \geq 0,$$

where M is the minimal Levi, B is finite, and a_i are positive integral multiples of the fundamental coweights in the split center of M. Taking multiples and the finite set B accounts for the lattice index and the compact part of M modulo its split center. For the opposite maximal radical U_i^-, conjugation by a_i^{-1} is strictly contracting, whereas the other a_j^{-1} are weakly contracting.

For clarity, the simultaneous contraction can be arranged inside an arbitrary D. The unipotent group U_i^- has equivariant affine coordinates of positive weight for a_i^{-1} and nonnegative weight for every a_j^{-1}. A sufficiently small integral coordinate neighborhood lies inside $D \cap U_i^-$ and is preserved by all these weak contractions. The compact set $b^{-1}(K \cap U_i^-)b$ is carried into this neighborhood by a sufficiently large power of a_i^{-1}, uniformly for the finitely many $b \in B$. Commutativity of the a_j and normality of D in K now give

$$K \cap U_i^- \subseteq aDa^{-1} \quad \text{whenever } n_i \text{ is sufficiently large.}$$

The associated Mackey summand $\tau^{K \cap aDa^{-1}}$ of $\mathcal{I}^D$ therefore vanishes. Only a bounded set of dominant exponents remains, which meets finitely many double cosets in $K\backslash G/D$. Each summand has dimension at most $\dim \tau$, so $\mathcal{I}^D$ is finite-dimensional. Arbitrarily small such D prove admissibility.

The representation $\mathcal{I}$ is generated by its finite-dimensional canonical fiber, so it has an irreducible quotient π. This quotient is admissible, has the prescribed central character, and has zero proper Jacquet modules; hence it is supercuspidal. Its unitary central character gives a unitary realization. Compact Frobenius reciprocity makes τ an occurring K-representation in π. Theorem 32.2 identifies π with $\mathcal{I}$, proving irreducibility as well. □

The compact conditions are finite tests, with a qualification concerning the center. Choose a compact open normal subgroup $D \subseteq \ker \tau$ of K. For $H = K \cap P_0$, the quotient $H/(Z \cdot (H \cap D))$ is finite. Matrices for representatives of this quotient determine whether $\tau|_H$ is irreducible; the omitted central elements act by scalars. Thus the relevant projective image is finite even when the unitary central character has infinite image. The images of the compact groups $K \cap U_Q$ are finite, so their invariant spaces are computed by finite averaging. There are only finitely many K-conjugacy classes of parabolics to test, one for each standard type, again by Iwasawa decomposition. No assertion here makes an arbitrary supercuspidal representation contain a type satisfying these hypotheses.

33. Compact induction without a tame splitting hypothesis

The conclusion that a supercuspidal representation is compactly induced requires less than the conclusion that it comes from a tame datum over the original field. The following consequence of the known exhaustion and equivalence theorems separates these conclusions.

Lemma 33.1. *Let E be a nonarchimedean local field of odd residue characteristic, and let H be a tamely ramified connected reductive E-group with simply connected derived group. Let (K, κ) be the full inducing pair of a generic cuspidal Yu datum, and put $\tau = \text{c-Ind}_K^{H(E)} \kappa$. If an E-automorphism α of H satisfies $\alpha\tau \simeq \tau$, there is $h \in H(E)$ such that*

$$\alpha(K) = hKh^{-1}, \qquad \alpha\kappa \simeq {}^h\kappa.$$

Here a representation is transported to the image of its group by precomposition with the inverse automorphism.

Proof. Transport the datum, its point in the building, and the choices in its Heisenberg constructions by α. Moy–Prasad groups, genericity, commutator pairings, and the finite Heisenberg and Weil representations are transported by these isomorphisms. Thus the transported datum has full inducing pair $(\alpha(K), \alpha\kappa)$ and produces $\alpha\tau$.

Hypothesis C for every twisted Levi sequence in H follows from [13, Lemma 3.5.2], since the derived group is simply connected. Hakim–Murnaghan's equivalence theorem [12, Theorem 6.6] therefore applies to these two extended generic cuspidal data. It relates them by conjugation, refactorization, and elementary transformations. Refactorization preserves the inducing subgroup and its representation [12, Proposition 4.24]; elementary transformations do the same. These operations commute with conjugation. Consequently the full inducing pairs are $H(E)$-conjugate, as also stated explicitly in the proof of [12, Theorem 6.6].

These references allow local fields of positive characteristic. Hakim–Murnaghan assume odd residue characteristic, and [13, Section 2.1] allows either characteristic in its Section 3.5. No characteristic-zero hypothesis enters this argument. □

Lemma 33.2. *Let Γ be a second countable locally profinite group, let Z be a closed central subgroup, and let $N \supset Z$ be an open normal subgroup of finite index. Let $K \supset Z$ be open and compact modulo Z, and let κ be a finite-dimensional irreducible smooth representation of K with central character. Suppose that $\nu = \mathrm{c\text{-}Ind}_K^N \kappa$ is irreducible. Define*

$$I_\nu = \{g \in \Gamma : {}^g\nu \simeq \nu\}, \qquad P = \{g \in \Gamma : gKg^{-1} = K,\ {}^g\kappa \simeq \kappa\}.$$

If $I_\nu = NP$, then $I_\Gamma(\kappa) = P$ and P/K is finite. For every irreducible constituent σ of $\mathrm{Ind}_K^P \kappa$, the representation $\mathrm{c\text{-}Ind}_P^\Gamma \sigma$ is irreducible. Every irreducible smooth representation of Γ containing ν upon restriction to N is obtained in this way.

Proof. Mackey decomposition and Schur's lemma give $I_N(\kappa) = K$: the identity double coset already contributes one dimension to $\mathrm{End}_N(\nu) = \mathbf{C}$, so no other double coset contributes. Here Schur's lemma over $\mathbf{C}$ applies because ν is irreducible and countable-dimensional. Every intertwiner of κ in Γ gives, by Mackey decomposition over N, a nonzero homomorphism between ν and its conjugate. Thus $I_\Gamma(\kappa) \subset I_\nu$. If $g = np \in I_\Gamma(\kappa)$ with $n \in N$ and $p \in P$, then n intertwines κ, whence $n \in K$. This proves $I_\Gamma(\kappa) = P$. In particular, $P \cap N = K$, and P/K embeds in the finite group Γ/N. The subgroup P is therefore open and compact modulo Z.

The finite-dimensional representation $\mathrm{Ind}_K^P \kappa$ is semisimple. Indeed, its restriction to K is a direct sum of copies of κ. A K-equivariant projection onto any P-invariant subspace can be averaged over P/K to give a P-equivariant projection. Every irreducible constituent σ consequently restricts to a direct sum of copies of κ. If g intertwines σ, restrict a nonzero intertwiner to $K \cap gKg^{-1}$ and take a nonzero matrix entry between these copies. This shows that g intertwines κ, hence $g \in P$. Thus $I_\Gamma(\sigma) = P$.

We recall the compact-induction irreducibility argument in this form. In $\mathrm{c\text{-}Ind}_P^\Gamma \sigma$, Mackey decomposition and $I_\Gamma(\sigma) = P$ show that the σ-isotypic subspace under P is exactly the inducing fiber supported on P. Projection onto that isotypic subspace is given by compact averaging modulo Z; the common central character makes the averaging well defined. Given a nonzero subrepresentation, translate one of its nonzero functions so that its value at 1 is nonzero. Evaluation at 1 is P-equivariant, so the isotypic projection preserves this nonzero value. The subrepresentation therefore meets the inducing fiber. Irreducibility of σ, followed by translation by Γ, gives the entire compact induction. Finally, Frobenius reciprocity and induction in stages show that every irreducible representation containing ν is a quotient of the direct sum of these irreducible compact inductions, and hence is one of them. □

Theorem 33.3. *Let G be a connected reductive group over a nonarchimedean local field F of residue characteristic p. Let W be its absolute Weyl group. If $p \nmid |W|$, every irreducible smooth complex supercuspidal representation of $G(F)$ is compactly induced from a finite-dimensional irreducible smooth representation of an open subgroup compact modulo $Z(G)(F)$. No tame splitting hypothesis on G is required.*

Proof. Write $\Gamma = G(F)$, $Z = Z(G)(F)$, and let $q_D : D \to G_{\mathrm{der}}$ be the simply connected cover. The case that G is a torus is immediate, so suppose that its root system is nonempty. Then p is odd. The standard decomposition of a simply connected semisimple group gives finite separable extensions E_i/F and absolutely almost simple simply connected groups H_i/E_i with

$$D = \prod_{i=1}^{r} \mathrm{Res}_{E_i/F} H_i, \qquad D(F) = \prod_{i=1}^{r} H_i(E_i).$$

Each absolute Weyl group W_i of H_i has order prime to p. By [11, Proposition 2.1], every maximal torus of H_i splits over a finite tame extension of E_i. Splitting such a maximal torus splits H_i, so H_i itself is tamely ramified over E_i. Fintzen's exhaustion theorem [10, Theorem 8.1] therefore applies to $H_i(E_i)$.

We first verify the finite-index assertion needed for Clifford theory. The multiplication map

$$q : D \times Z(G)^\circ \longrightarrow G, \qquad (d, z) \longmapsto q_D(d)z,$$

is a central isogeny. Its kernel C embeds, by projection to D, in $Z(D)$. The order of the center of each absolutely almost simple simply connected factor divides the order of its Weyl group. Thus C has order prime to p and is finite etale. The image $N_0 = q(D(F) \times Z(G)^\circ(F))$ is open, because an etale isogeny is a

local analytic isomorphism on rational points. The boundary map embeds Γ/N_0 in the finite abelian group $H^1(F, C)$.

For completeness, finiteness here does not require tame splitting of C. Choose a finite Galois extension L/F which splits C and contains the roots of unity of its exponent n. Inflation–restriction reduces finiteness to $H^1(L, C)$ and cohomology of the finite group $\mathrm{Gal}(L/F)$ with finite coefficients. Over L, the group C is a product of groups μ_m with $m \mid n$, and $H^1(L, \mu_m) = L^\times/(L^\times)^m$ is finite. Indeed, m is prime to p, so the mth-power map is an automorphism on principal units; the valuation and residue-unit quotients are finite.

Set $N = q_D(D(F))Z$. This is an open normal subgroup of finite index containing N_0 and Z. Let π be an irreducible supercuspidal representation of Γ, with central character ω. Choose an irreducible constituent ν_0 of $\pi|_{N_0}$. Such a constituent exists: restriction to a subgroup of finite index preserves finite generation and admissibility, and pullback along q gives a finitely generated admissible representation of the reductive group $(D \times Z(G)^\circ)(F)$, hence one of finite length. The finitely many Γ-translates of its socle generate π, so restriction is semisimple. Since Z acts by ω, the subspace carrying ν_0 is stable under N and gives an irreducible representation ν of N.

The pullback of ν_0 along q is an external tensor product

$$q^*\nu_0 \simeq (\boxtimes_i \tau_i) \boxtimes \omega|_{Z(G)^\circ(F)}.$$

Each τ_i is irreducible and supercuspidal. To see the latter assertion directly, parabolics correspond under a central isogeny and the map on their unipotent radicals is an isomorphism. The Jacquet modules of $q^*(\pi|_{N_0})$ are therefore the corresponding Jacquet modules of π, and vanish for proper parabolics. This remains true for its direct summands, and the product formula for Jacquet modules gives supercuspidality of every τ_i.

Choose a Yu inducing pair (K_i, κ_i) for τ_i, and put $K_D = \prod_i K_i$ and $\kappa_D = \boxtimes_i \kappa_i$. Each K_i is compact, since H_i has finite center. The continuous open surjection

$$q_Z : D(F) \times Z \longrightarrow N, \qquad (d, z) \longmapsto q_D(d)z,$$

has finite central kernel, and

$$q_Z^*\nu \simeq \text{c-Ind}_{K_D \times Z}^{D(F)\times Z}(\kappa_D \boxtimes \omega).$$

Every element of $\ker(q_Z)$ acts trivially on this compact induction. If such an element did not lie in $K_D \times Z$, it would translate the nonzero inducing fiber to functions on a disjoint coset, a contradiction. Its action on that fiber is therefore trivial as well. The inducing representation descends, and the identification of the two coset spaces identifies compactly supported equivariant functions. We obtain a pair (K, κ) with

$$K = q_D(K_D)Z, \qquad \nu \simeq \text{c-Ind}_K^N \kappa.$$

Here K is open and compact modulo Z and κ is irreducible and finite-dimensional.

It remains to check full inertia, rather than to assume that this pair extends. Put $I = \{g \in \Gamma : {}^g\nu \simeq \nu\}$. Conjugation by $g \in \Gamma$ lifts uniquely to an automorphism α_g of D. The identification

$$G/Z(G) \simeq D/Z(D) \simeq \prod_i \mathrm{Res}_{E_i/F}(H_i/Z(H_i))$$

shows that α_g acts separately on each $H_i(E_i)$ through an E_i-rational inner automorphism $\alpha_{g,i}$. In particular, it does not permute the factors or introduce a field automorphism. If $g \in I$, pullback along q_Z, followed by uniqueness of external tensor factors, gives $\alpha_{g,i}\tau_i \simeq \tau_i$ for every i. Lemma 33.1 supplies $h_i \in H_i(E_i)$ with

$$(\alpha_{g,i}(K_i), \alpha_{g,i}\kappa_i) \simeq (h_i K_i h_i^{-1}, {}^{h_i}\kappa_i).$$

Let $h = (h_i)$ and $n = q_D(h) \in N$. The element $n^{-1}g$ normalizes K and preserves κ: this is the displayed factorwise equality after descent, and conjugation fixes the central factor ω. Therefore, for

$$P = \{g \in \Gamma : gKg^{-1} = K,\ {}^g\kappa \simeq \kappa\},$$

we have $I = NP$. Lemma 33.2 now gives

$$\pi \simeq \text{c-Ind}_P^\Gamma \sigma,$$

where P is open and compact modulo Z, and σ is an irreducible constituent of the finite induction $\mathrm{Ind}_K^P \kappa$. This proves the assertion. □

Corollary 33.4. *Under the hypotheses of Theorem 33.3, use its notation D, H_i, E_i, q_D, Z, and N. Choose, for each i, a generic cuspidal Yu datum over E_i with full inducing pair (K_i, κ_i), representation τ_i, and central character ω_i. Put $K_D = \prod_i K_i$, $\kappa_D = \boxtimes_i \kappa_i$, and $\omega_D = \boxtimes_i \omega_i$ on $Z(D)(F)$. Choose a smooth character $\chi : Z \to \mathbf{C}^\times$ satisfying the finite compatibility condition*

$$\omega_D(c) = \chi(q_D(c)) \qquad (c \in Z(D)(F)).$$

Then $\kappa_D \boxtimes \chi$ descends to a representation κ of $K = q_D(K_D)Z$. Let P be the stabilizer of this pair under conjugation by $G(F)$, and choose an irreducible constituent σ of $\operatorname{Ind}_K^P \kappa$. The representation

$$\text{c-Ind}_P^{G(F)} \sigma$$

is irreducible and supercuspidal. These data give every irreducible supercuspidal representation of $G(F)$. Equivalence of the resulting representations is determined by the finite intertwining criterion of Theorem 30.3 applied to (P, σ). For a torus, the data consist simply of the character χ.

Proof. Every K_i contains $Z(H_i)(E_i)$ and κ_i has central character ω_i, as follows either from the construction or from the inducing-fiber support argument. The kernel of $q_Z : D(F) \times Z \to N$ consists exactly of $(c, q_D(c)^{-1})$ with $c \in Z(D)(F)$. The stated compatibility therefore makes $\kappa_D \boxtimes \chi$ trivial on this kernel. Its descent satisfies

$$q_Z^*\big(\text{c-Ind}_K^N \kappa\big) \simeq (\boxtimes_i \tau_i) \boxtimes \chi.$$

The right side is irreducible, so $\nu = \text{c-Ind}_K^N \kappa$ is irreducible. The factorwise argument using Lemma 33.1 in the proof of Theorem 33.3 gives $I_\nu = NP$ for these supplied data. Lemma 33.2 proves finiteness of P/K and irreducibility of the displayed compact induction.

An irreducible smooth complex representation of $G(F)$ is admissible. The inducing fiber and its dual supply a nonzero matrix coefficient supported on P, which is compact modulo Z. Their translates generate the representation and its smooth contragredient, so all matrix coefficients are compactly supported modulo Z. This proves supercuspidality. Conversely, the proof of Theorem 33.3 obtains precisely these data from an arbitrary supercuspidal, with χ equal to its central character. The cited finite intertwining criterion compares the resulting inducing pairs without identifying their auxiliary choices. □

The proof uses tame Yu data over the fields E_i and finite Clifford theory over F. It does not assert that G or its maximal tori are tamely ramified over F. For example, over $F = \mathbf{F}_3((t))$, let E/F be the Artin–Schreier extension defined by $u^3 - u = t^{-1}$. The group $\operatorname{Res}_{E/F} \mathrm{SL}_2$ has absolute Weyl group of order 8, but wild inertia permutes its three absolute factors, so it cannot split over a tame extension of F. Its rational points are nevertheless $\mathrm{SL}_2(E)$, and the theorem applies. Arbitrary central tori, including wildly ramified ones, are likewise allowed. This is a consequence of the cited exhaustion and equivalence theorems, not an exhaustion claim at primes dividing $|W|$.

34. The remaining extraction problem

For $G = \mathrm{PGL}_2$ in mixed characteristic with residue characteristic two, write h for the class of $\operatorname{diag}(1, 0)$ in its integral toral Lie algebra. The coroot differential is $2h$. If $\beta(h) = 1$, the coroot-lattice Smith exponent is $e = \mathrm{val}(2)$. Proposition 2.1 applies with $m \geq 2e + 1$, giving

$$P/H \simeq (\mathcal{O}/\varpi^{e+1}\mathcal{O})^2, \qquad \dim \omega_\theta = q^{e+1}.$$

This repairs the finite pairing by changing the actual root subgroups.

It does not give a generic representative in the original filtration coset. Identify the dual of $\mathfrak{pgl}_2(\mathcal{O})$ with trace-zero matrices by the trace pairing, and take $B = \operatorname{diag}(1, -1)$. Every matrix $B' \in B + \varpi\mathfrak{sl}_2(\mathcal{O})$ has unit determinant and eigenvalues $\lambda, -\lambda$ of valuation zero over a splitting field. Its coroot value is 2λ, of valuation e, whereas its depth at the original vertex is zero. Thus no such representative satisfies the toral genericity equality at depth zero. Scaling the coset shifts the two valuations equally and leaves their gap unchanged. The correction margin in Lemma 1.1 cannot be discarded.

In equal characteristic two the same coroot differential is zero. Replacing it by the formally divided vector h does not change the zero differential of the actual root commutator. More generally, for a central homomorphism $f : \widetilde{G} \to G$ and a pulled-back character, one has the exact identity

$$B_{\theta \circ f}(u, v) = B_\theta(f(u), f(v))$$

whenever the indicated finite commutator quotients are defined. A pulled-back pairing cannot acquire directions absent from the actual commutator by a change of terminology. Changing filtrations or retaining additional central and Clifford data is a further operation that needs its own occurrence argument.

Propositions 6.1 and 6.2 supply compactness, the actual inducing data, and exhaustion of the specified characteristic-two sectors. Their proof uses the entire character coset and the precision bound $m \geq 2\delta + 1$. An arbitrary occurring minimal-depth character need not satisfy that bound. Theorem 11.1 removes the group and splitting restrictions for a nonempty construction. Its central twist is applied after constructing a representation and can change both depth and the frequency of a congruence character. It therefore does not assert a correction within the original character coset.

Proposition 12.2 proves that no choice of the single deep linear character in Theorem 11.1 can cover the representation (61); Corollary 12.3 excludes repair by its central extensions and smooth twists. Thus universal occurrence in this family is false. The depth-zero cuspidal datum already gives the correct compact induction for that example. A full classification must allow a larger collection of data, including cuspidal representations of finite reductive quotients and further stages at positive depth. Extracting such data while preserving occurrence, and proving that extraction terminates for every supercuspidal, remain unresolved here.

One sufficient relative step would retain a smaller twisted Levi group M, a compact inducing representation τ on $K_M \subset M(F)$, and an occurring representation λ on a subgroup $K = JK_M \subset G(F)$, open and compact modulo $Z(G)(F)$. Here J denotes the additional compact subgroup supplied by the step. The essential intertwining calculation would be

$$I_{G(F)}(\lambda) \subseteq JI_{M(F)}(\tau)J.$$

If the lower compact induction is irreducible, so that $I_{M(F)}(\tau) = K_M$, this inclusion confines all intertwining to K. For irreducible λ, the compact-induction argument then identifies the given representation with $\text{c-Ind}_K^{G(F)} \lambda$. Depth-zero exhaustion supplies a terminal case. What is missing is extraction of the occurring data and this relative intertwining comparison in small characteristic. The existence of a suitable reductive M is itself part of that problem; the coordinate calculations above do not establish it.

Proposition 28.3 supplies a genuine local part of extraction: it chooses a character that occurs, then corrects it by an actual group element. Lemma 28.4 computes the exact residual commutator form, including its quadratic correction. Neither result forces the precision hypothesis for an arbitrary input or gives a compatible sequence of residual groups and representations. Proposition 28.5 shows that geometric closedness of the orbit cannot be imposed on every occurring frequency coset in characteristic two.

Proposition 29.1 describes rationally cocharacter-closed covectors using elliptic covectors on F-Levi subgroups. The description holds in every characteristic. The Levi subgroup in this description need not be the stabilizer of the covector. In Proposition 29.2, the stabilizer is noncompact modulo center and contains a nontrivial smooth connected unipotent normal subgroup. Thus replacing geometric closedness by rational cocharacter closure requires a further representation-theoretic descent argument; the orbit description does not supply one.

The finite equivalence theorem applies once inducing pairs have been obtained, including pairs arising through different central extensions and twists. For a unitary supercuspidal π, Proposition 31.1 expresses the extraction target as an occurring pair (J, τ) satisfying

$$\dim \tau = d_\pi \operatorname{vol}(J/Z).$$

Every occurring pair satisfies the opposite weak inequality. Proving that equality occurs somewhere is an existence assertion; it does not follow from the inequality or from finite-dimensionality of an endomorphism algebra. Proposition 14.2 also tests a geometric compression on an actual supercuspidal representation. Removing the maximally descending Busemann blocks leaves the nonzero defect $QPQ - (QPQ)^2 = \frac{2}{9}Q$. A geometric extraction must produce a subspace that reduces the coefficient projection, rather than only annihilating its extreme matrix blocks.

Theorem 15.2 bounds the closure search for an arbitrary open subgroup compact modulo center. It terminates even when $Z(\mathcal{A}) = 0$: in that case it returns a finite certificate of failure for the supplied coordinate model. The representation can still have compact inducing data in a different model. Theorem 32.2 proves a positive result under a finite restriction condition on an occurring special-vertex type. Neither the existence of such a type for an arbitrary input nor a change of model guaranteed to succeed is established.

For a fixed occurring local type, Theorem 16.1 replaces the search over embeddings by finitely many equations. Proposition 16.2 gives a full multiplicity space whose successful locus consists of two points. Proposition 16.4 gives another occurring type whose entire locus is empty, although its supercuspidal representation has an explicit edge inducing pair. The compatibility criterion in Proposition 16.3 explains the distinction: the chosen local type must be a full induction from its intersection with an inducing subgroup, not just a constituent of that induction. A universal procedure must therefore change local types as well as embeddings and prove that a successful type is reached.

The global complement identity in Theorem 18.1 shows that $n < R < 2n$ is impossible, and every model with $R < 2n$ already extracts. Both endpoints of the diagonal bounds give finite coordinate blocks by Proposition 18.2. At $R = 2n$, a reducible complement gives a smaller occurring type. Theorem 19.1 instead changes the embedding explicitly when $c = 1/2$ and the complement is isomorphic to the original type. The rank of the coefficient blocks after a general change of type is not controlled by these statements. At the same rank boundary, Theorems 20.1 and 20.2 handle every model with two active neighbor orbits. The equality argument in Lemma 21.1 gives further extractions within the original coefficient model. Theorems 21.2, 21.3, and 22.4, together with Corollary 22.6, handle full central support, degree three, individual block ranks at most two, and prime fiber dimensions. Theorem 23.1 handles every commutative local range algebra. Theorem 24.2 also treats noncommutative algebras acting with multiplicity one under the incidence spanning condition (122). Consequently an uncovered boundary model has a single orbit preserved by reversal, degree at least four, individual block rank at least three, and composite fiber dimension at least twelve, by Corollary 26.2. Its local range algebra is noncommutative and has more than one central summand, some of which are omitted by an active range projection. In multiplicity one it also fails the more general recovery condition of Theorem 24.7. Under the spanning condition, Corollary 25.5 further excludes prime multiplicities and multiplicity four. These restrictions do not exclude every remaining local algebra.

The individual-orbit moves of Proposition 27.1 are defined for every total rank. Their strict lexicographic descent terminates within the explicit bound of Theorem 27.3. A terminal model with $R \leq 2n$ gives compact induction, but a terminal model with $R > 2n$ is not excluded. Corollary 27.5 proves the same necessary rank inequality for a pair minimizing the ratio and then the fiber dimension. The finite descent is not shown to attain that global minimum or to produce an inducing pair for every representation.

When $p \nmid |W|$, Theorem 33.3 proves exhaustion without assuming tame splitting over F, and Corollary 33.4 specifies the inducing data. Its proof uses the full-pair conjugacy theorem for Yu data to control the finite central-isogeny quotient. At primes dividing $|W|$, neither exhaustion on the absolute factors nor this conjugacy mechanism is supplied by that argument.

The unresolved assertion is universal extraction of such a pair. Even the underlying compact-induction statement remains Conjecture 1.1 in [14]; that work proves a geometric extraction theorem in relative rank one and formulates higher-rank conjectures. The local results here do not establish unrestricted higher-rank extraction or classification.

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